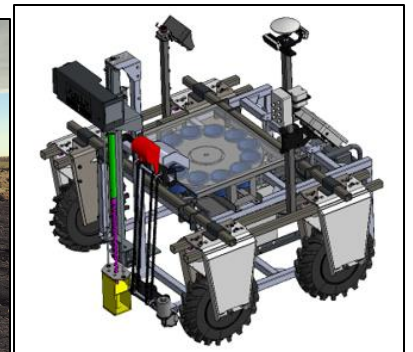
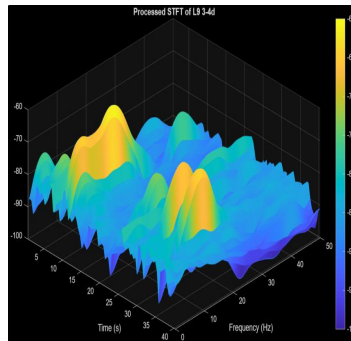
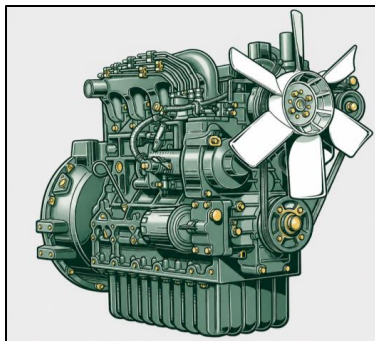
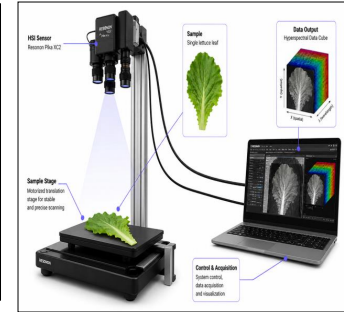
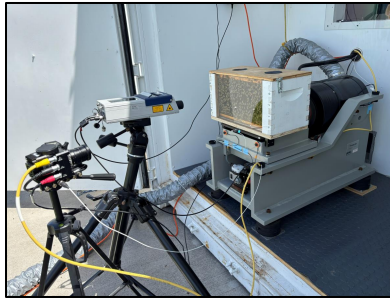
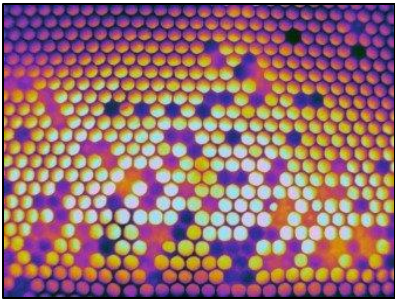
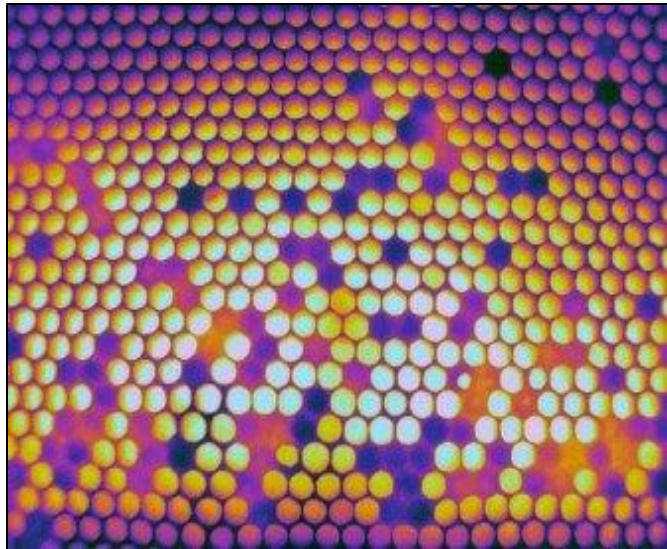
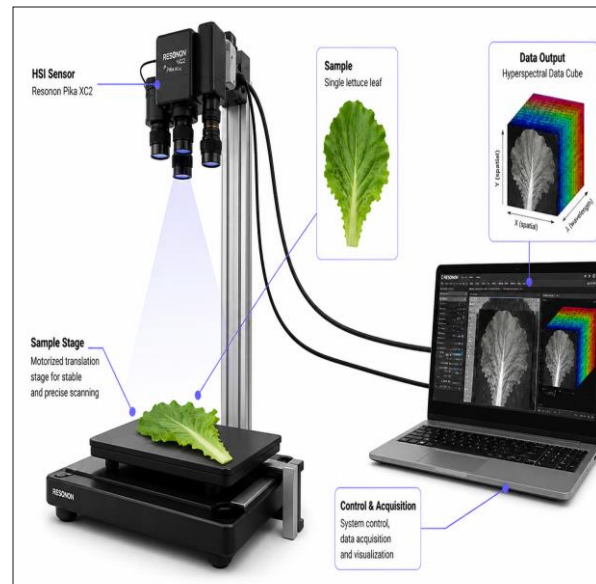




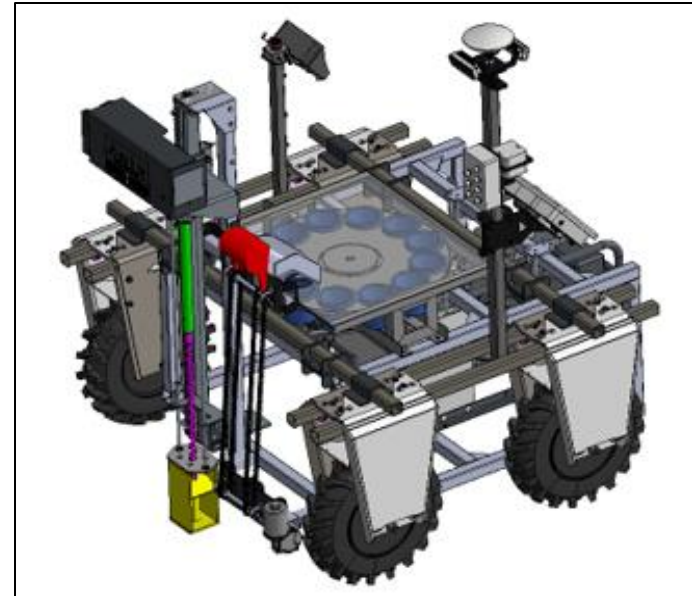
Domain # 1: Honeybees – Modeling, Identification, and Prediction

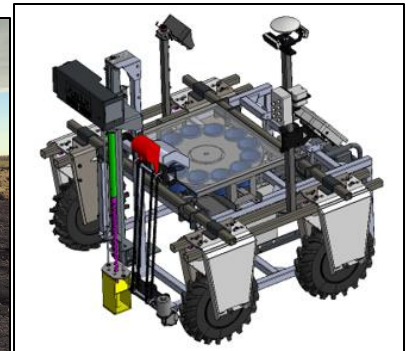
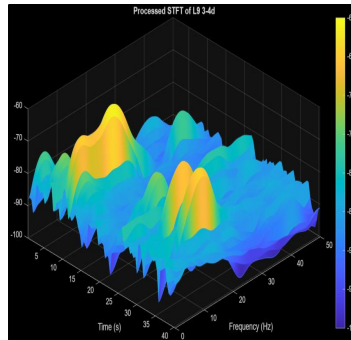
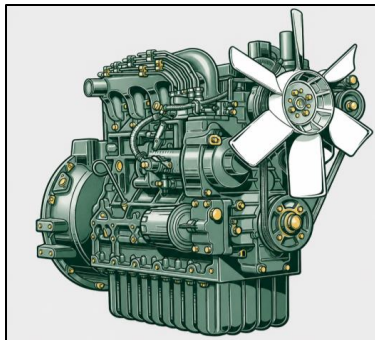


Domain # 2: Controlled Environment Agriculture



Domain # 3: Robotics and Robot Applications





Domain # 1: Save Honeybees – Modeling, Identification, and Prediction

Brady Lindsey, Jewel woodcock, Shoaib Ahmmad, Joanna Daniella Fajardo, Joha Md. Ibne,
Tyan Onnolu

Supervisor: Dr. Sulaymon Eshkabilov

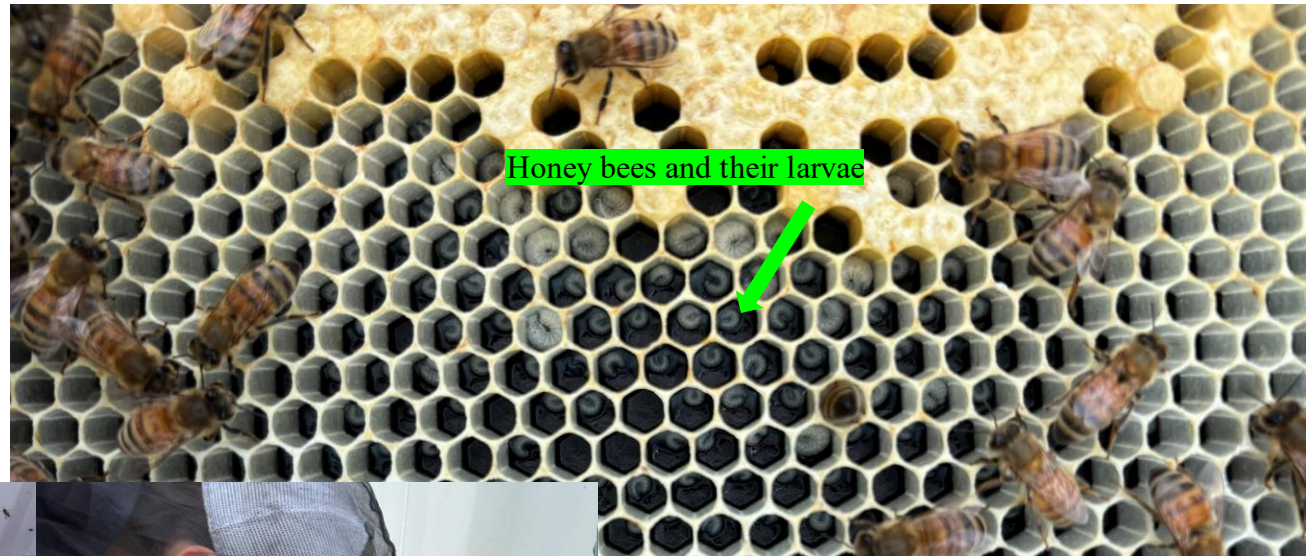
Research Questions:

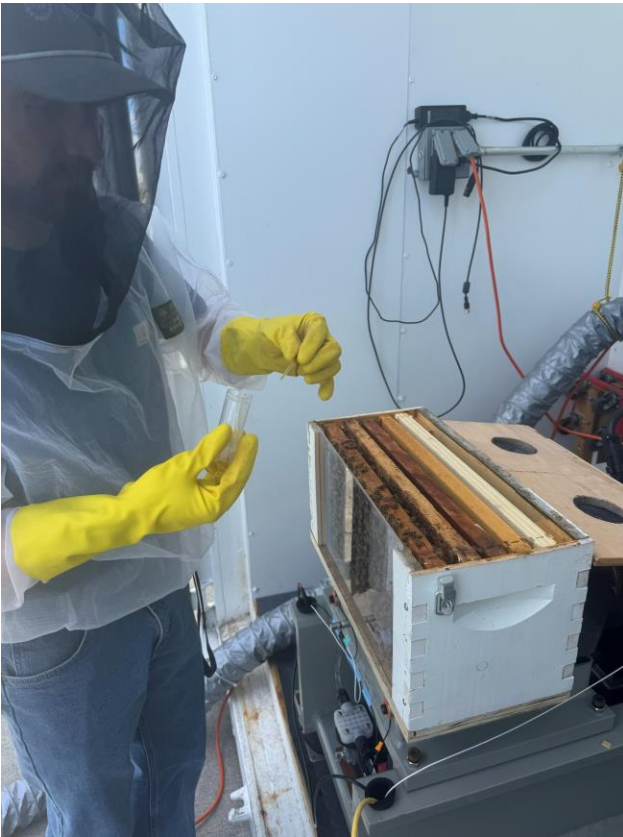
1. Can we non-invasively measure the vibrational response characteristics of honey bees' and their larvae using a laser vibrometer?
2. What are the frequency response characteristics of honey bees' and their larvae under both normal and excited conditions?
3. During excitation, what frequencies and magnitudes are most stressful to honey bees' and their larvae?

Research Objectives:

1. To measure and establish ground truth honey bee vibrational response characteristics.
2. Classify when honey bee's and their larvae become stressed from excitation.
3. Reduce/eliminate these frequencies from interfering with honey bee behavior during transport.

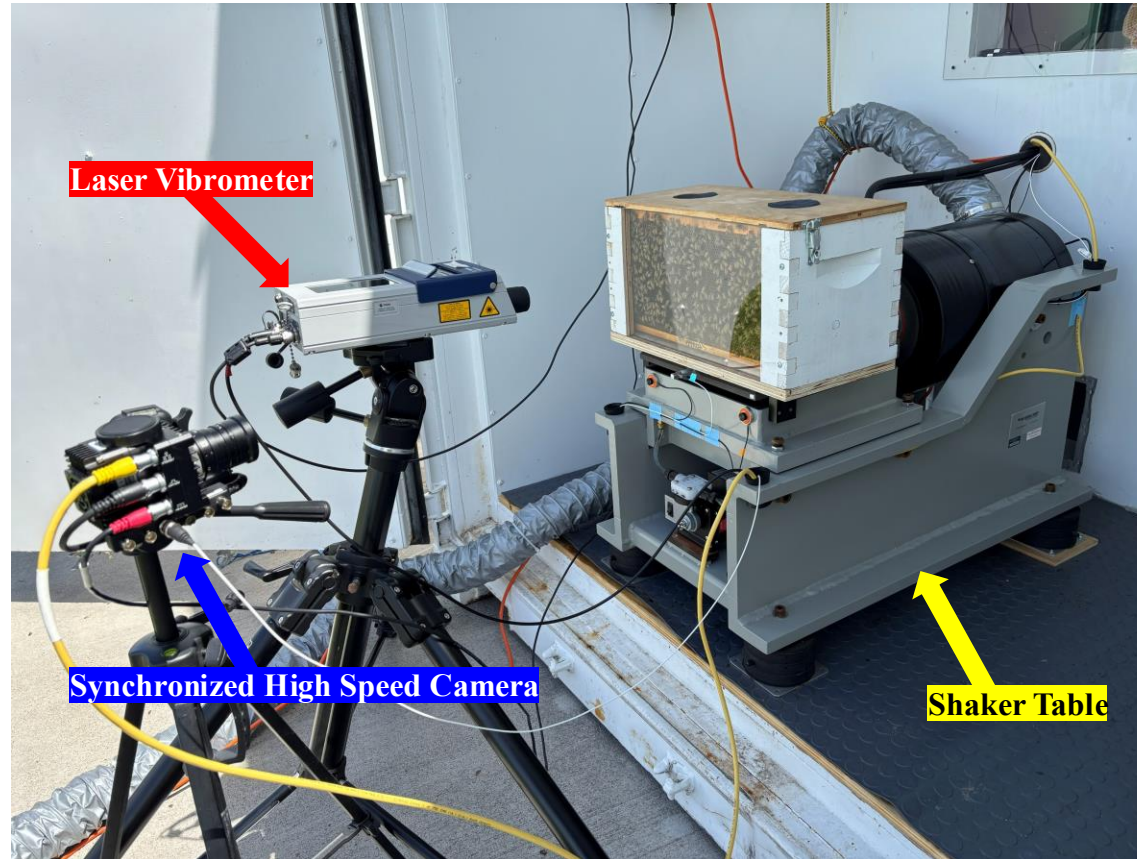






Once the frame is ready, we moved it into a custom display nuke!

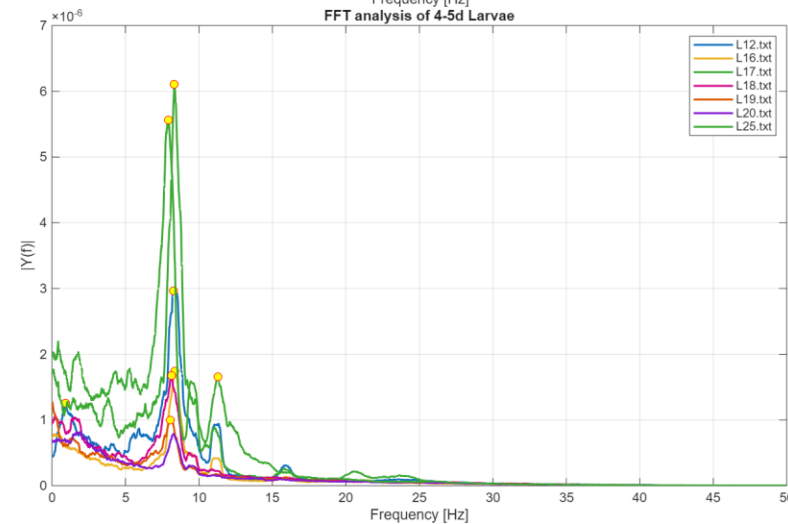
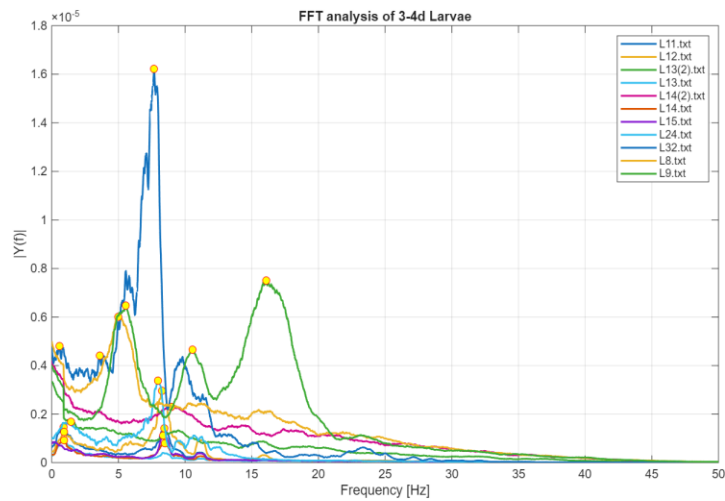
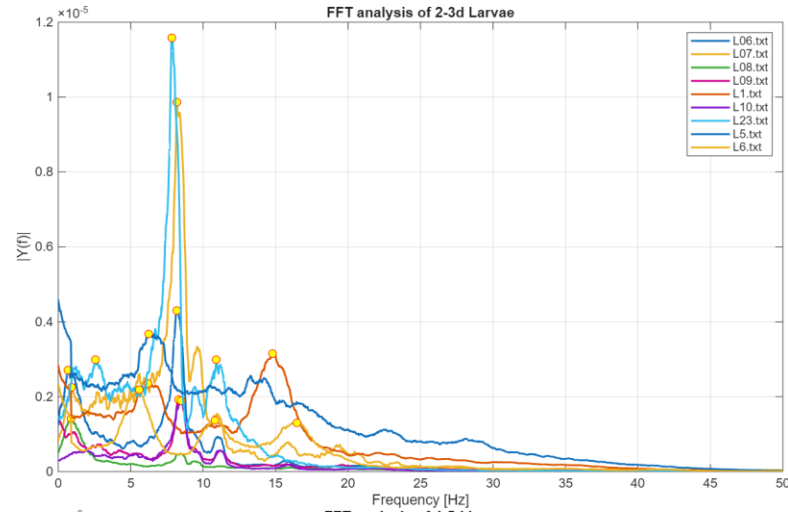
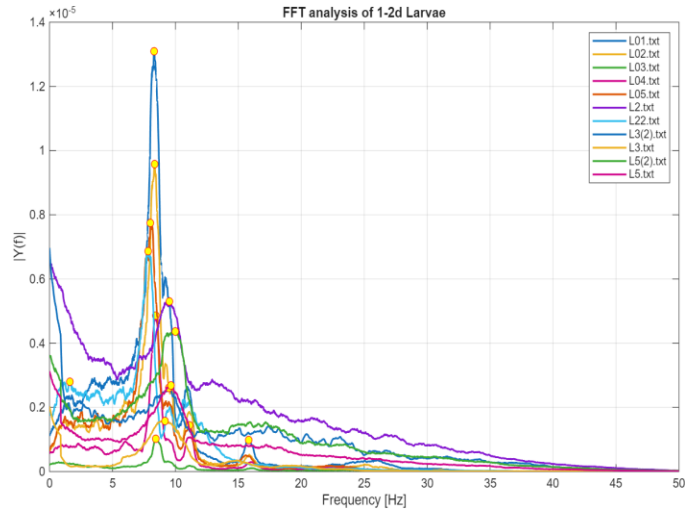
Placing queen pheromone stick in nuke to simulate queen presence and normal worker bee behavior.



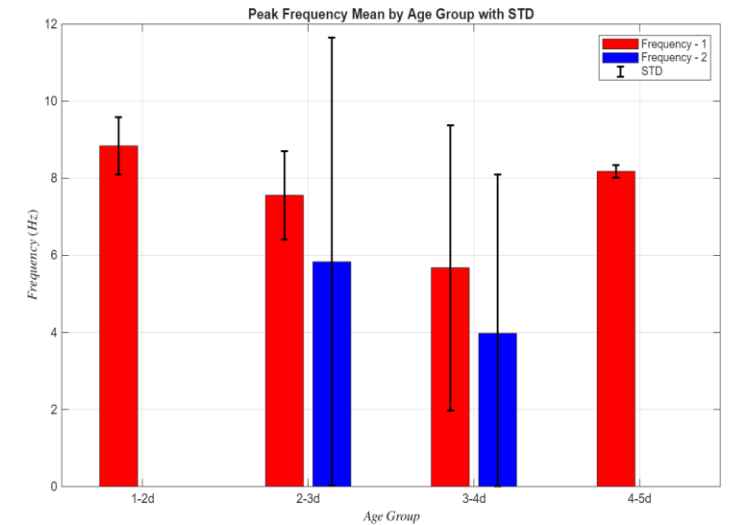
Current Nuke Colony on Test Bed – Established 6.22.2026



Current Nuke (empty) – Initial Shaker Test @ 10Hz Video



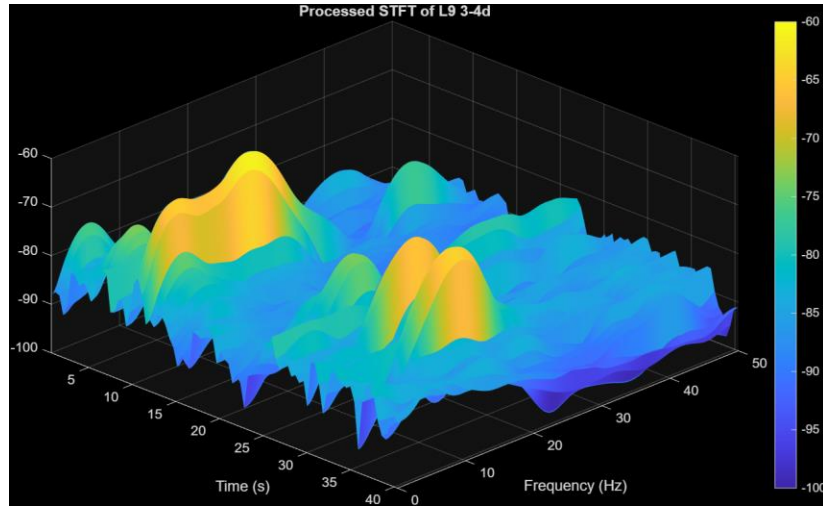
Peak Analysis of FFT results



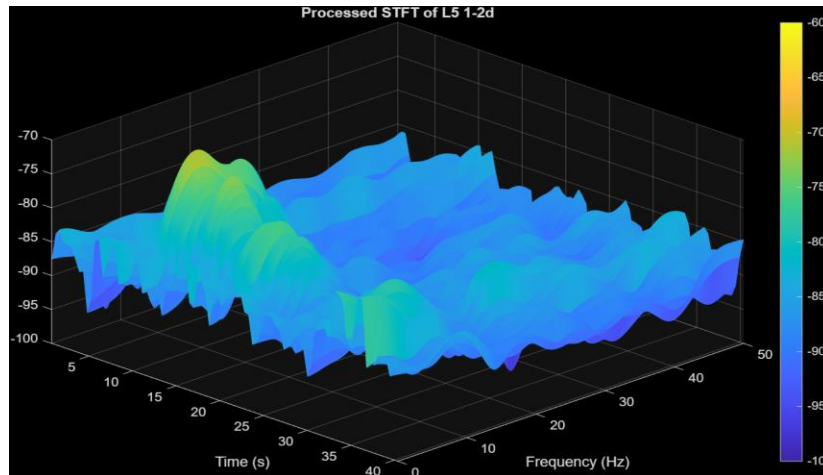
8Hz appeared to be a shared frequency amongst age groups

FFT results from October and December 2025 Larvae Testing

- Categorized by age groups



STFT Results of Initial Larvae Testing



We aim to improve our methodology and results:

Nuke Colony:

- We simulate a colony that has a queen, as well as worker bees that will take care of the larvae throughout development.
- This is because previous test was capped to 2 hours, because after this larvae begins to starve, which therefore could change their normal behavior.

Data Acquisition:

- We have improved the mounting to the shaker table.
- We aim to include time and frequency domain results (STFT) that will provide more insight into the duration and patterns of signals.
- As well as synchronizing other sensors together using an IMC DAQ – Laser, Microphone, and High Speed Camera.

Novel Approach in Honeybee Fanning Detection with Only A Monocular Camera, without any Computer Vision Foundational Model

Jewel Woodcock

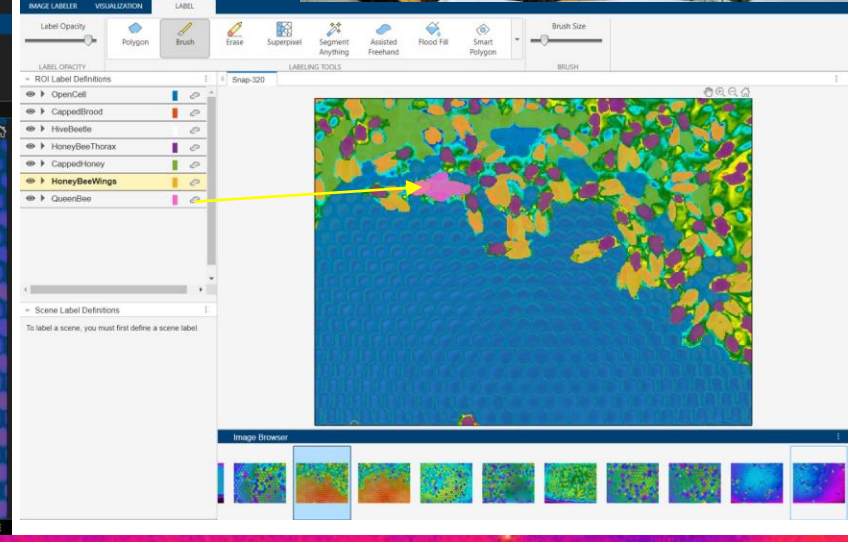
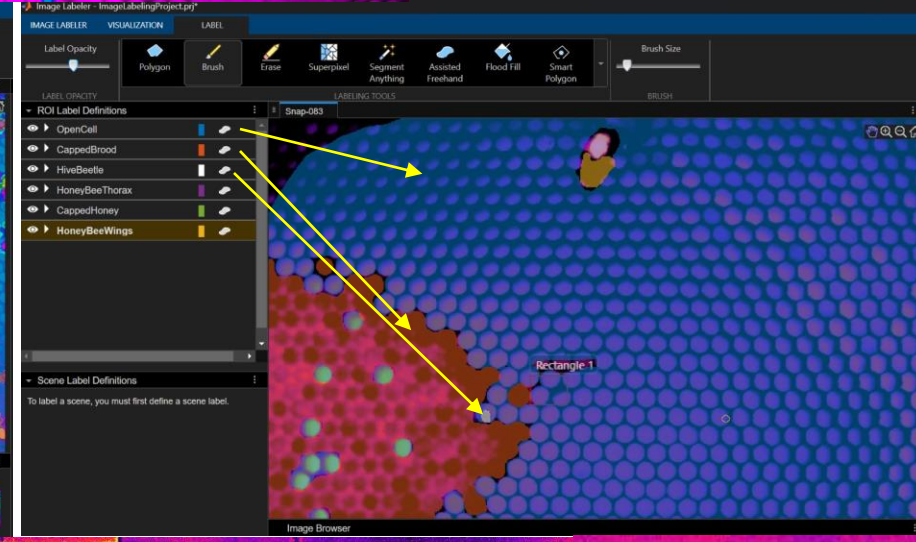
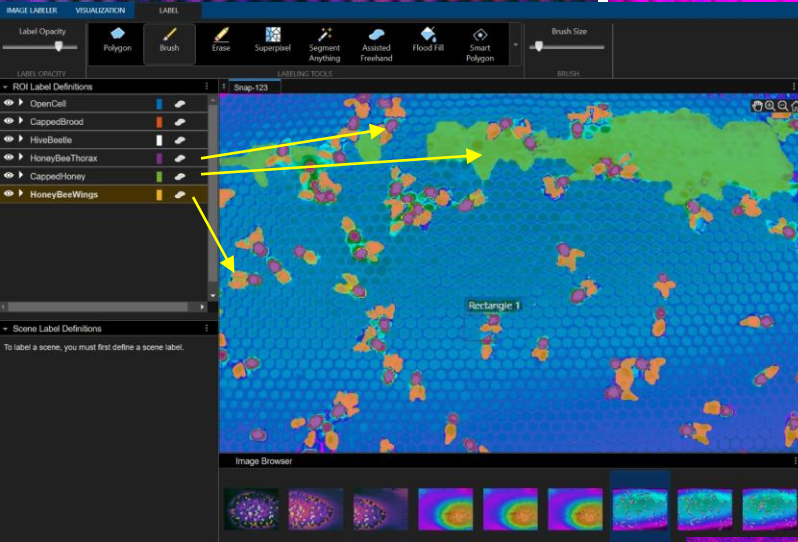
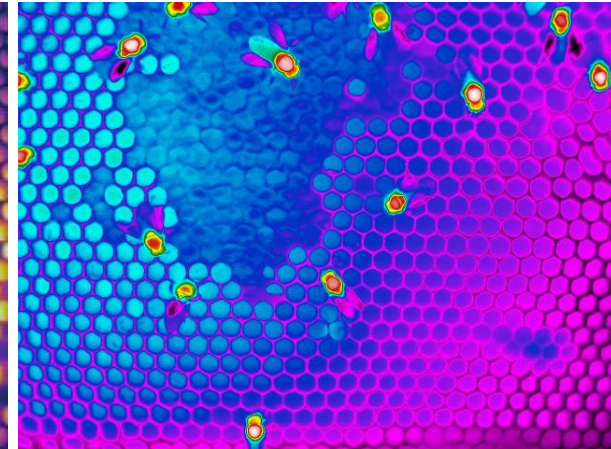
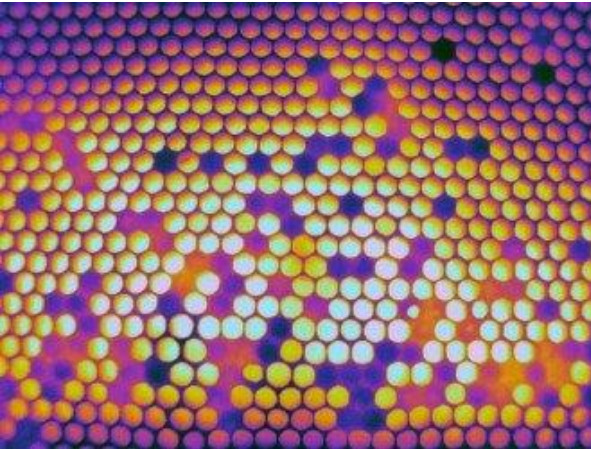
Supervisor: Dr. Sulaymon Eshkabilov

Objectives

Research Objective: To develop and evaluate an AI-based thermal imaging system capable of identifying and classifying honey bee hive components and colony conditions for non-invasive hive monitoring.

- Can artificial intelligence accurately identify and differentiate honey bees, hive structures, hive pests, and colony resources from thermal images of honey bee hives?
- Can thermal imaging and artificial intelligence be used to assess honey bee colony health and activity without opening the hive?





Novel Approach in Honeybee Fanning Detection with Only A Monocular Camera, without any Computer Vision Foundational Model

Shoaib Ahmmad

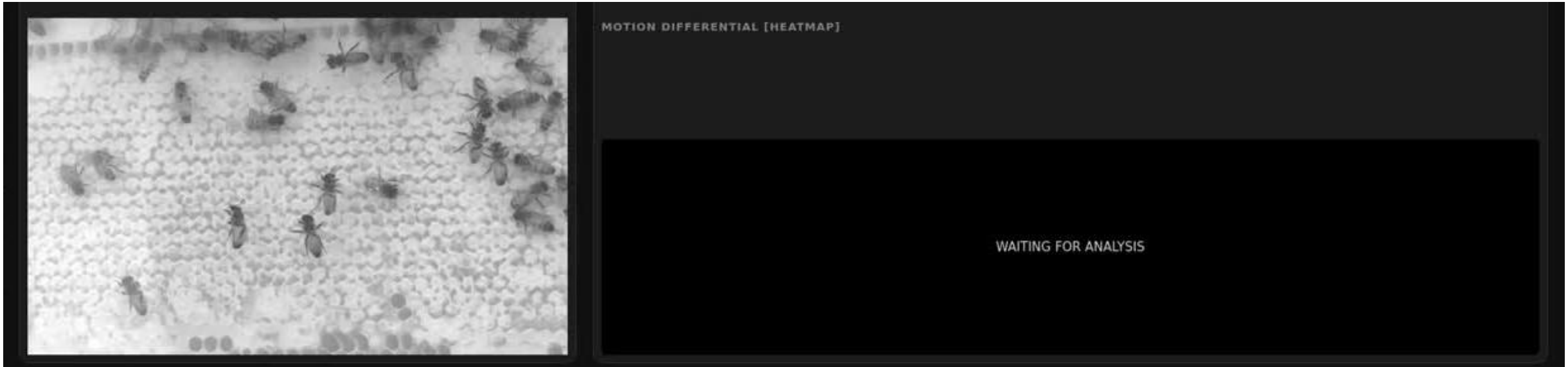
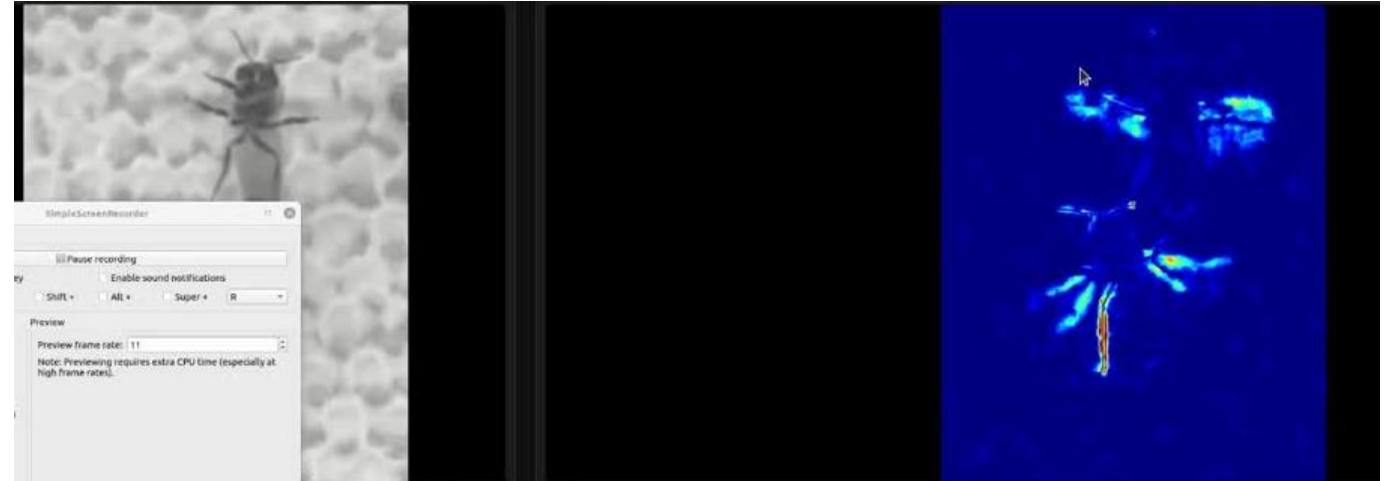
Supervisor: Dr. Sulaymon Eshkabilov

Objectives

- Latest Computer Vision Foundational Models are Very Bad in Detecting Wingbeats of Honeybees When Fanning. Annotating a dataset of this kind is difficult and most often inaccurate.
- Even with careful extensive annotating and strong training, we are only training the foundational model to learn that blurry pixels around bees mean fanning. And there are instances that this turns into false negatives.
- Foundational models rely heavily on the quality and lighting and color of the image and therefore, changing the scene for various beehives means annotating and training the a new model all over again
- Even after quantized, these models perform sluggish when introduced to consumer CPU and Edge Devices

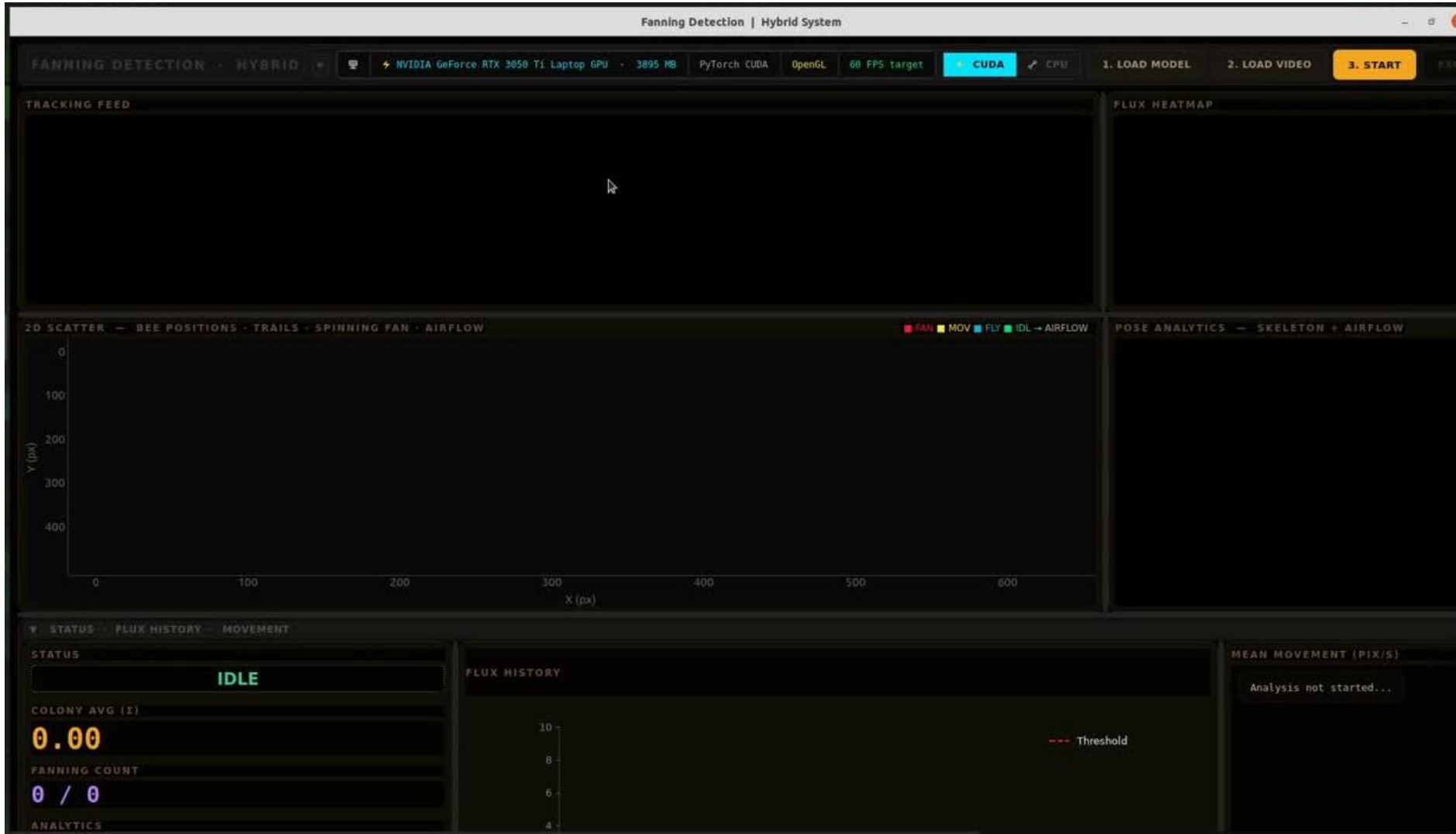
Video 1: Fanning

Motion Flux of Honeybee Wing Movements Shown By Heatmaps

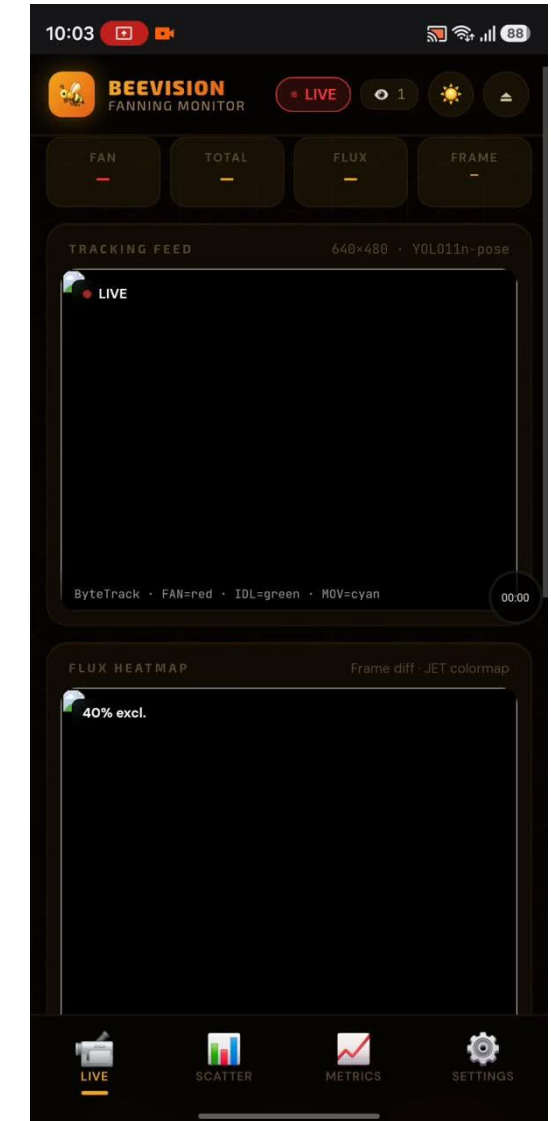


Video 2: Fanning Hive Detection Realtime

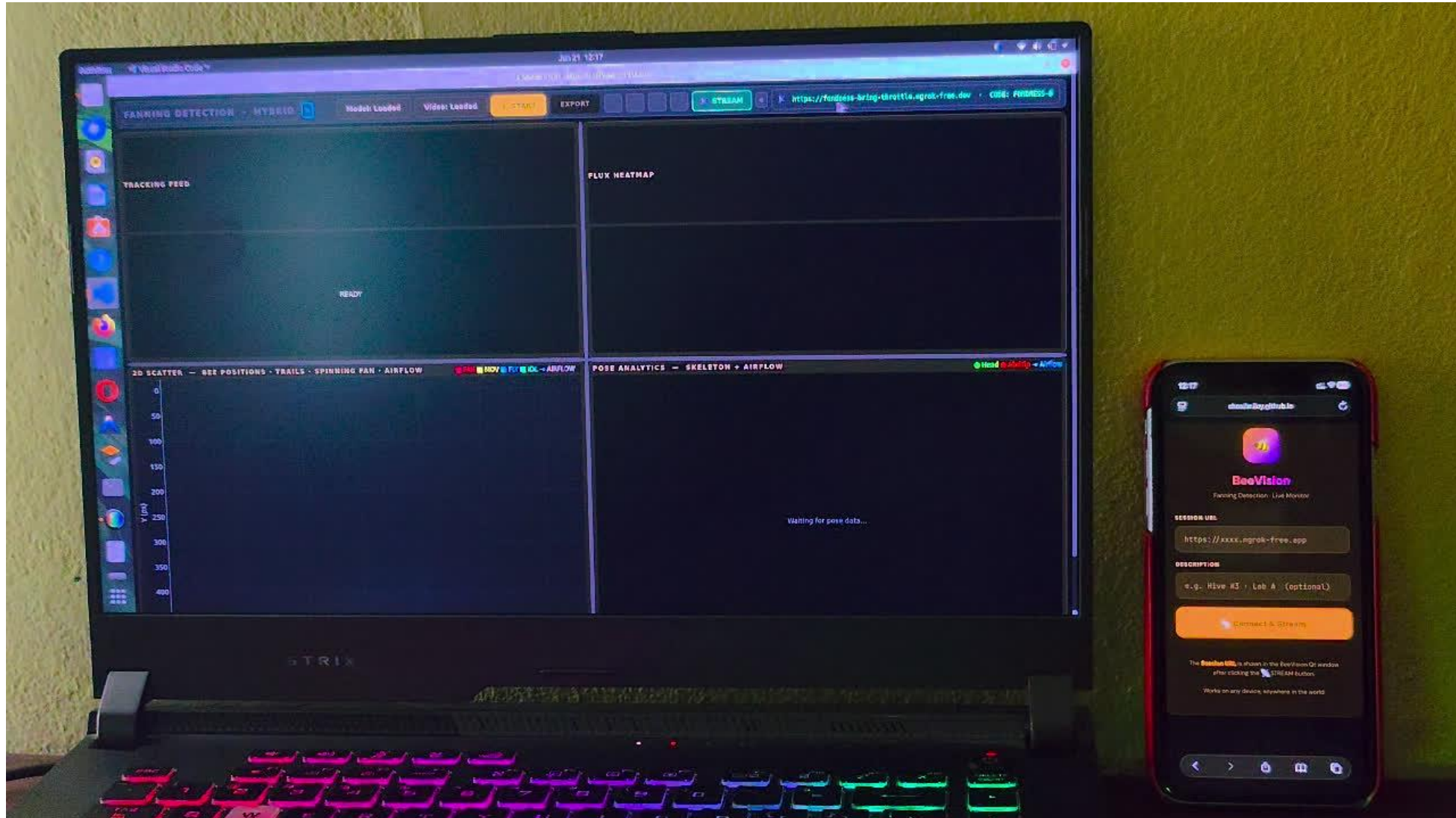
Video 3: Desktop/Edge Application



Video 4: Web Application



Video 5: Beevision fanning both apps in Action



Getting Rid of Foundational Computer Vision Model for Real Time Fanning Detection of Honeybees

Pros

- Significantly Higher Accuracy
- Super Lightweight
- Versatile Lighting Conditions
- Very Low Dataset Labeling
- Very Easy to Implement in Edge Devices
- Near Zero False Positives

Cons

- Tuning required in various lighting conditions
- Still needed to annotate datasets for tagging bees with foundational detection and tracking models, but significantly lower (50-100 enough)

Bee-Hive Monitoring Using Wireless Sensor Network

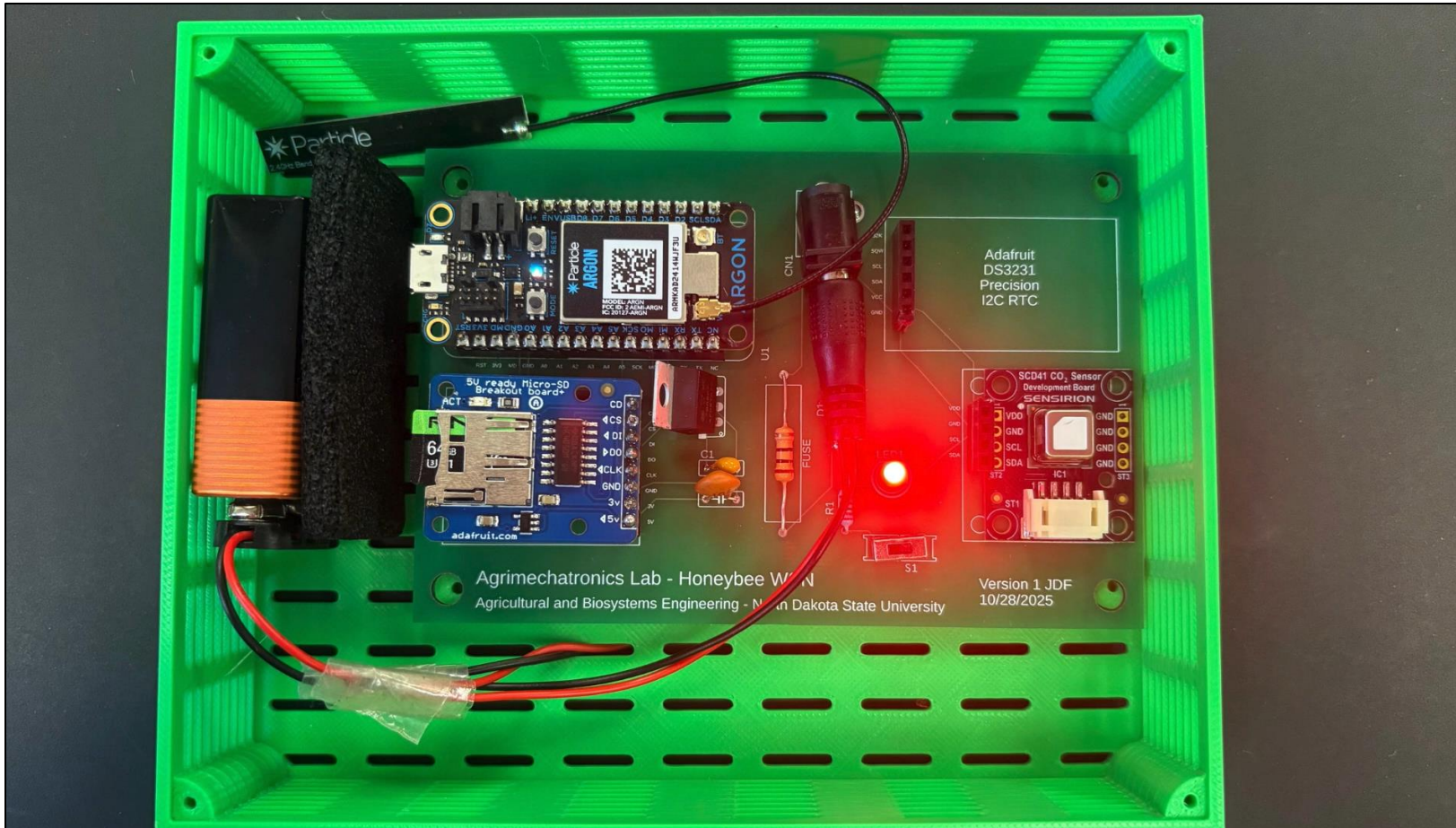
Joanna Daniella Fajardo

Supervisor: Dr. Sulaymon Eshkabilov

Objectives

- Design and evaluate a smart honey bee monitoring system that integrates CO₂ and piezoelectric sensors with wireless communication for real-time hive health assessment
- Assess the influence of honey fill level on sensor response and establish the optimal placement of piezoelectric sensors for reliable hive vibration monitoring





Video Demonstration of the Developed Smart Hive Monitoring System
and Piezoelectric Sensor Testing

Bee Excitor Project

Md. Ibne Joha

Supervised By: Dr. Sulaymon Eshkabilov

Objectives:

- A non-invasive active vibration excitation and sensing system for quantifying honey bee colony behavioral responses and health status.
- A multimodal sensing framework combining environmental, acoustic, and vibrational measurements for comprehensive honey bee colony monitoring.
- An Edge-IoT-enabled precision beekeeping platform supporting real-time analytics, anomaly detection, remote monitoring and control, and intelligent decision-making.

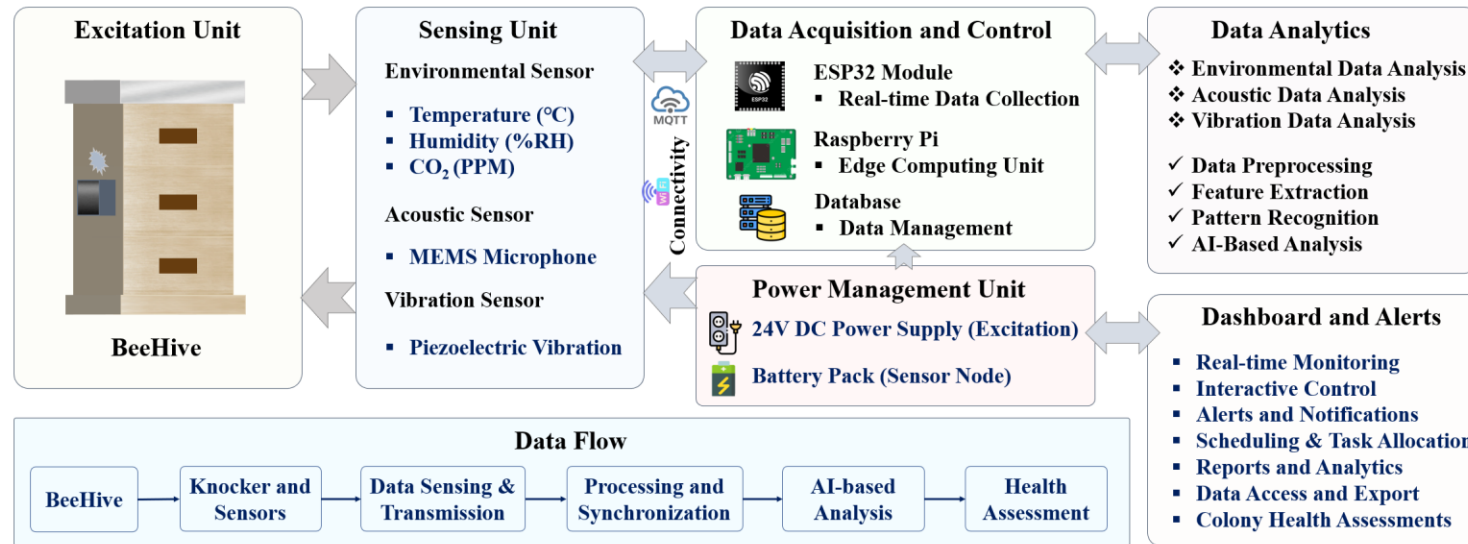
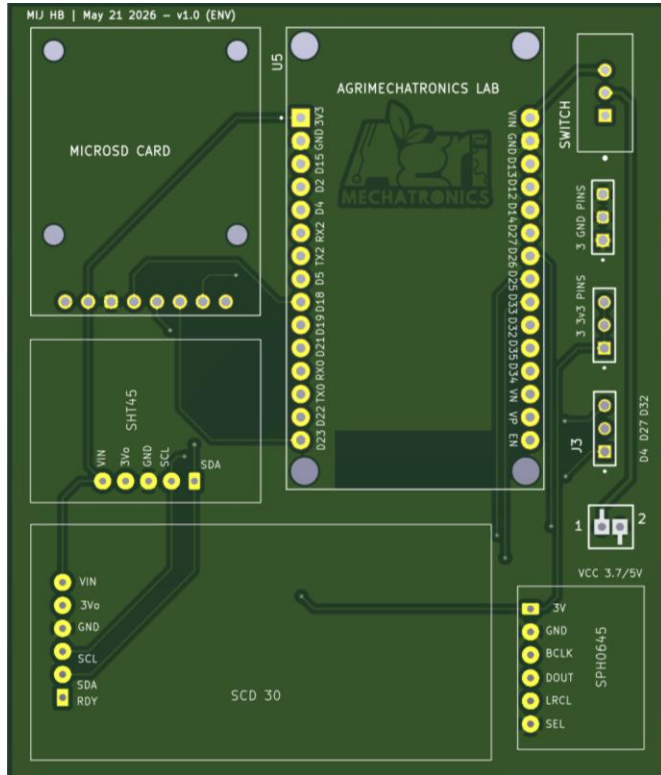
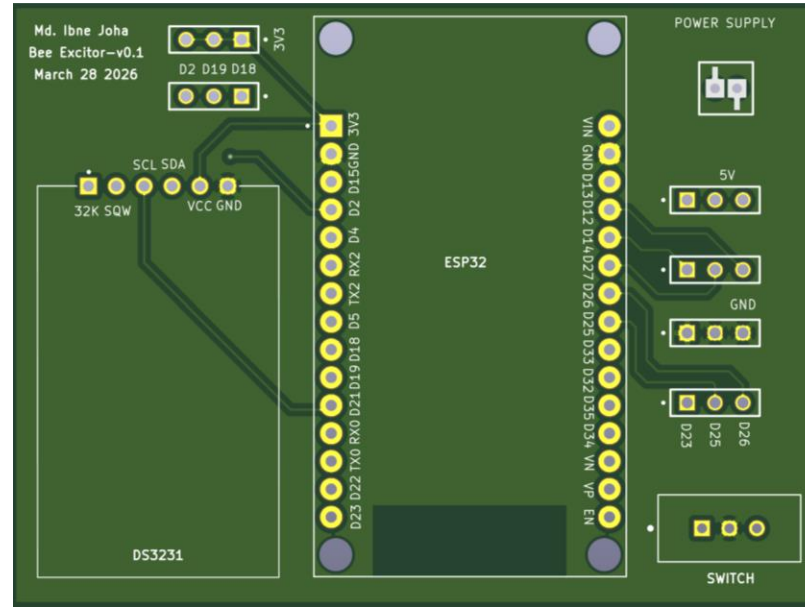


Figure 1: Proposed IoT-driven framework for non-invasive honeybee colony monitoring.

Experimental Setup:



(a)



(b)

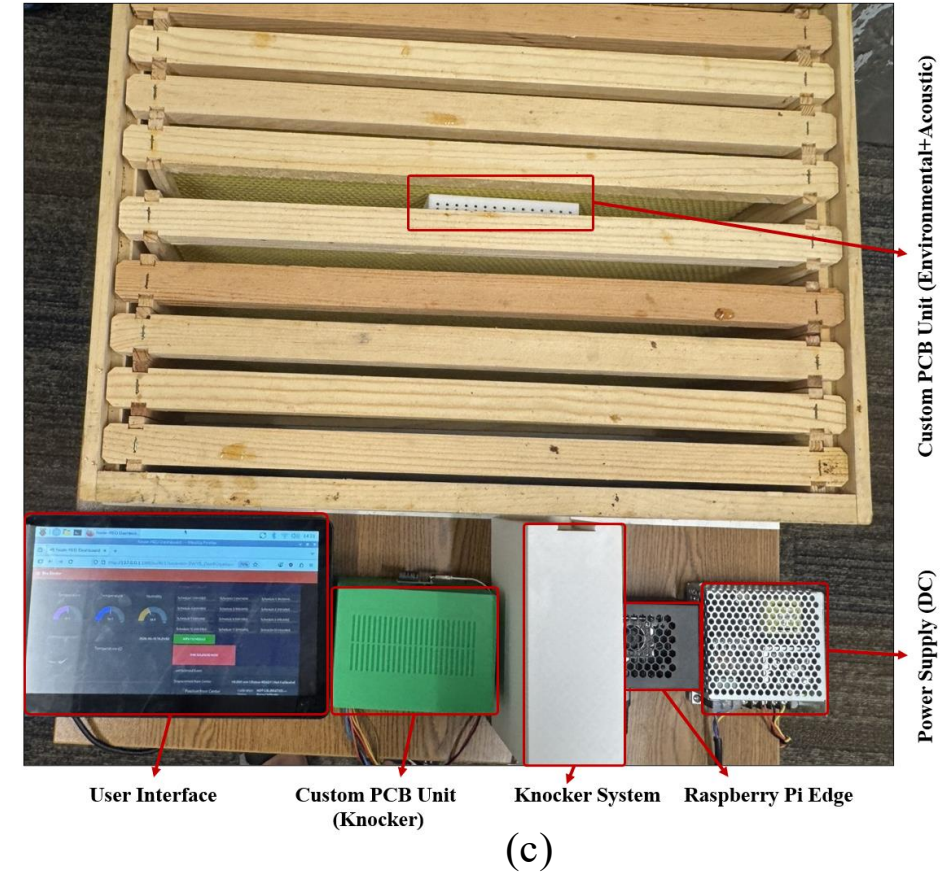
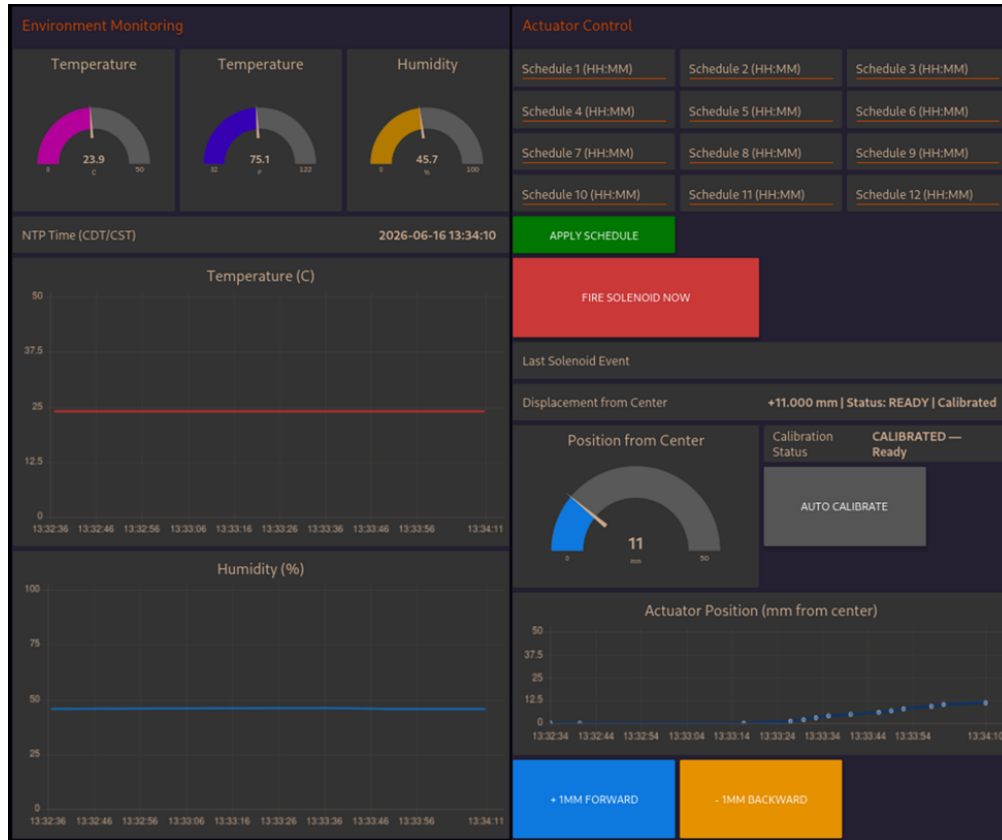
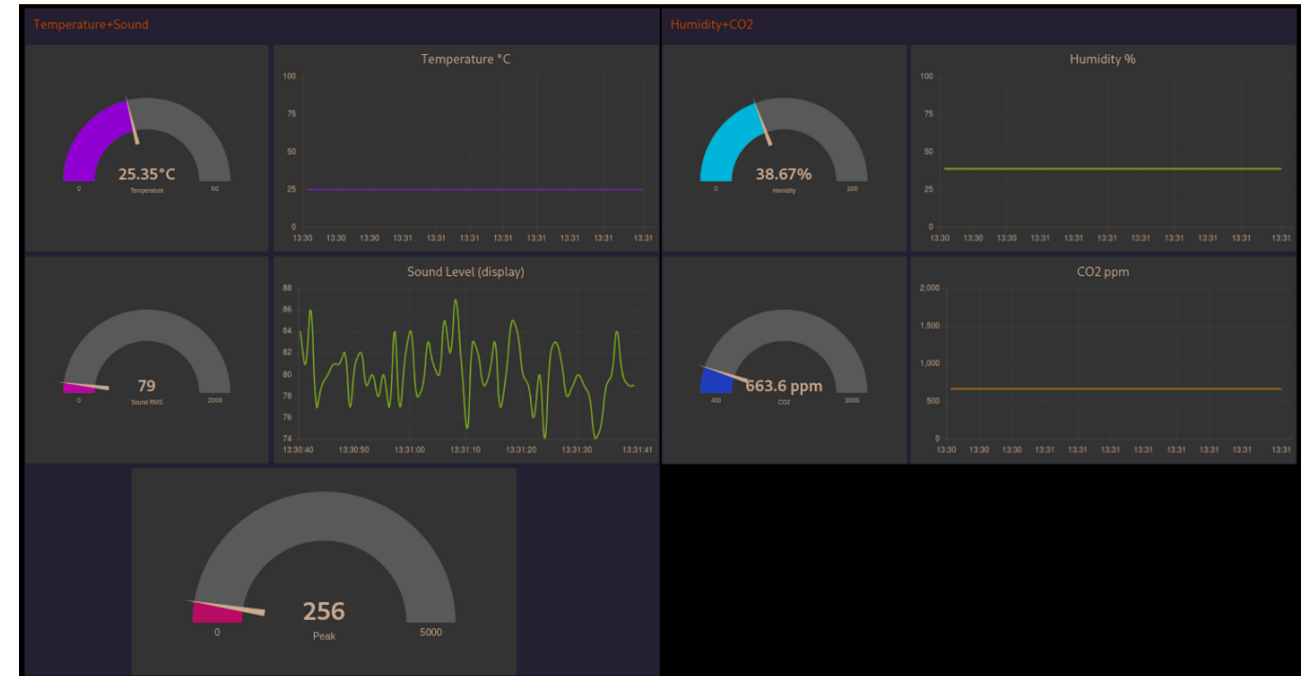


Figure 2. (a) PCB design for the environmental sensing module, (b) PCB design for the vibration excitation (knocker) module, and (c) prototype of the proposed smart beehive system.

Results and Discussion:



(a)



(b)

Figure 3: User interface for real-time monitoring and control of the proposed knocker system.

Conclusion:

- The proposed framework demonstrates the feasibility of integrating active colony excitation, multimodal sensing, and IoT-based edge computing within a unified precision beekeeping platform for comprehensive honey bee monitoring.
- The incorporation of AI-driven analytics, real-time monitoring and control, and automated alert mechanisms is expected to enhance the early identification of colony stress, behavioral analysis, and potential colony health-related assessments.
- The developed architecture provides a scalable foundation for future intelligent apiculture systems, supporting data-driven decision-making, early health intervention, and sustainable beekeeping practices.

Honey Concentration Detection in Bee Combs Using Image Analysis and UWB Radar

Mohammad Aftabi Talami

Supervisor: Dr. Sulaymon Eshkabilov

Objective

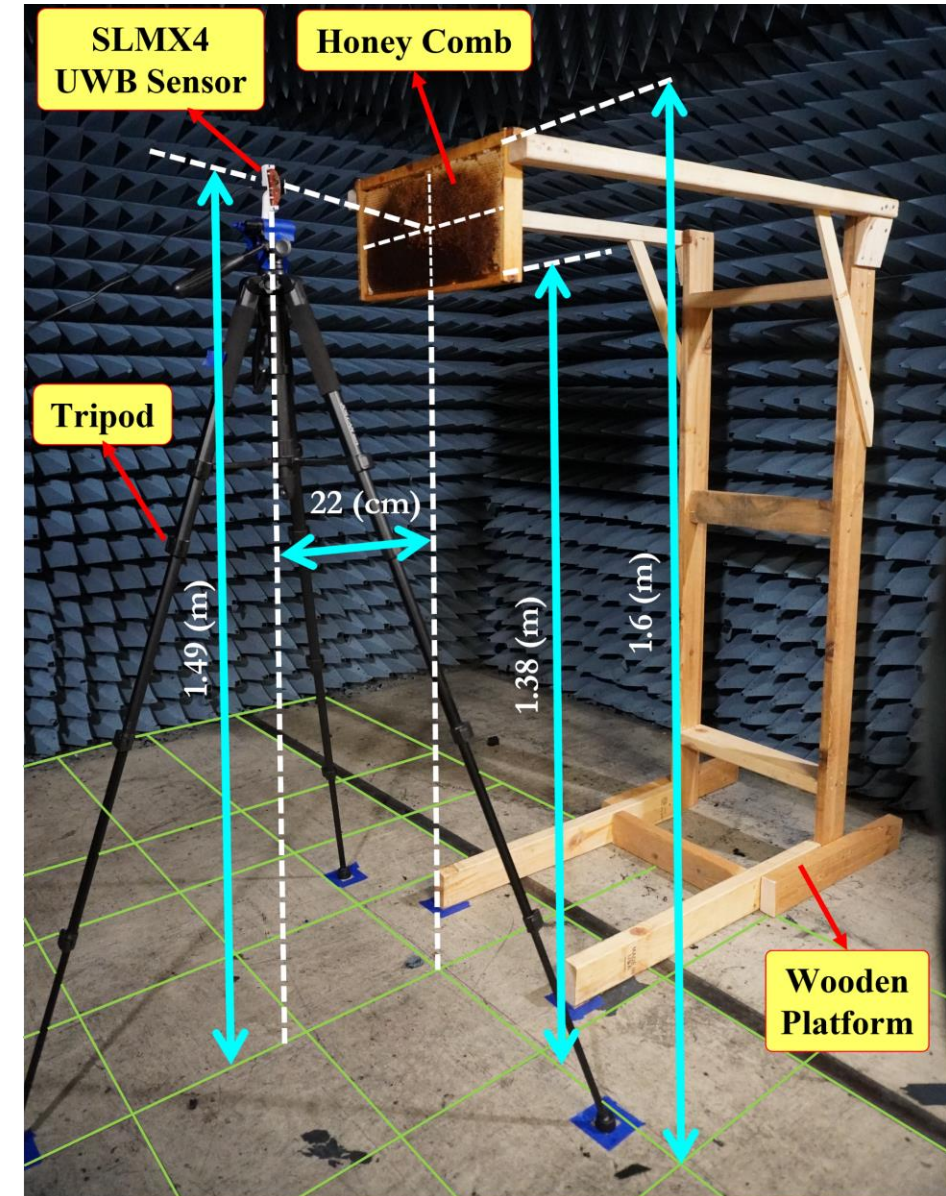
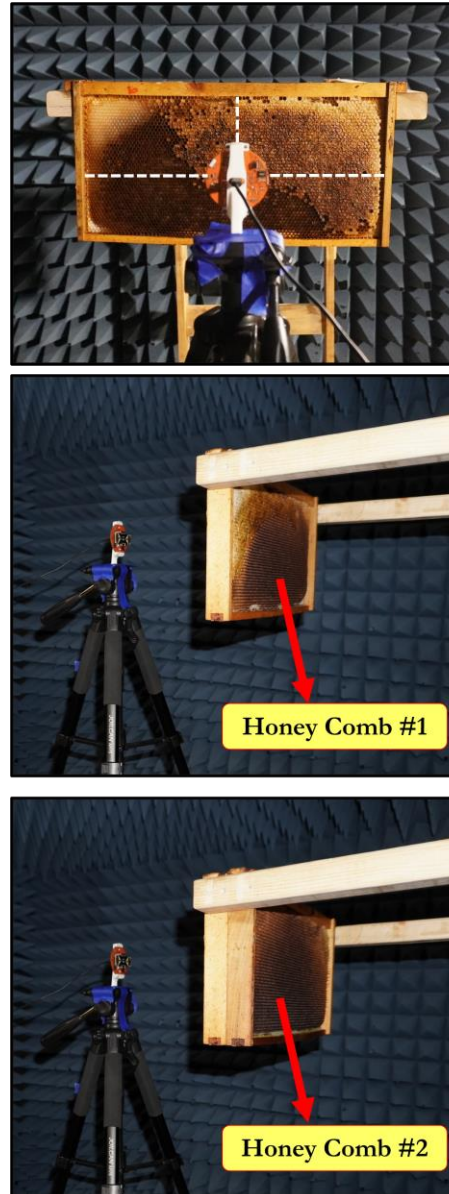
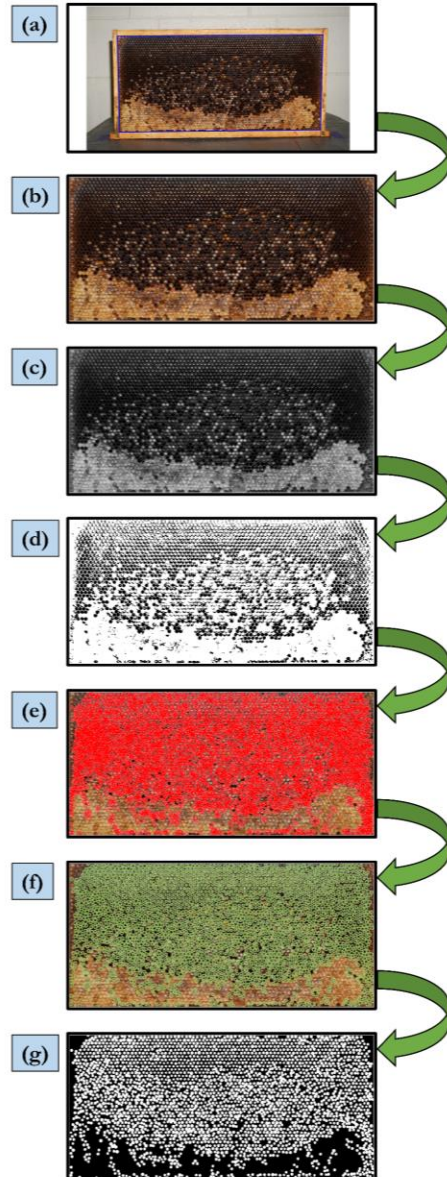
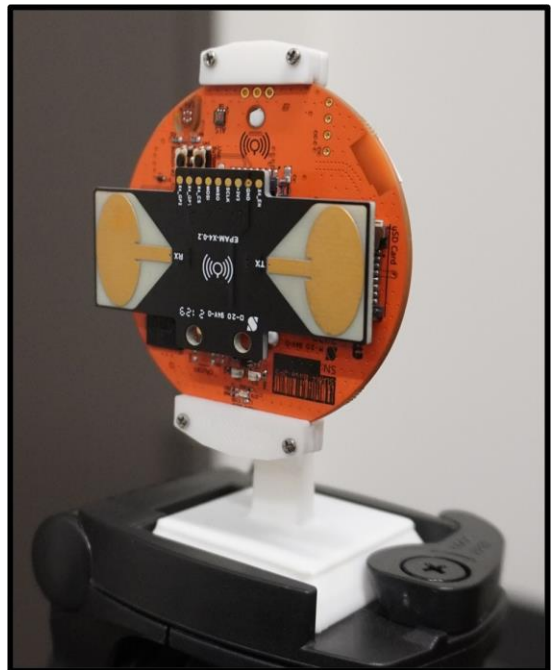
One of the critical tasks requiring a significant amount of manual labor is to monitor the rate of stored honey inside honeycombs. This study introduces a novel application of UWB sensing technology along with the image analysis approach to estimate the percentage of honey stored in honeycombs.

➤ The specific objectives of this study are

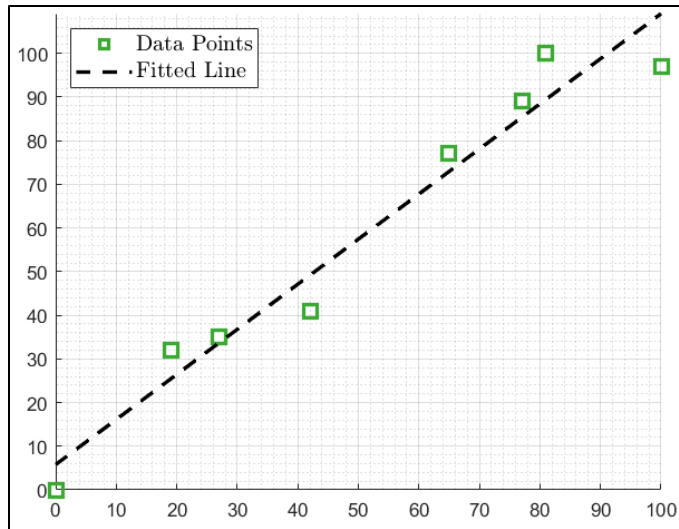
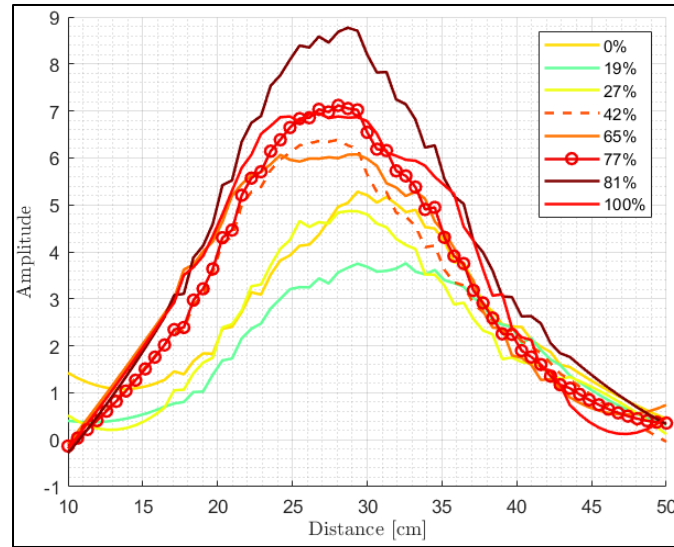
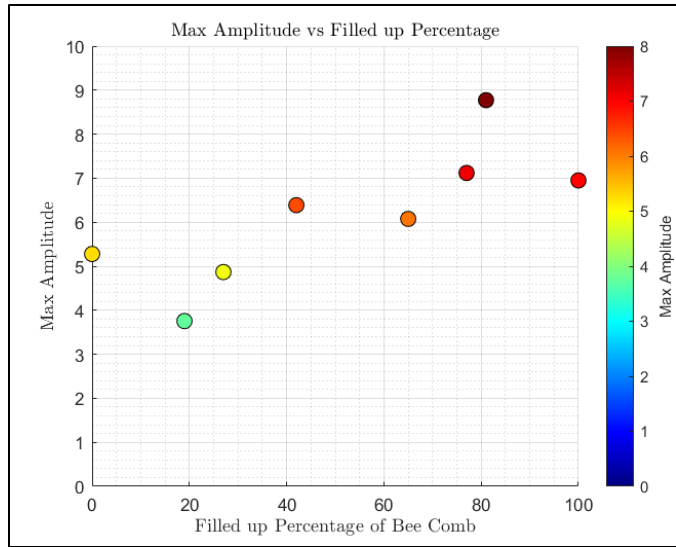
- 1) to estimate the honey amount in combs using true color images and image analysis techniques and
- 2) to evaluate the SLMX4 sensor performance in detecting honey amount in a single honey comb.



Methodology



Results

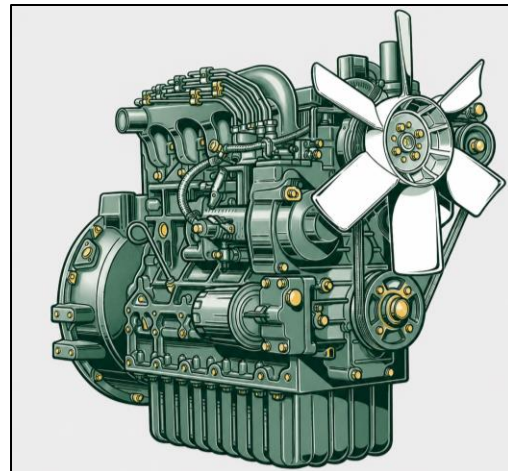


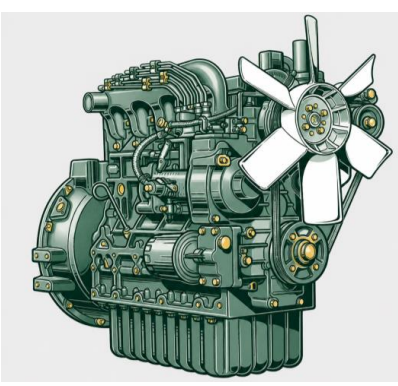
	Measured	Estimated with image analysis	Estimated with SLMX4 UWB readings
Measured	100.0	98.1	82.9
Estimated with image analysis	98.1	100.0	77.3
Estimated with SLMX4 UWB readings	82.9	77.3	100.0

Social Implications of Clean Energy Tractors for Farming

Omid Nazempour, Mohammad Aftabi Talami, Luke Bordeaux

Supervisor: Dr. Sulaymon Eshkabilov





Application of AI in Engine Modeling

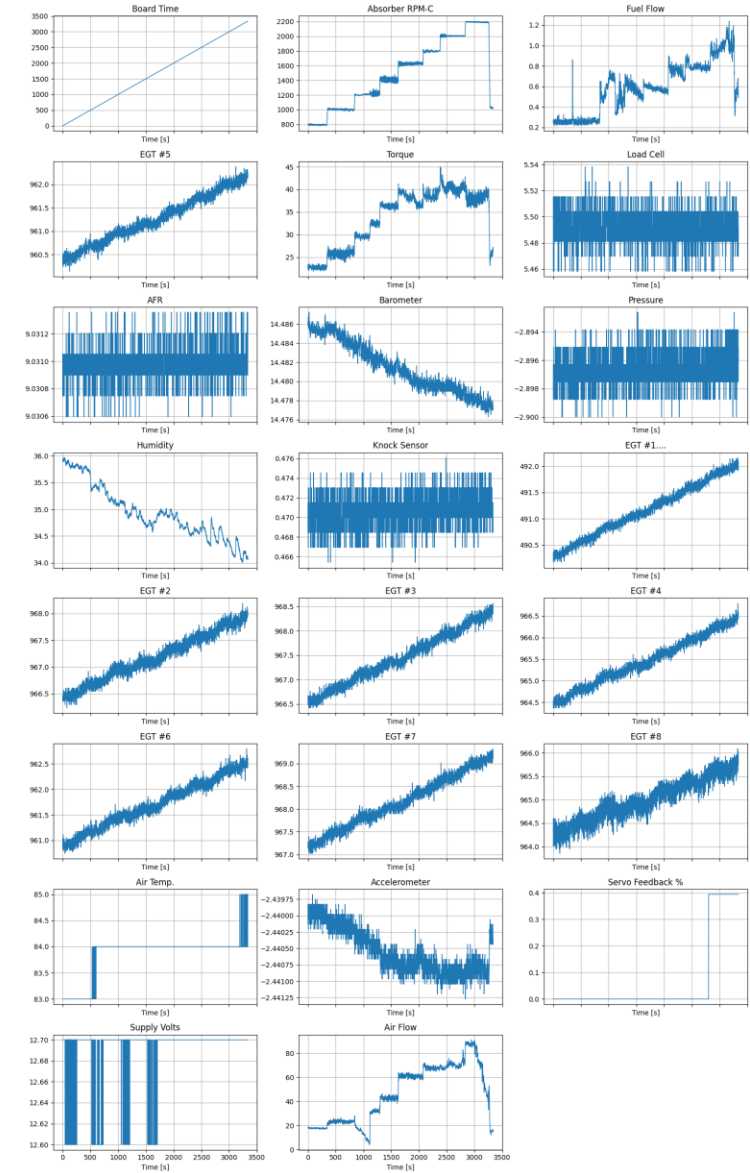
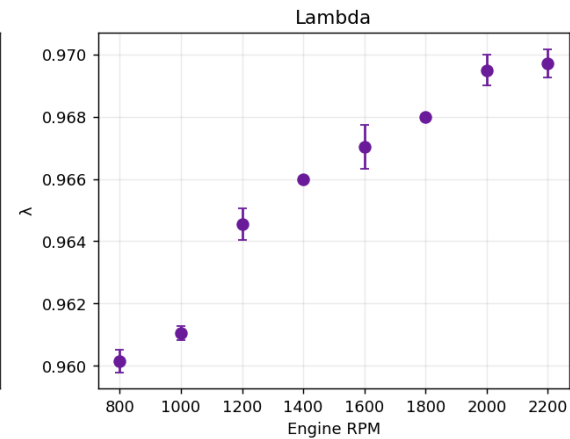
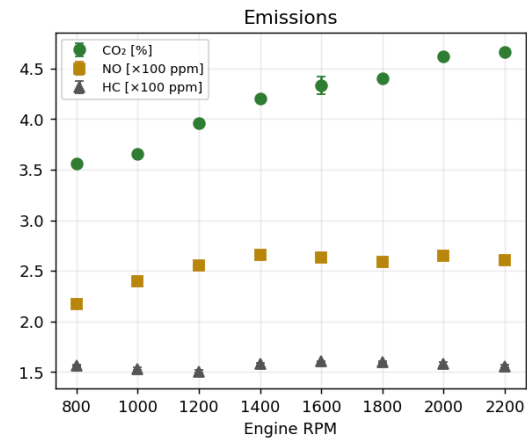
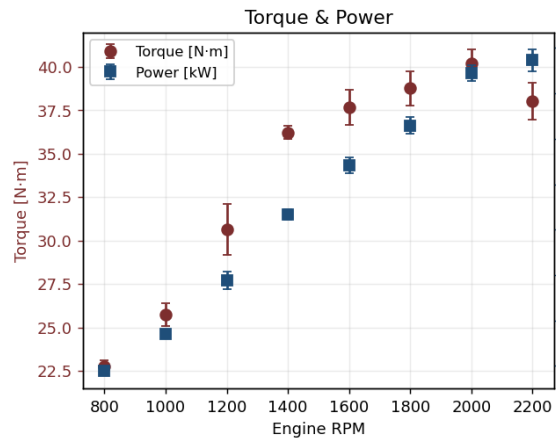
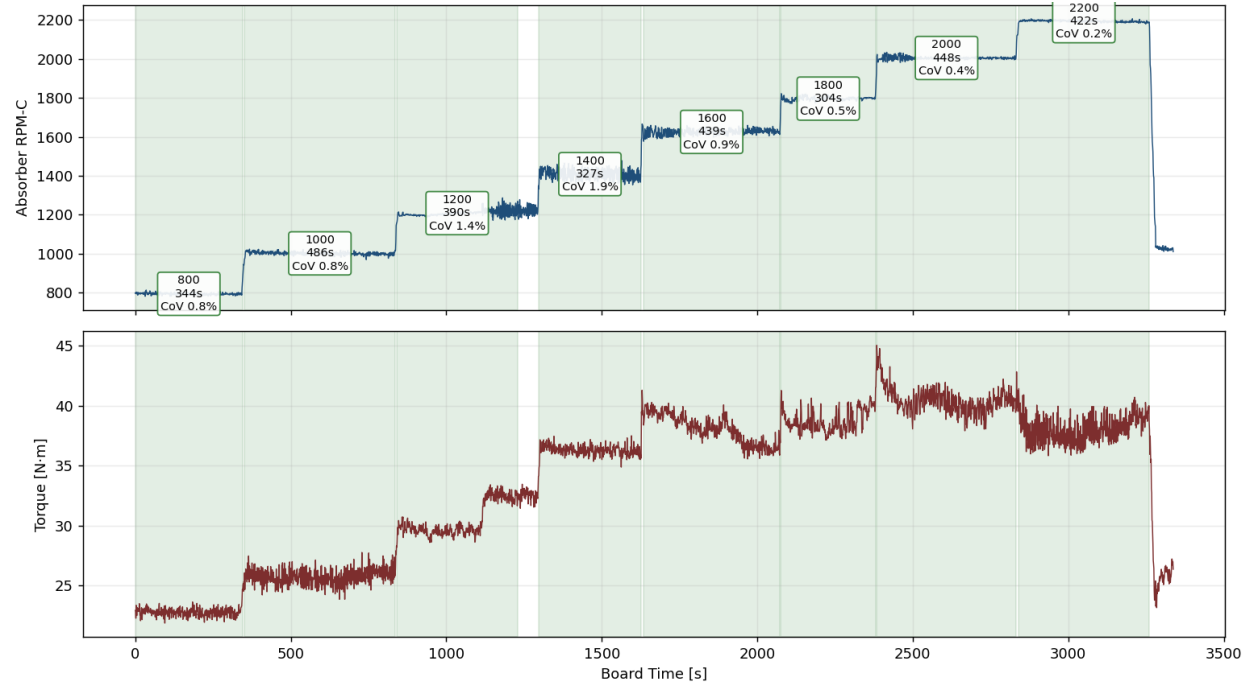
Omid Nazempour, Mohammad Aftabi Talami, Luke Bordeaux

Supervisor: Dr. Sulaymon Eshkabilov

Objectives



Results





Tractor Operation Simulator Software

Mohammad Aftabi Talami

Supervisor: Dr. Sulaymon Eshkabilov

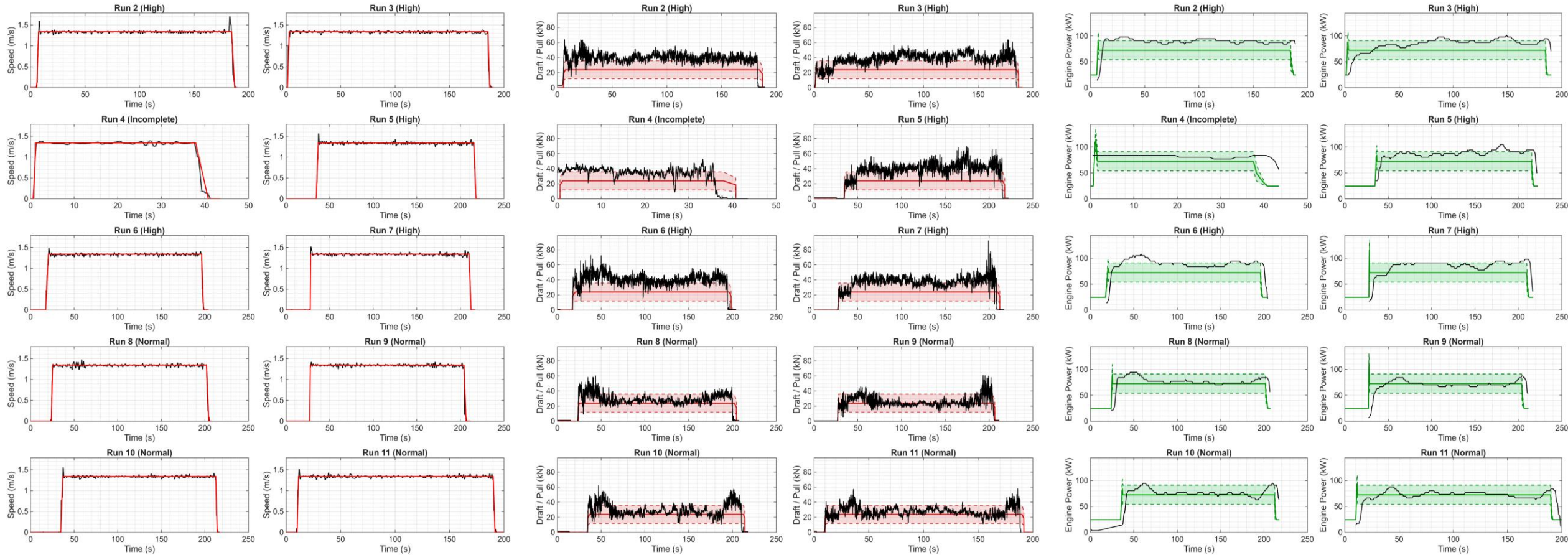
Objectives

The Goal of this study is Revisiting Becker's equation and tractor modeling performance,

The objectives of this study are:

- Comparing the implement's model results with real field test results,
- Comparing the tractor model results with real field test results,
- Developing an application that estimate required power, draft forces, and emission for performing an operation related to a selected tractor.

Results



Results

Complete

Profile: Trapezoidal

Vehicle / Drive

Implement

Traction

Motion

Simulation and Results

About

Model Inputs

Vehicle

01 Case IH Steiger Series R...

Type

tractor

Active

Case IH Steiger Series Rowtrac ...

Apply Vehicle Defaults

Mass (kg)

2.464e+04

Wheelbase L (m)

4.064

Static front fraction

0.56

CG height zG (m)

1.5

Drawbar height (m)

0.52

Drawbar x offset (m)

0

Drawbar angle alpha (deg)

0

Rolling resistance c_rr

0.03

Drivetrain efficiency

0.88

Auxiliary power (kW)

24.55

Gravity (m/s^2)

9.81

Vehicle Context

Complete

Profile: Trapezoidal

Vehicle / Drive

Implement

Traction

Motion

Simulation and Results

About

Implement Inputs

Implement

D4 Chisel plow - 5 cm str...

Unit system

SI

Soil texture

Medium-textured

W = number of tools/shanks

12

TD = tillage depth in cm

20

Draft A

91

Draft B

5.39

Draft C

0

Soil factor FI

0.85

Draft range (%)

50

Equation

$$D = FI * (A + B * S + (C * S^2) * W * TD)$$

Unit: N/tool/cm Speed

S: km/h Tool element: Chisel plow...

Draft A, B, C, and FI are locked to the Harrigan-Rotz table values. FI updates from Soil texture; W, TD, and draft range remain editable.

Selected Implement

Complete

Profile: Trapezoidal

Vehicle / Drive

Implement

Traction

Motion

Simulation and Results

About

Traction Inputs

Cone index CI (kPa)

1500

Device width b (m)

0.762

Diameter/contact d (m)

2.5

Driven devices

4

Driven front devices

2

Driven rear devices

2

Maximum traction mu

1

Use acceleration

☒ Include inertial force

Slip model

$slip = -\log(1 - mu_{eff} / (0.3 * C_n))$

Cn is recalculated at each time step from normal load per driven tire/track. Static Cn is shown below as a quick setup check.

Front/rear static 135.335/106.335 kN | per driven device 60.417 kN | Cn 47.296 | driven 4 = 2 + 2

Traction Context

Traction Relationships

$$C_n = \frac{CI * b * d}{W_{device}}$$

Dimensionless Bearing Capacity

$$p = \frac{1.2}{C_n} + 0.04$$

Average Contact Pressure (kPa)

$$\mu_g = \mu_{max} (1 - e^{-0.3 * C_n})$$

Gross Traction Coefficient

$$S = \frac{-\ln(1 - \mu_{req}/\mu_{max})}{0.3 * C_n}$$

Slip Required for Required Traction

$$C_n \uparrow \rightarrow C_n \uparrow \rightarrow \mu_g \uparrow \rightarrow \mu_{req} \uparrow$$

$$b, d \uparrow \rightarrow C_n \uparrow \rightarrow \mu_g \uparrow \rightarrow \mu_{req} \uparrow$$

Complete

Profile: Trapezoidal

Vehicle / Drive

Implement

Traction

Motion

Simulation and Results

About

Simulation Summary

Table Summary

Metric	Value	Unit	Note
1 Vehicle	Case IH Stei...	tractor	
2 Profile type	Trapezoidal		
3 Total time	76.600	s	
4 Peak speed	1.3400	m/s	
5 Mean speed	1.3038	m/s	

Vehicle: Case IH Steiger Series Rowtrac 470 (tractor)
 Implement: Chisel plow - 5 cm straight point
 Category: Major tillage | Unit system: SI | Table unit: N/tool/cm
 Soil: medium-textured soil | FI: 0.850 | Draft range: +/- 50.0%
 Path: 100.000 m | Target: 1.3400 m/s | Total time: 76.600 s
 Target speed reached.

The Results panel is reserved for plots. Export CSV saves the full time-series table.

Results

Speed

Draft

Slip

Power

Distance

Acceleration

Traction

Cn

Engine Power Band

Power band

Lower

Mean

Upper

40

Social Implications of Clean Energy Tractors for Farming

Mohammad Aftabi Talami, Istiak Md. Shafi

Supervisor: Dr. Sulaymon Eshkabilov





Control and Navigation of UGV in the Fields

Mohammad Aftabi Talami

Supervisor: Dr. Sulaymon Eshkabilov

Control and Navigation of Unmanned Ground Vehicles:

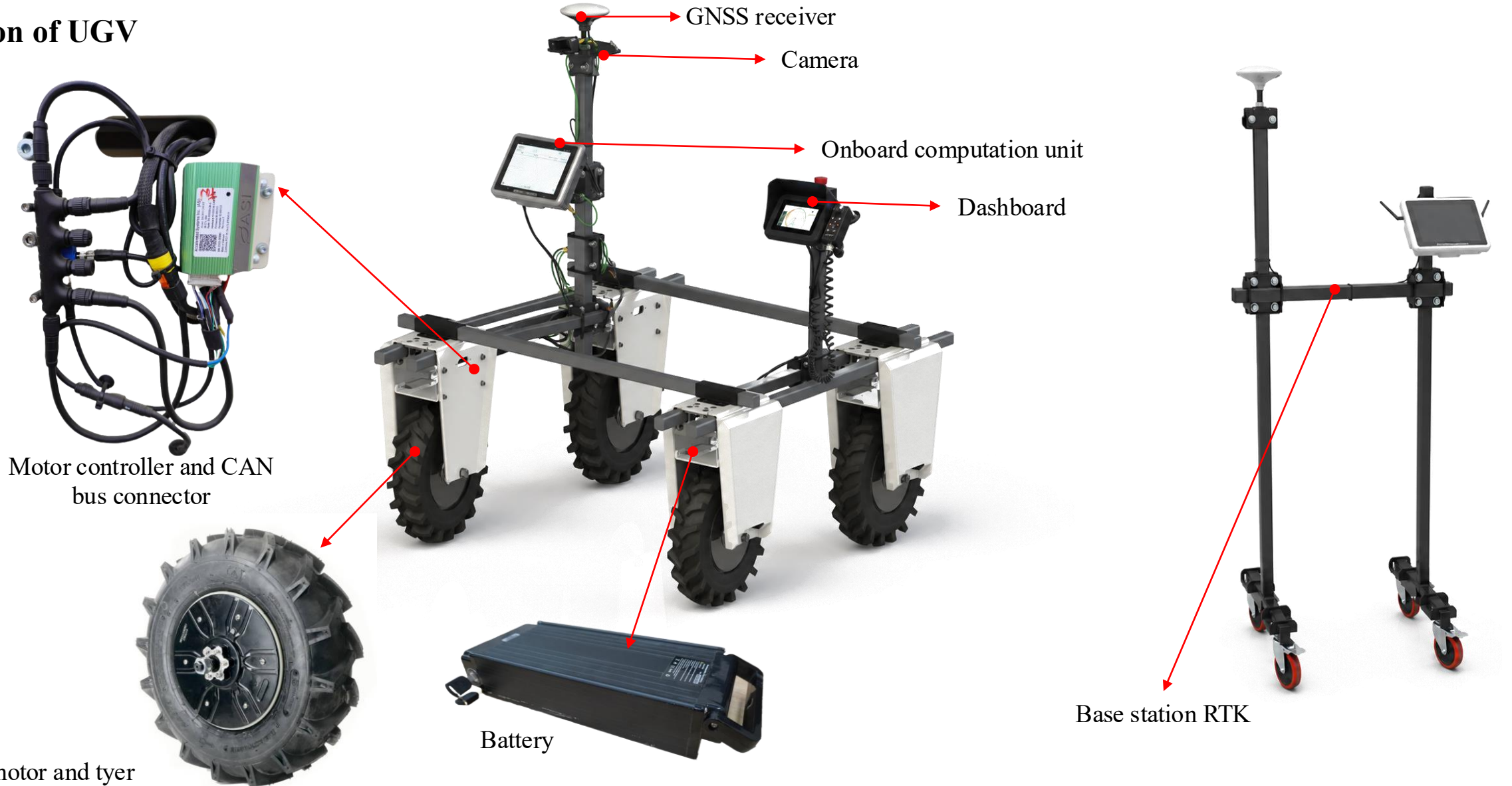
The objective of this research is to design the navigation, and speed control for UGV to be used in the different agricultural applications to reduce labor intensive works (like weed control etc.)

- loss of traction is an important parameter to be considered in controlling the movement of wheeled robot, which will cause:
 - loss of energy,
 - becoming less precise in farming operations,
 - reduce the efficiency of smart farming.

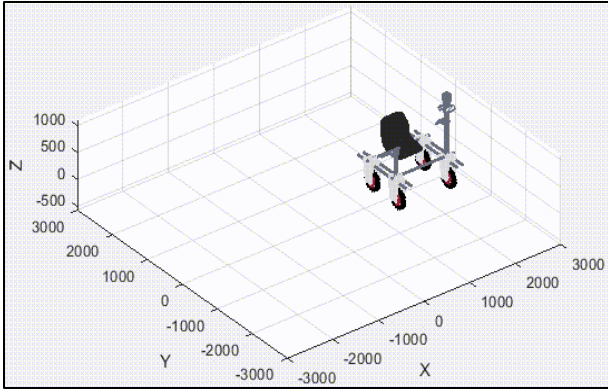
Therefore, we need to Implement and validate the controllers and sensors for UGV for self-correction in driving between the rows and reducing the loss of traction in these vehicles.



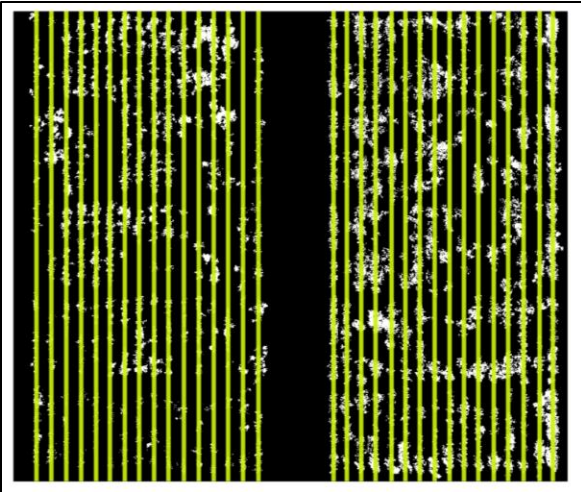
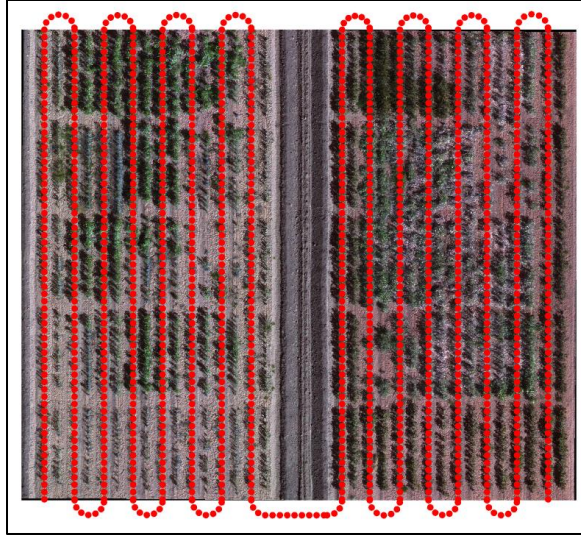
Motion of UGV



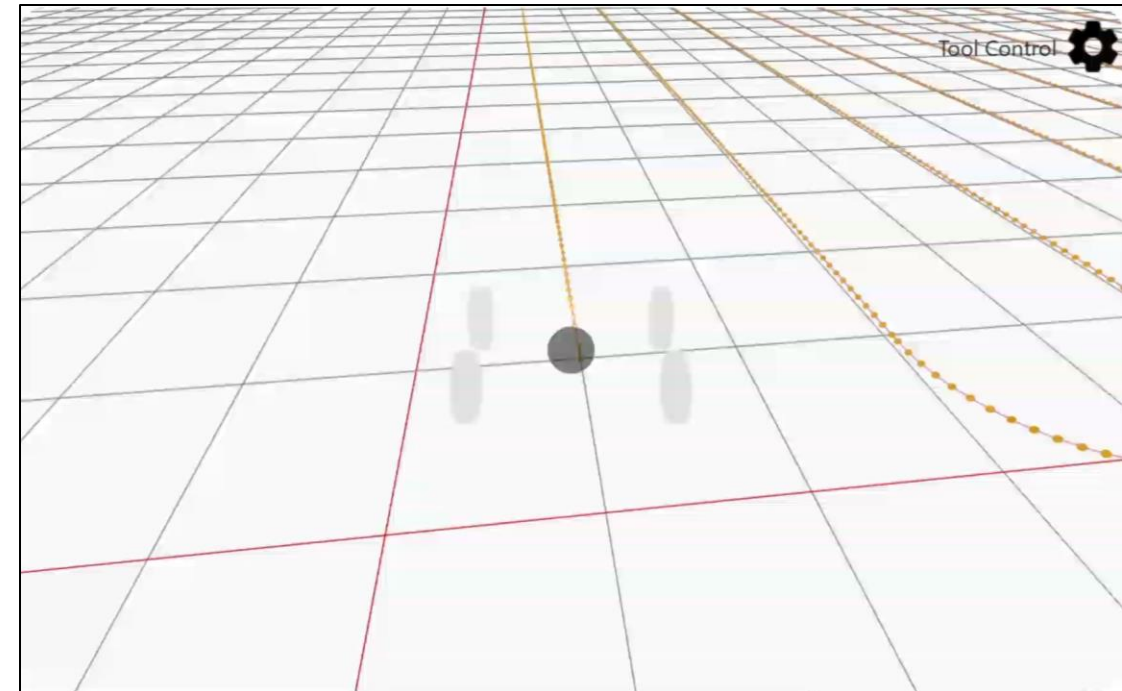
Motion of UGV on different situations



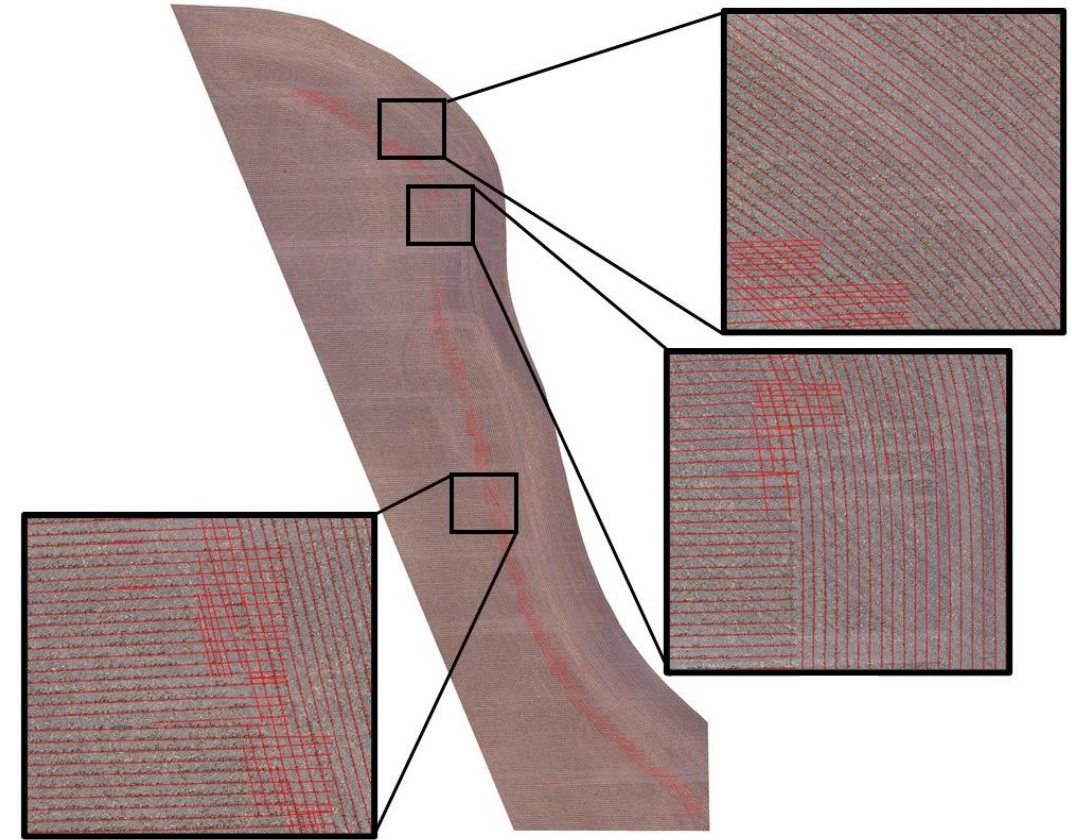
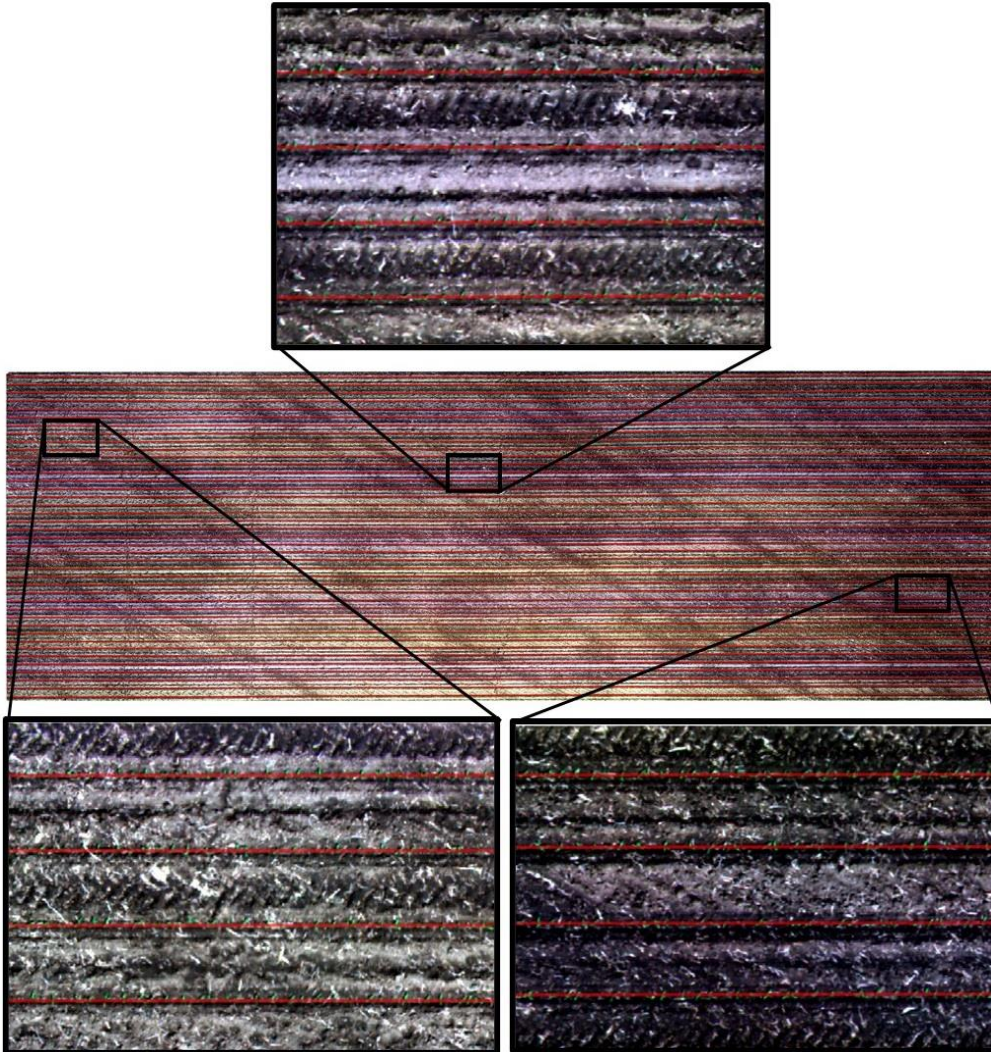
Waypoint Generation



- The straight crop rows were successfully detected.
- A path for UGV based on the implementation was generated.
- This test demonstrated accurate tracking performance by the UGV, with the error of 6.85" (17.39 cm).

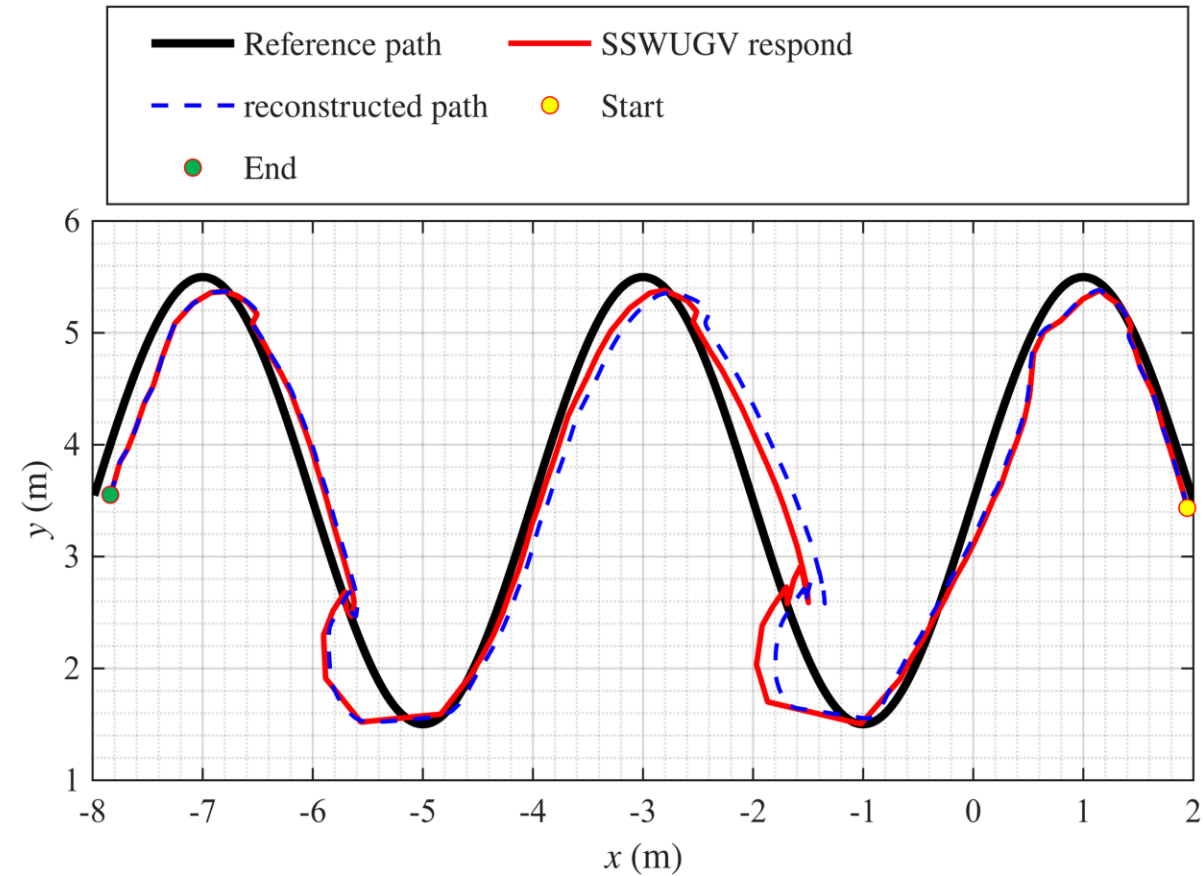
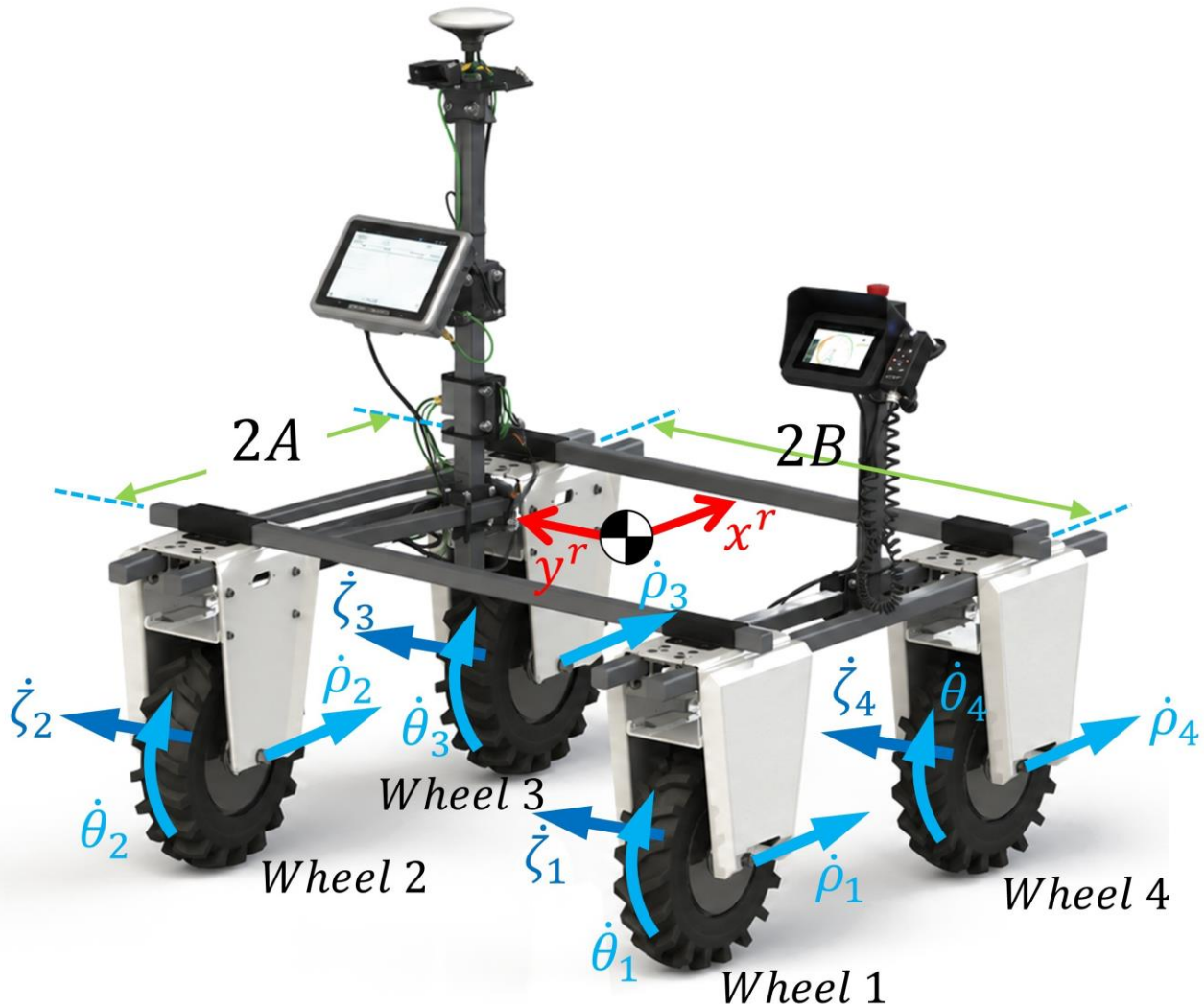


Crop Rows Detection from UAV Imagery



- A rule based crop row detection method for straight and curved crop rows is introduced.
- Quantitative evaluation against manually delineated reference rows showed high accuracy across all tested datasets, with ROR exceeding 99% and RMSE values below 5 cm.

Tire Forces Estimation and UGV Modeling



Experimental results show that the proposed USSWGV model reproduces the measured vehicle motion using reconstructed tire forces and agrees with real test data collected from the robotic platform ($RMSE < 1.5\%$).



Site-Specific Mechanical Weeding System (SSMW)

Shafi Md. Istiak

Supervisor: Dr. Sulaymon Eshkabilov

Motivation



Weeds significantly **reduce crop yield and Productivity**
(Image by Science Photo Library)



Sugar beet yield losses **\$1.25 billion** annually
(Image by Sola Seed Company)

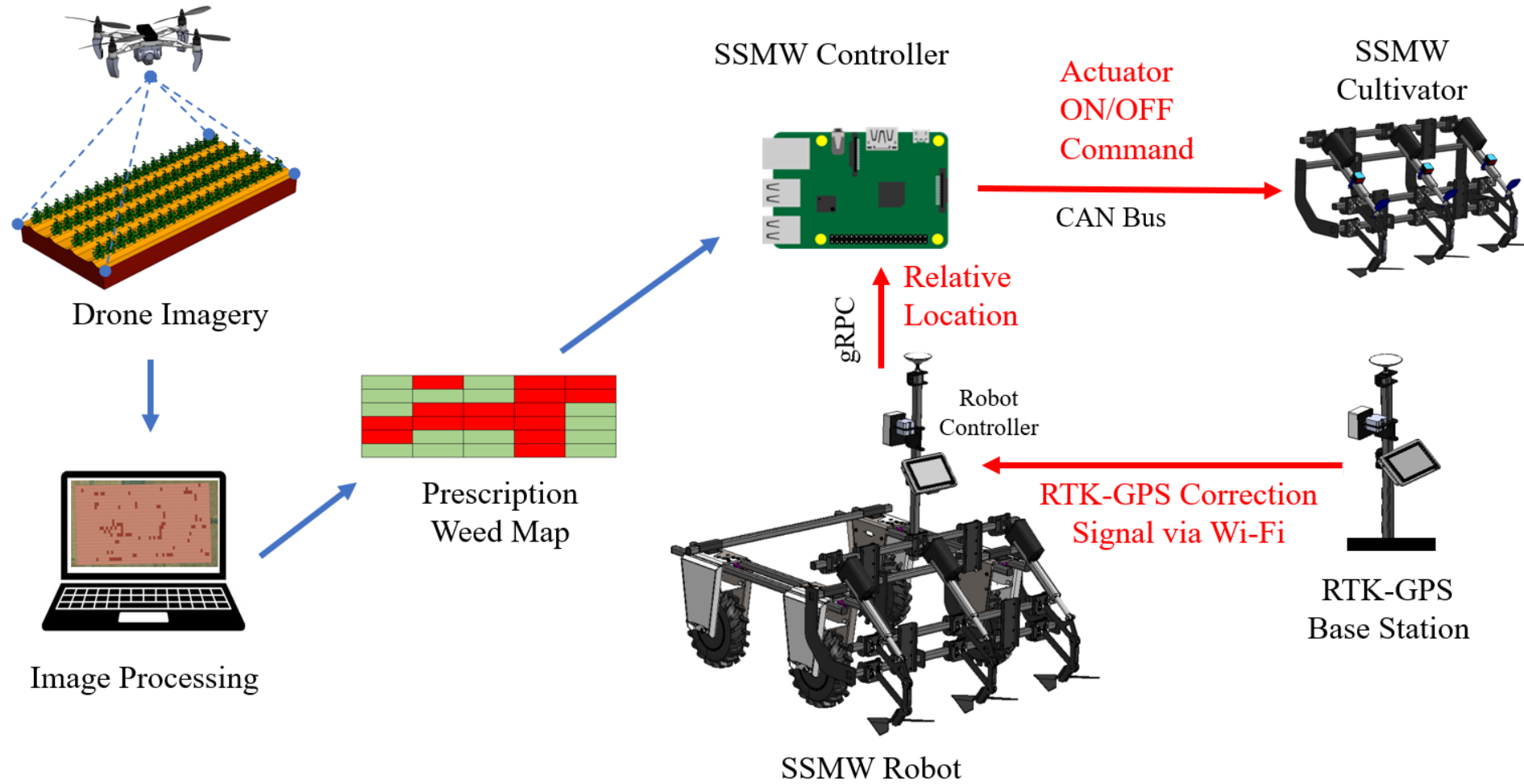


Ineffective herbicide application due to **herbicide-resistant weeds** (Image by Canva)

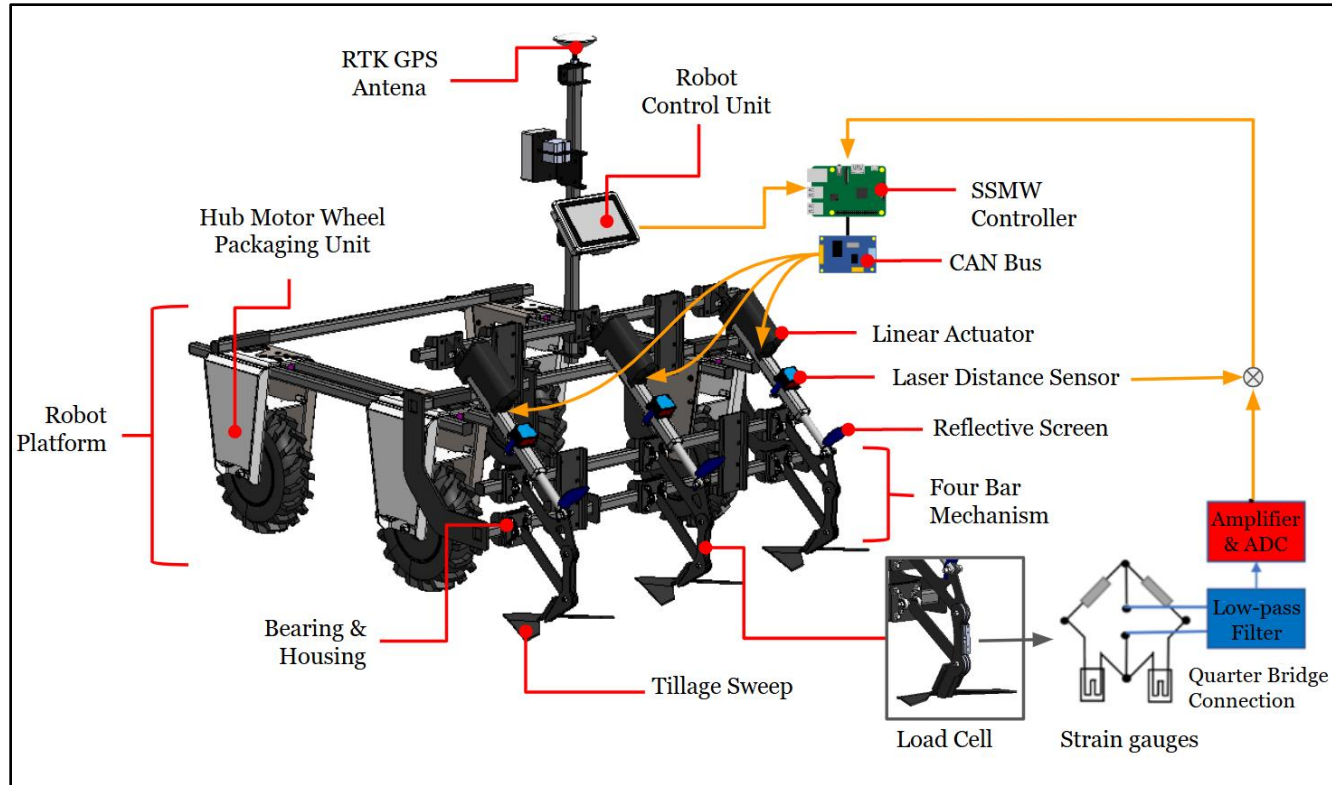
Objectives

- To design an individually controlled and self-adjustable shank-based three-row cultivator.
- To integrate the cultivator with an autonomous robotic platform for RTK-GPS guided site-specific weeding.

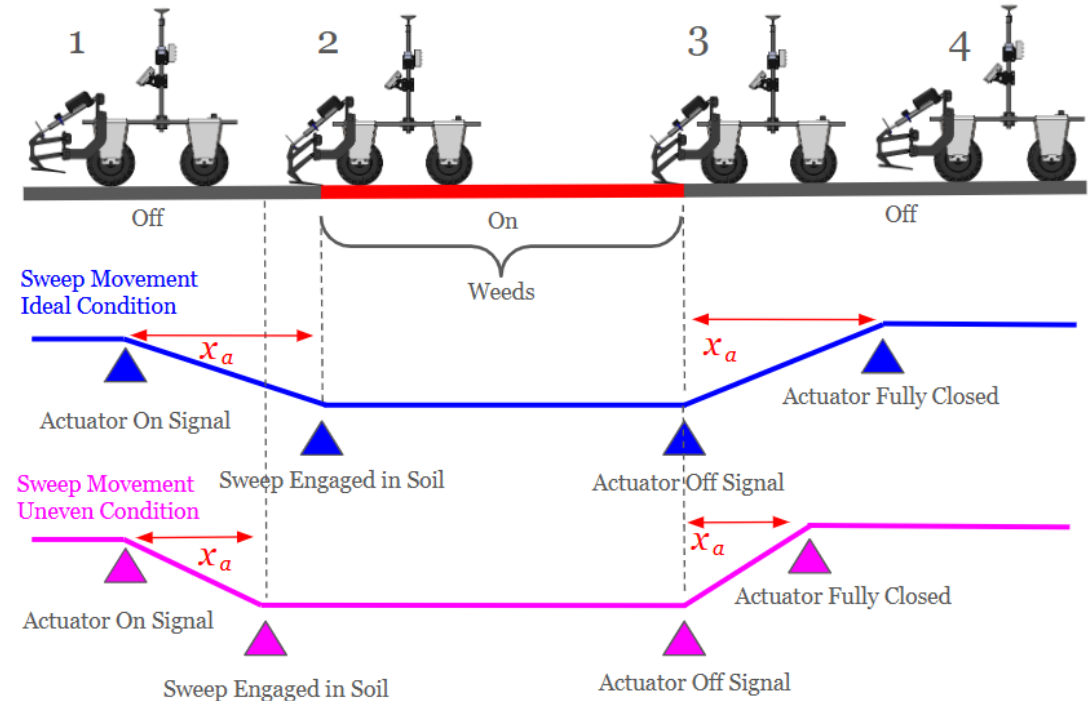
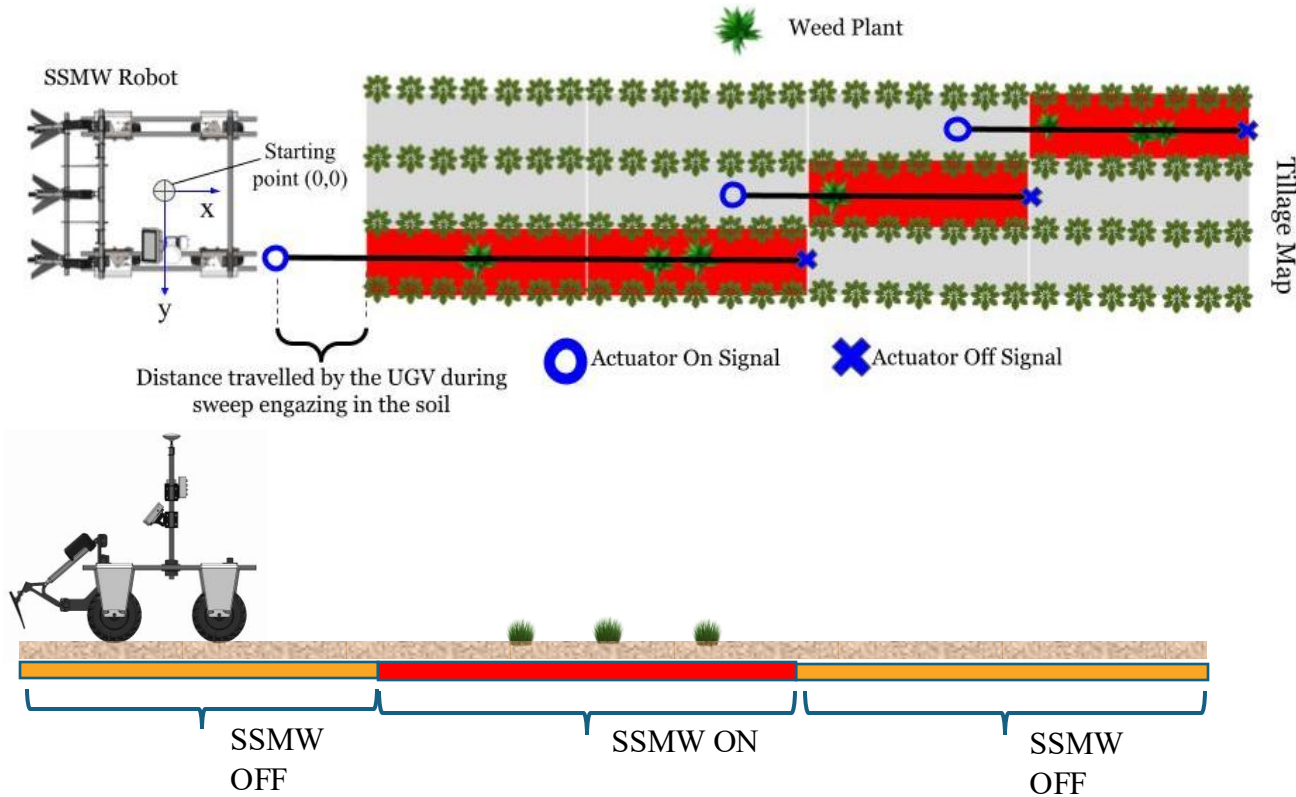
Site-Specific Mechanical Weeding



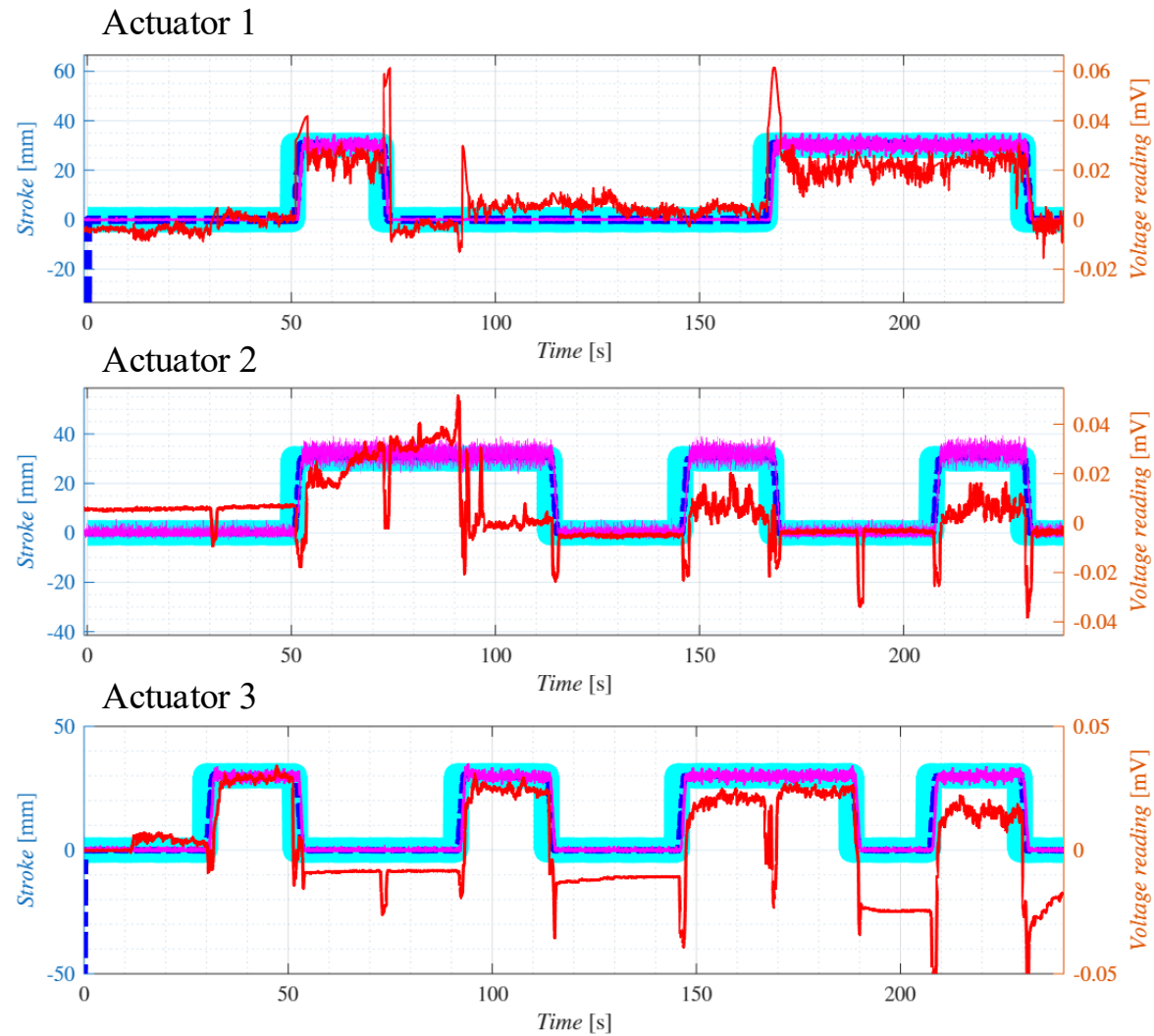
Site-Specific Weeding (SSMW) Robot



Site-Specific Weeding Operation



Field Test



Results

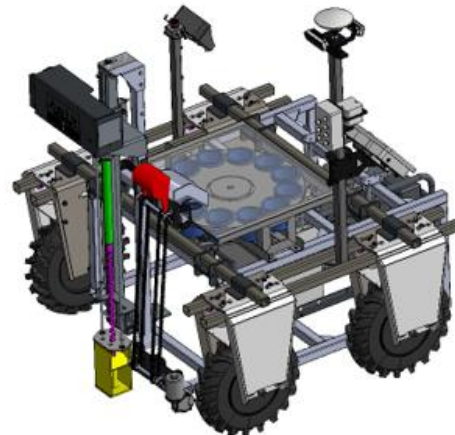
Parameters	Value
Total length of the rows, m	219.45
Total length for tillage, m	106.91
Length of tillage by the SSMW robot, m	107.08
Total positional error, m	3.93
Effective length of tillage by the robot, m	103.15
Efficacy by length, %	96.5
Draft force required for tilling 1 m of soil, unit force (UF)	1
Total draft force required for tilling whole test plot row, UF	219.45
Total draft force required for tilling with SSMW robot, UF	107.08
Energy savings, %	48.7
Soil disturbance reduced, %	48.7

- The System was achieved 96.5 % efficiency by length.
- Energy Savings was 48.7 % for 50% weed presence.
- Total positional error 3.93 m.
- Soil disturbance reduced by 48.7% for 50% weed presence.

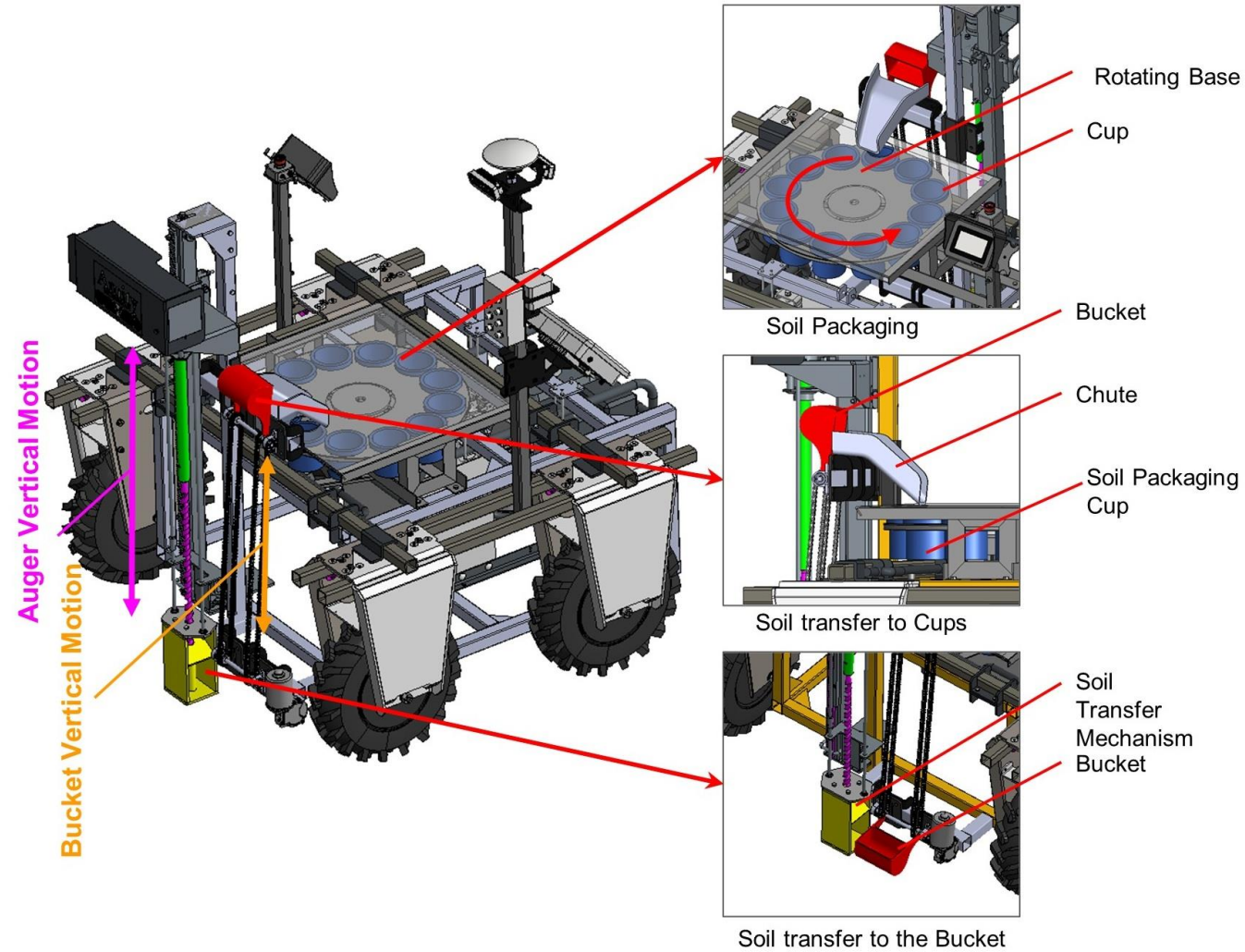
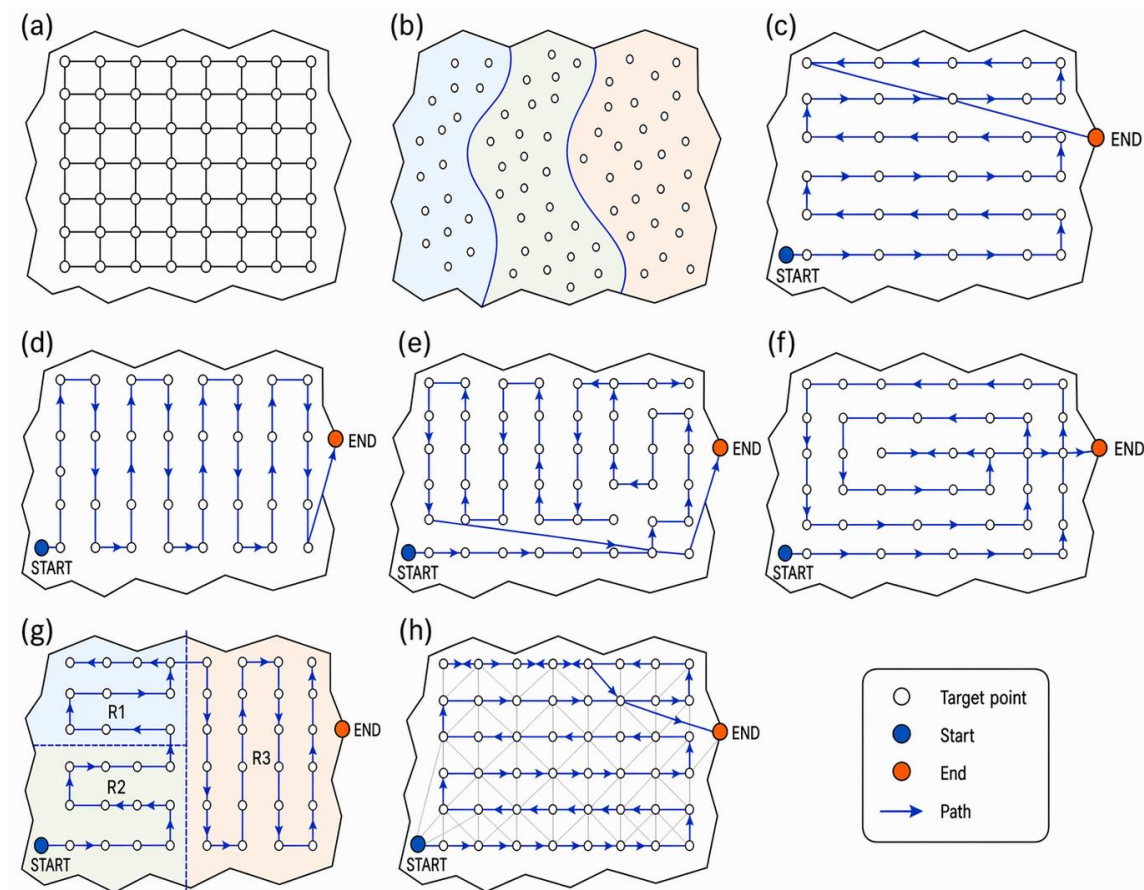
Farm Robotic Challenge 2026

Joanna Daniella Fajardo, Landen Krause, Shafi Md. Istiak, Mohammad Aftabi Talami, Joha Md. Ibn, Joshua Bruemmer, Artin Sadeghi

Supervisor: Dr. Sulaymon Eshkabilov



Autonomous Soil Sampler



Autonomous Soil Sampler



Top 5 Finalists



IOT AND WSN DEVELOPMENT FOR SUGAR BEET STORAGE

Bakht Alam Khan

Supervisor: Dr. Sulaymon Eshkabilov



Project Vision



Gas Analysis

Monitoring VOCs and Ethanol concentrations as early-warning markers of microbial rot and anaerobic fermentation.



Atmospheric

Precise tracking of Carbon Dioxide (CO₂) levels to assess respiration rates and metabolic activity in storage.



Conditioning

Real-time sensing of Temperature and Humidity to identify hotspots and high-moisture zones prone to fungal growth.

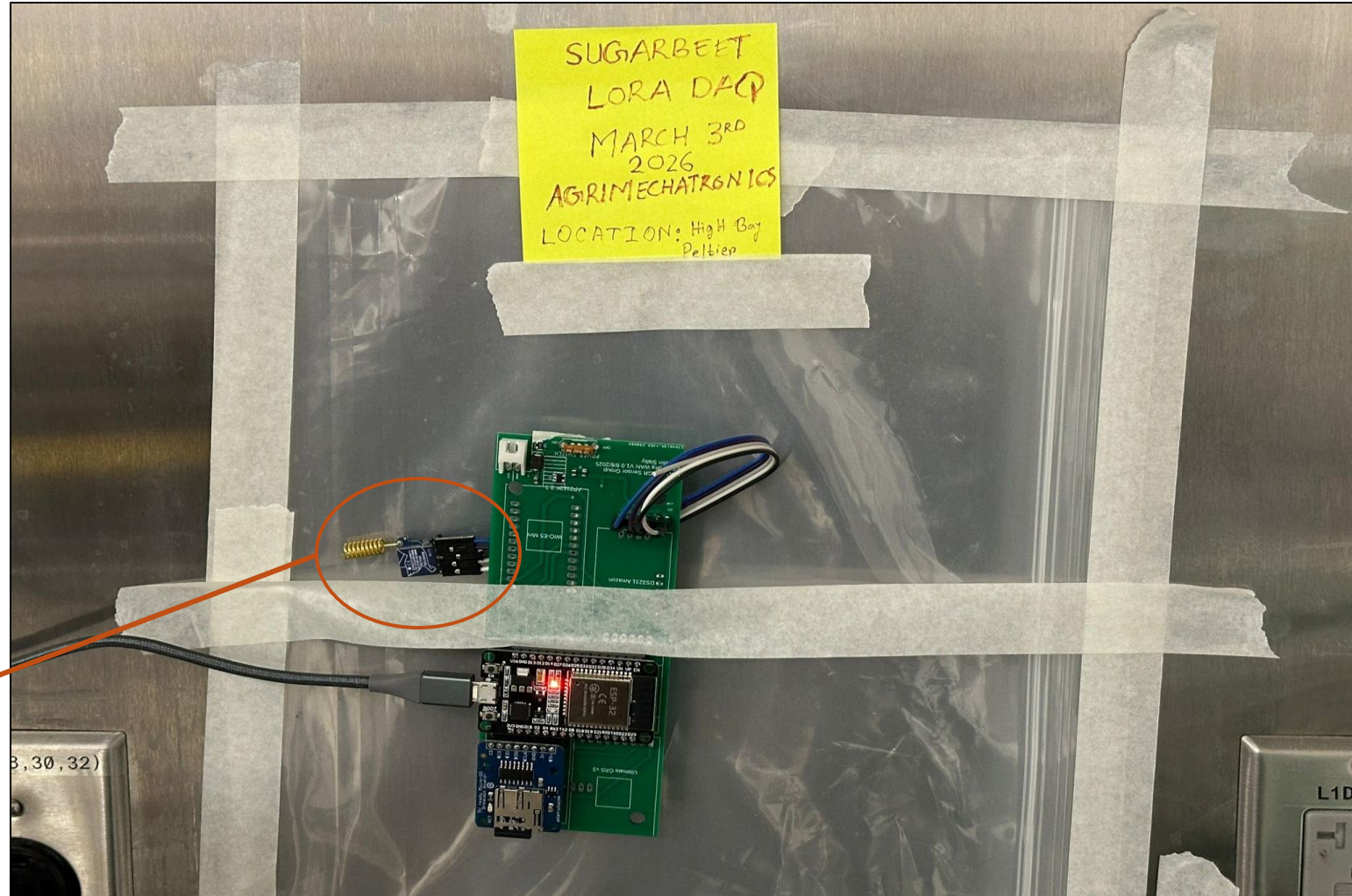
Deployment of PCBs and Its Explanation

- 6 PCBs were prepared in total.
- 3 PCB'S were kept at 4 degree Celsius.
- 3 PCBs were kept at room temperature.
- Buckets were kept non ventilated to replicate the beet pile storage in the field.
- Data was collected continuously from the beets.
- The studied variables are ethanol, co2, temperature and humidity.



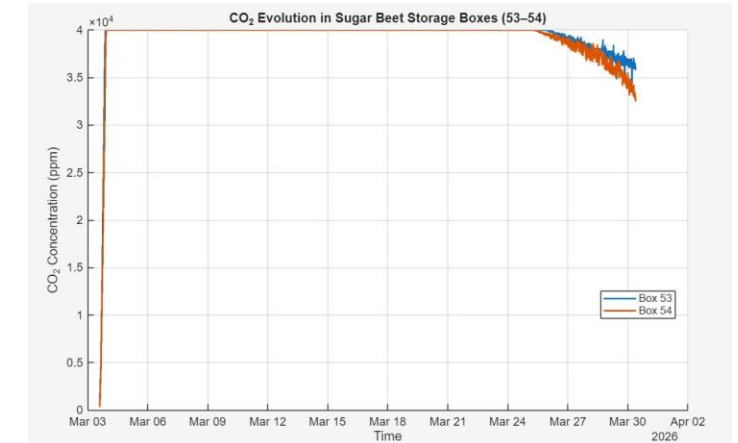
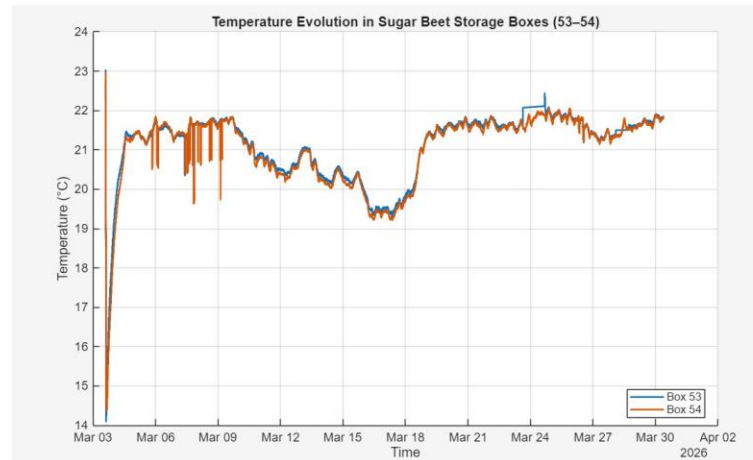
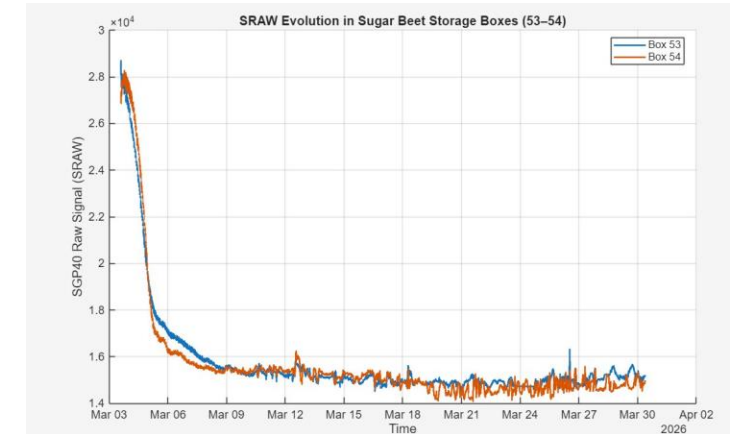
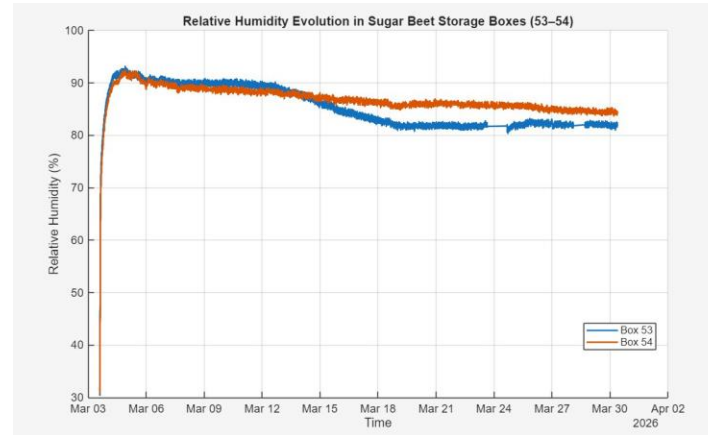
Deployment of PCBs

- Made a LORA station each bucket named differently has a different address.
- Lora will receive data from each box for 30 seconds and the sensor/PCB will sleep for 30 minutes,
- Noted the flash time so PCB each box don't clash for sending the data.

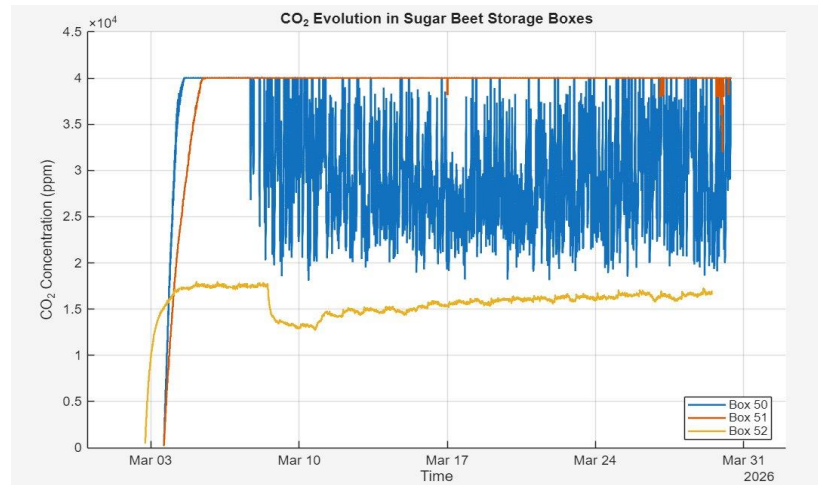
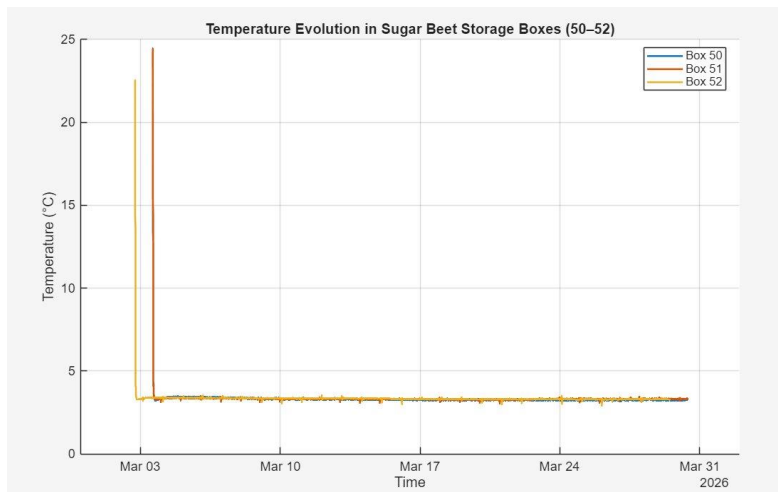
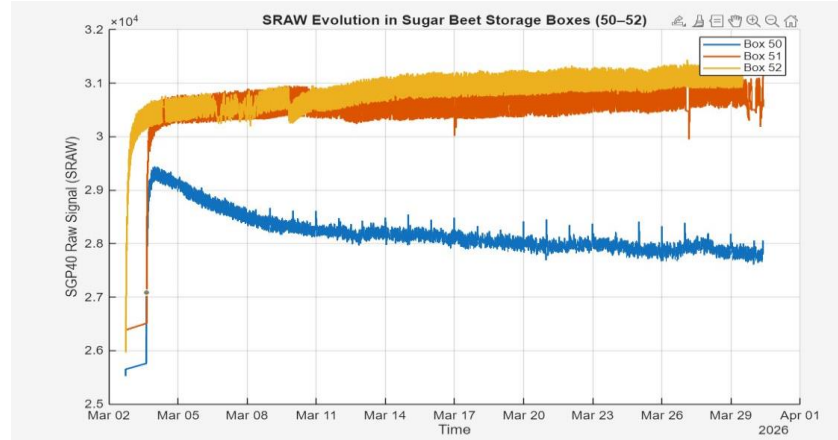
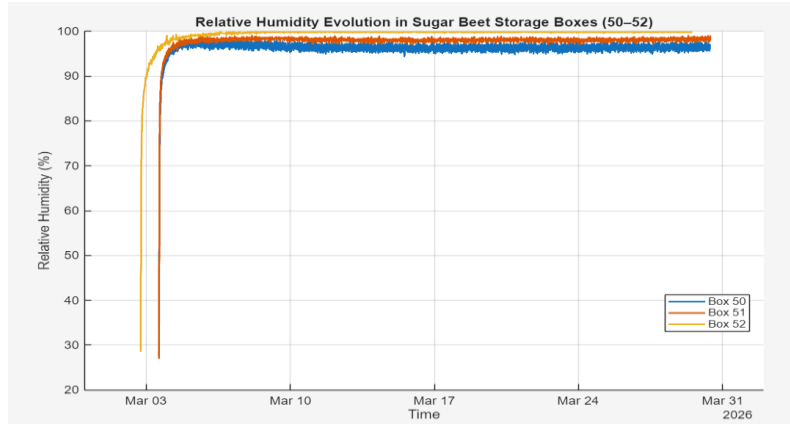


Data Visualization for Boxes at Room Temperature

- Boxes 53-54 consisted of 3D-printed enclosures housing the PCBs. These units were maintained in a controlled environment at 4°C, and the collected data was visualized as shown.
- The room temperature buckets were showing signs of high CO₂, humidity and ethanol build up immediately after being deployed.
- Box 55 stopped as the power rail was short circuited due to the condensation and shortening the power rail in turn.



Data Visualization for Boxes at 4 °C



- Boxes 50-52 consisted of 3D-printed enclosures housing the PCBs. These units were maintained in a controlled environment at 4°C, and the collected data was visualized as shown.
- The ethanol build up in the 4 degree Celsius was comparatively less than the room temperature buckets.
- Sugar beets were healthy and were showing visibly less signs of rotting or mold after the buckets were opened.

Future Steps

- Wireless sensor network setup for data collection and analysis.
- Use coating for PCB's to limit humidity and moisture issue that the PCB's are facing.
- Working on better sensors for TVOC and ethanol with proper calibration steps followed.
- Working on Signal Loss issue from LORA.
- Working on designing a PCB with sensors setup to target TVOC and ethanol at initial stage after harvesting the beets.
- Finding a CO₂ sensor that has a threshold more than 50,000ppm to observe higher CO₂ values as beets in storage tend to higher CO₂ value at room temperature.

Controlled Environment Agriculture in North Dakota

Shafi Md. Istiak

Supervisor: Dr. Sulaymon Eshkabilov



Motivation



9.7 billion global population
by 2050



4 million hectares
of farmland are being lost to the development
of infrastructures



**Soil erosion and depletes essential
nutrients**
in traditional farming methods



Year-round crop production



Growing controlled conditions
through soilless farming methods

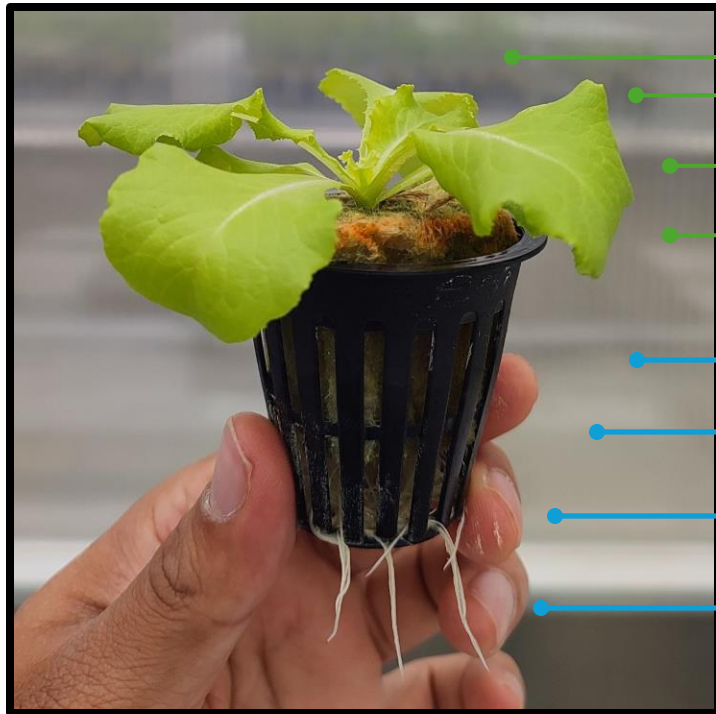


Enhances biosecurity, mitigates
pests and drought



Plants in Hydroponic CEA system grow
faster than traditional soil system

Monitoring Parameters



Environment Temperature (**DHT22**)

Relative Humidity (RH) (**DHT22**)

Light Intensity (**GY-39**)

CO2 (**SCD30**)

Environment
Parameters

PH (**Electrode PH Probe**)

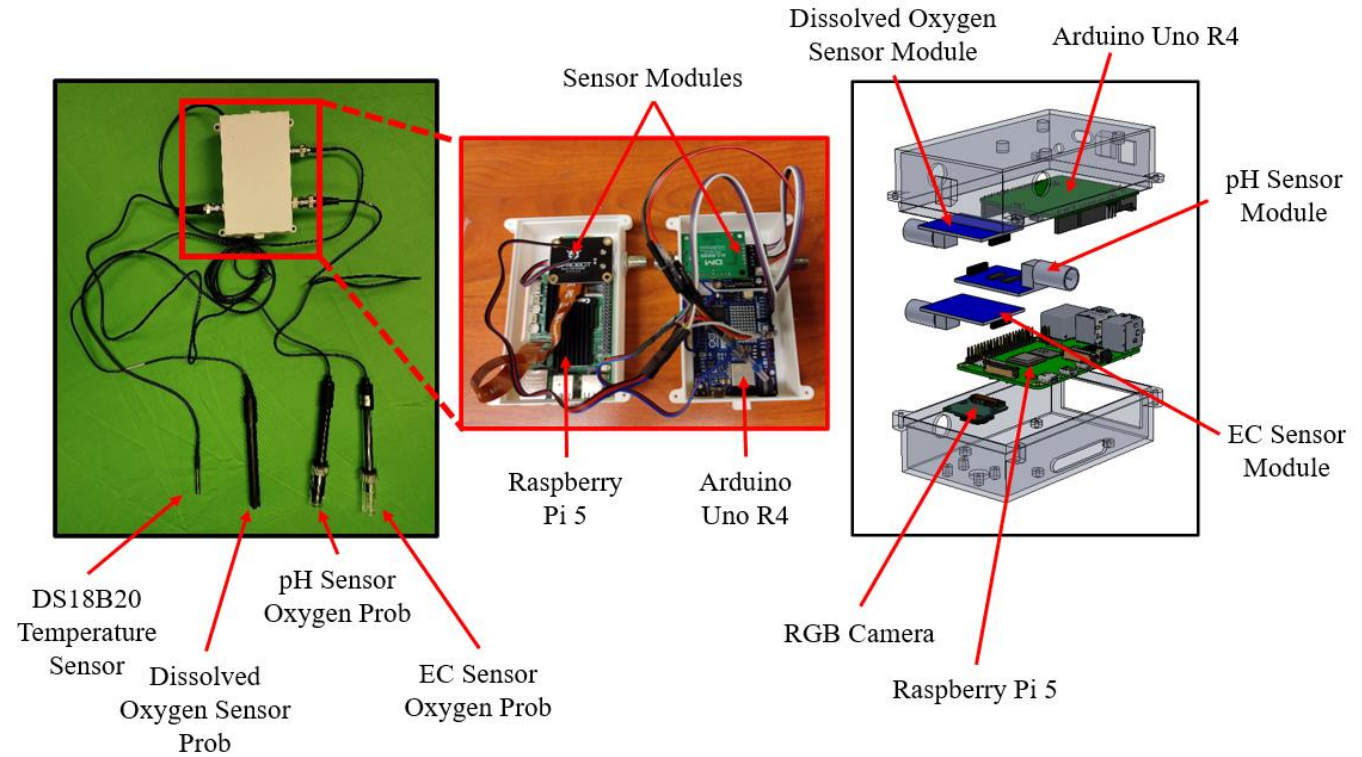
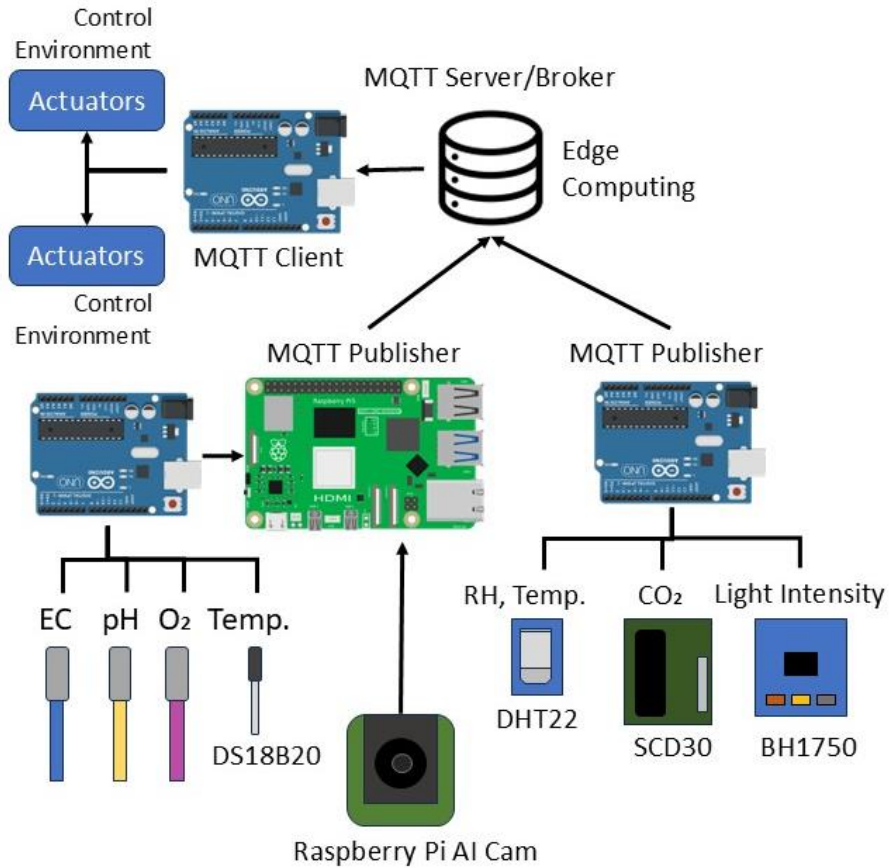
Electric Conductivity (EC) (**TDS Sensor**)

Water Temperature (**DS18B20**)

Dissolved Oxygen

Hydroponic
Parameters

System Architecture



Experimental Setup



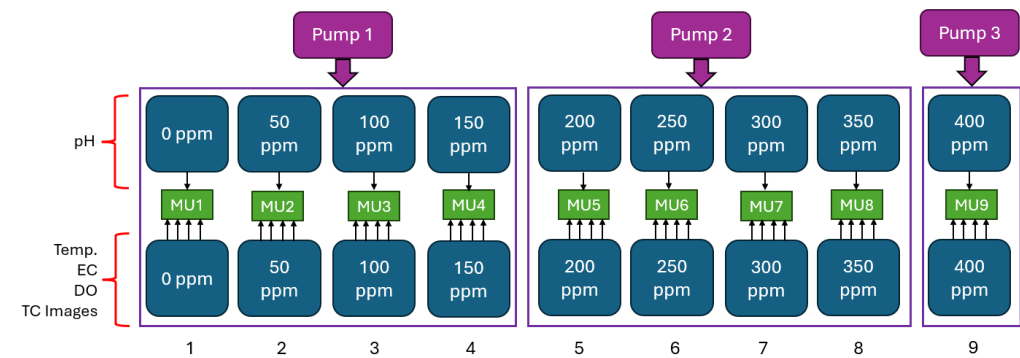
Day Light



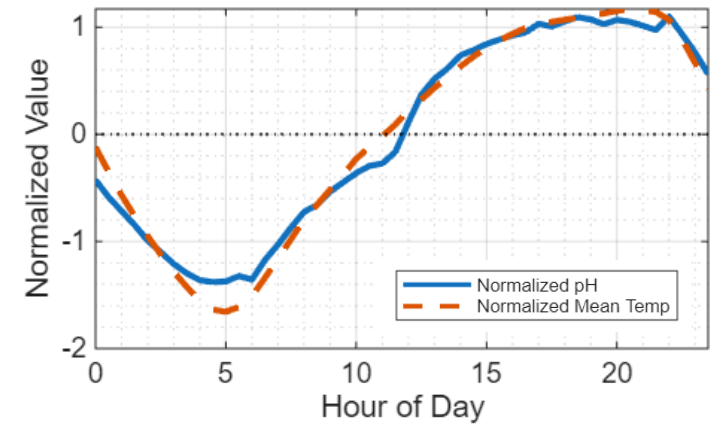
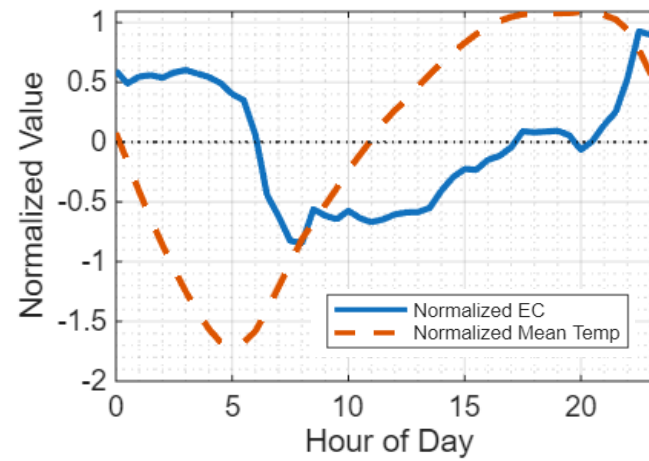
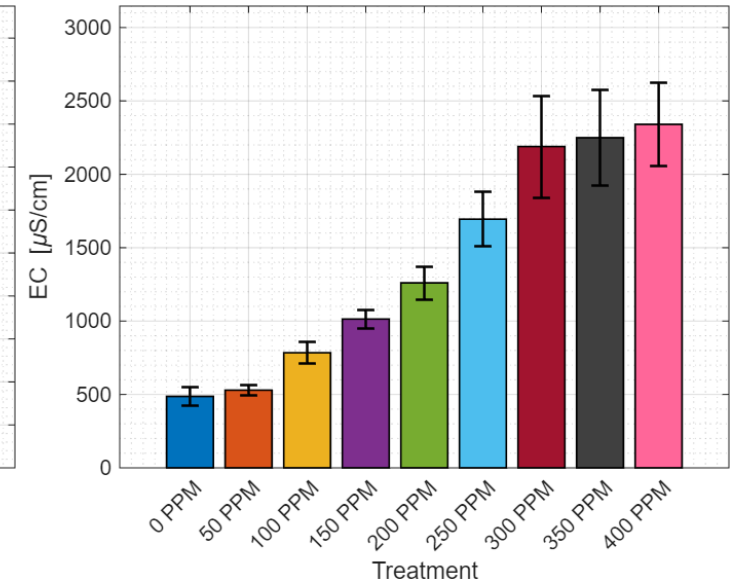
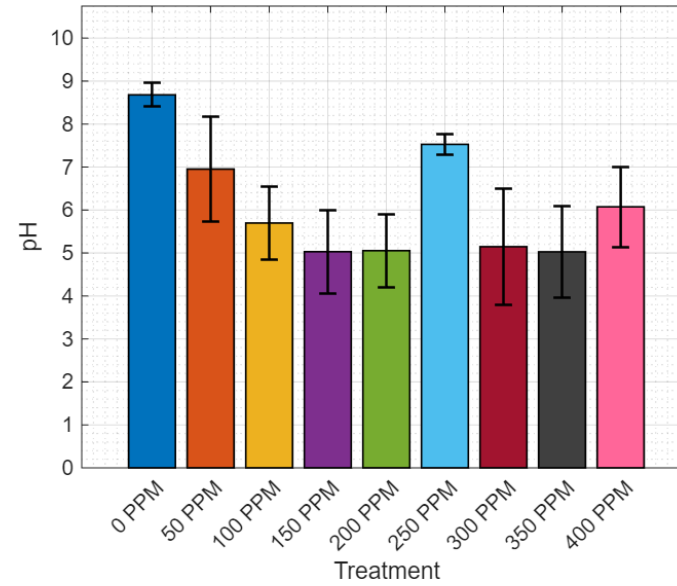
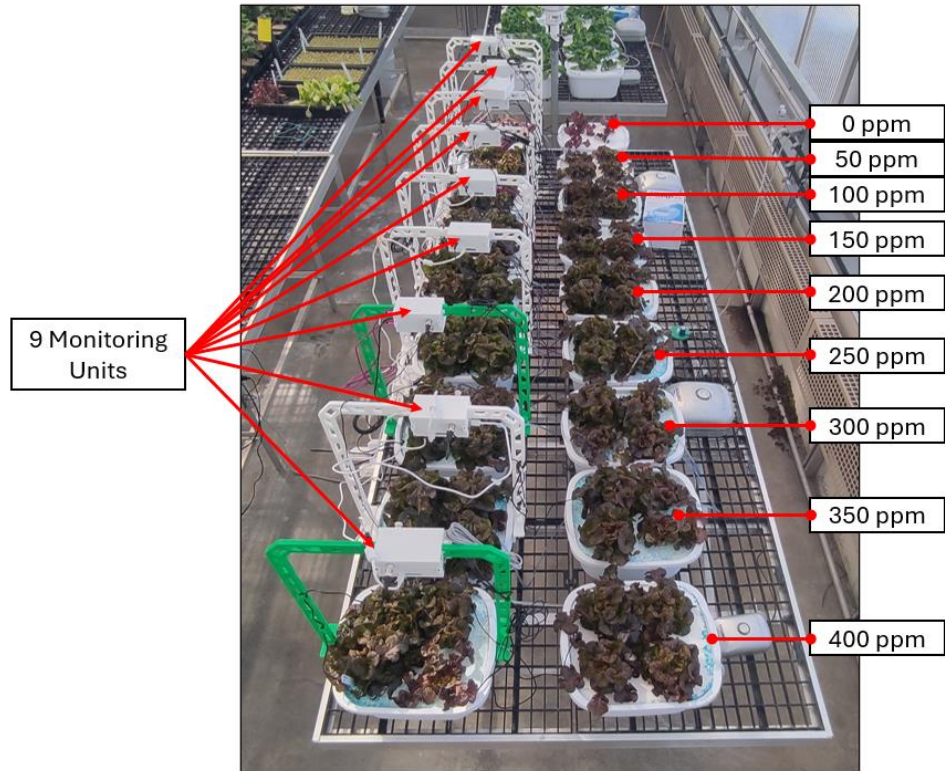
Artificial Light



Dark

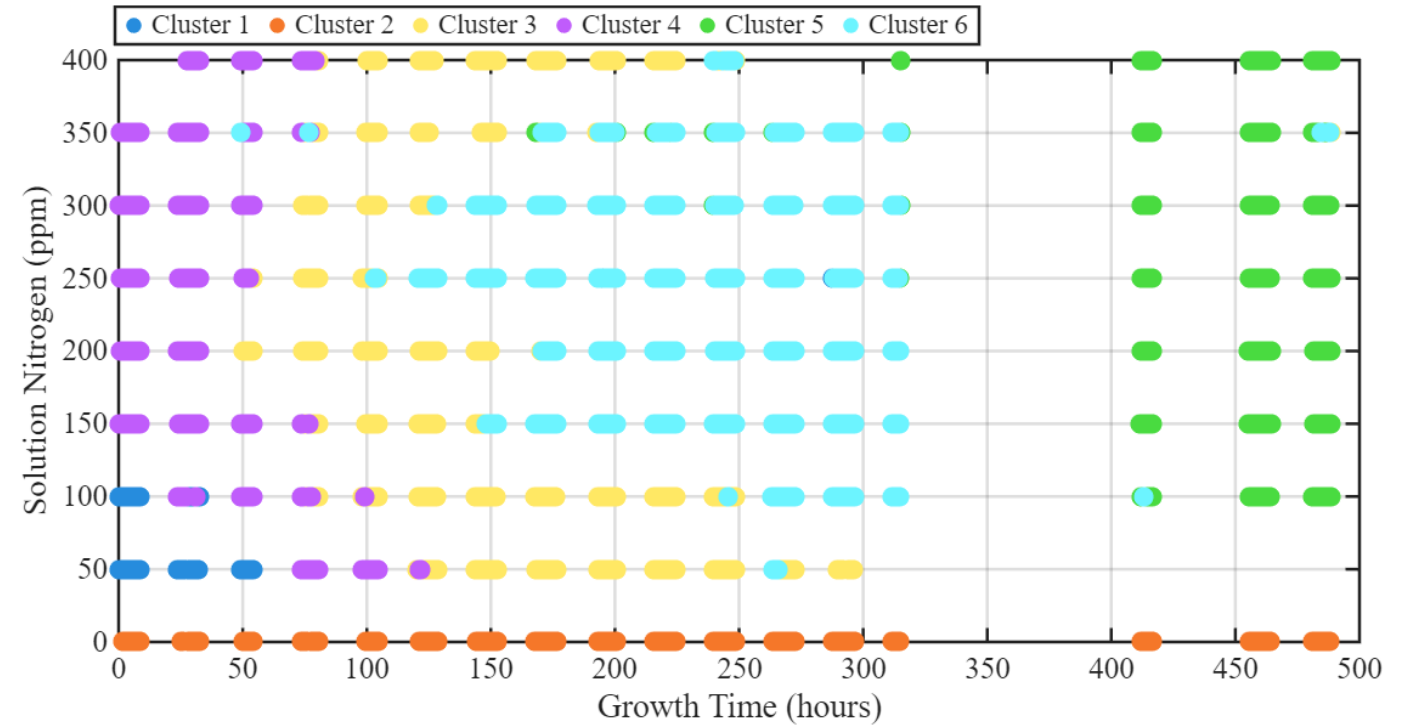
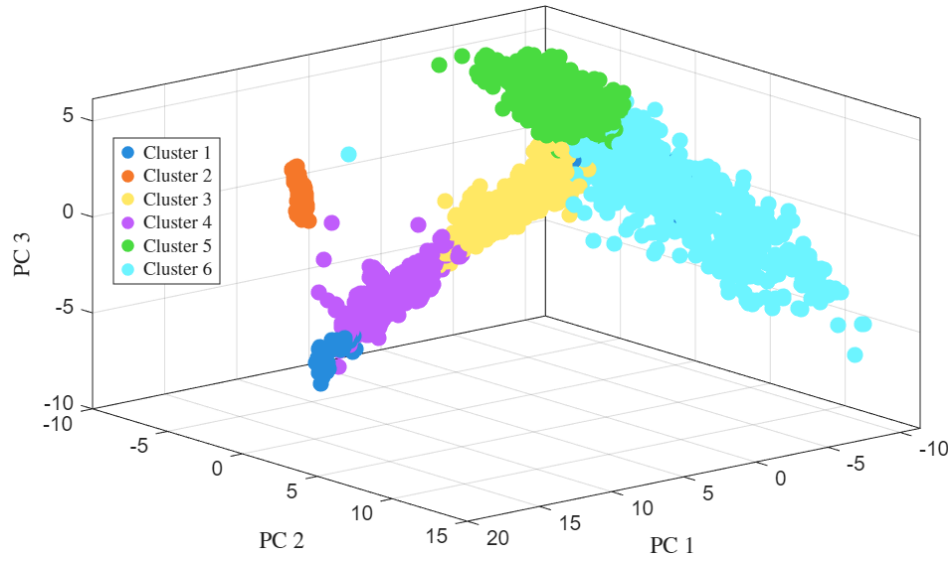


Greenhouse Data Collection

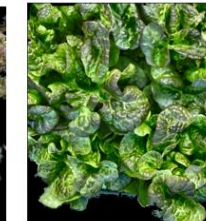
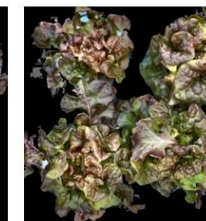
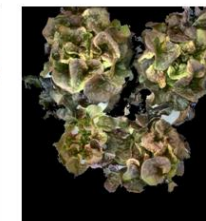
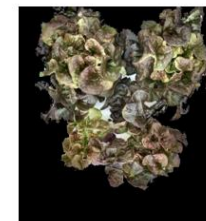
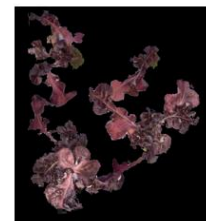


Machine Learning Application

Gaussian Mixed Model



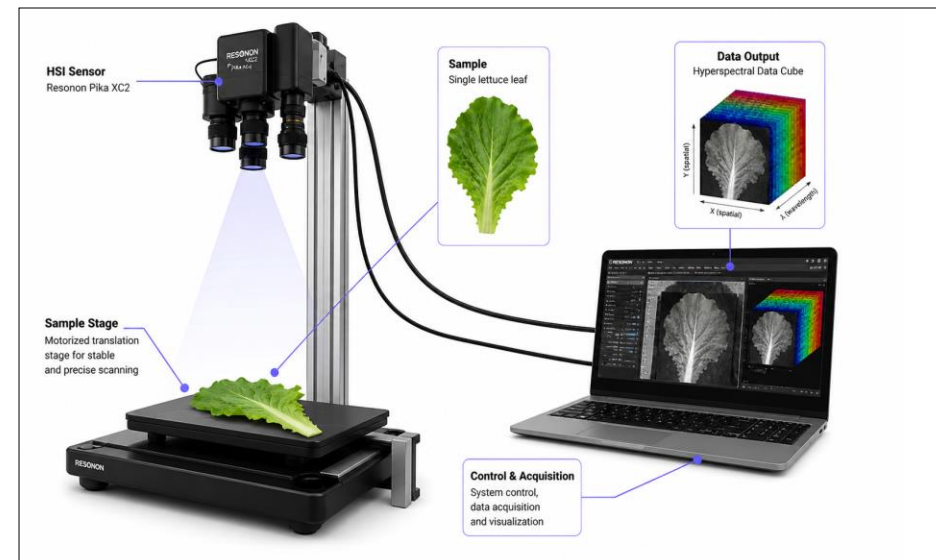
- Image based plant growth level prediction.
- Growth level based hydroponic water nutrients control from Temp., EC and pH sensor feedback.

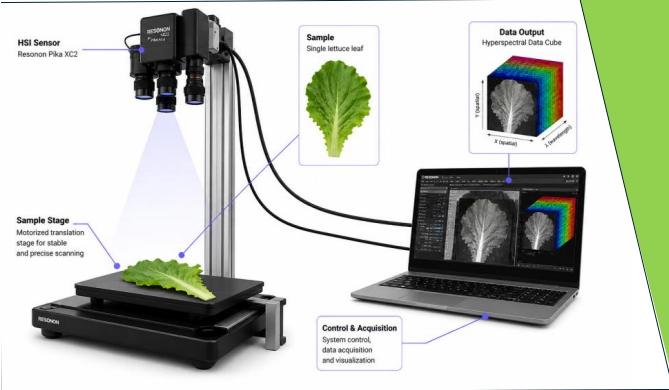


Using machine learning algorithms for optimizing lettuce production

Farhin Faiza Neha, Mahmoud Pranto Alam

Supervisor: Dr. Sulaymon Eshkabilov





Application of Hyperspectral Imaging and Machine Learning for Nutrient and Disease Monitoring of Lettuce in Controlled Environment Agriculture

Farhin Faiza Neha

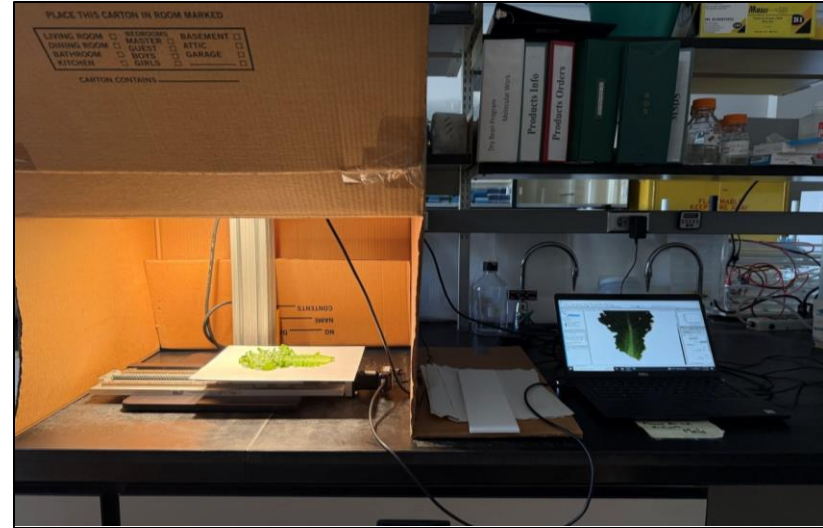
Supervisor: Dr. Sulaymon Eshkabilov

Objectives

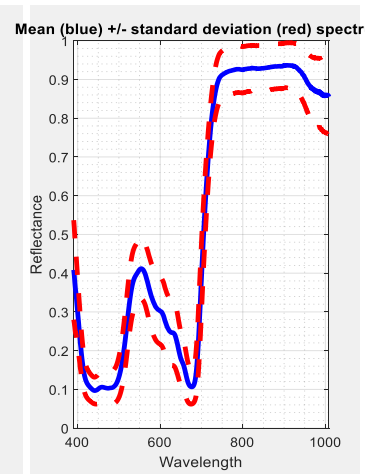
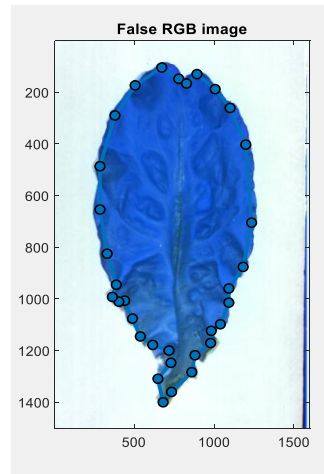
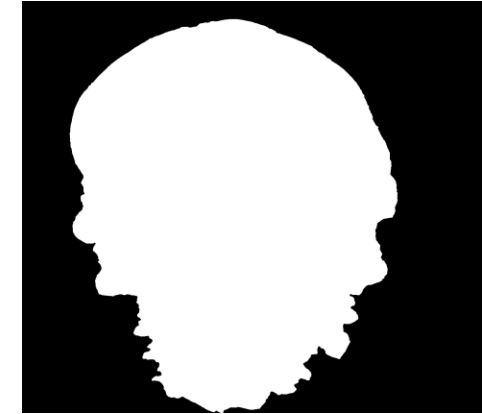
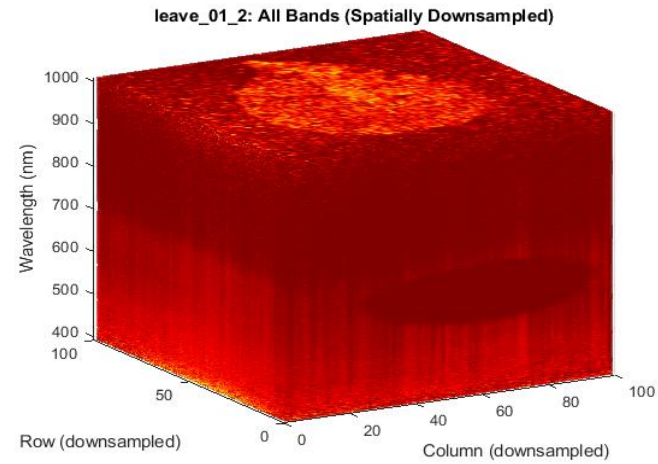
- **Leaf, root, whole lettuce plant integration**
- **Non-destructive, real-time monitoring** of lettuce grown with egg-washing wastewater.
- **Sustainability angle and food-safety** concerns related to wastewater usage in agriculture. Wastewater is considered as a **contamination risk** for vegetables, including leafy greens [8]
- **Non-invasive nutrient and pathogenic disease** monitoring.
- **Feature selection** and robust ML framework



Pictures from experiments



HSI Leaf Analysis (Results: plots and graphs)

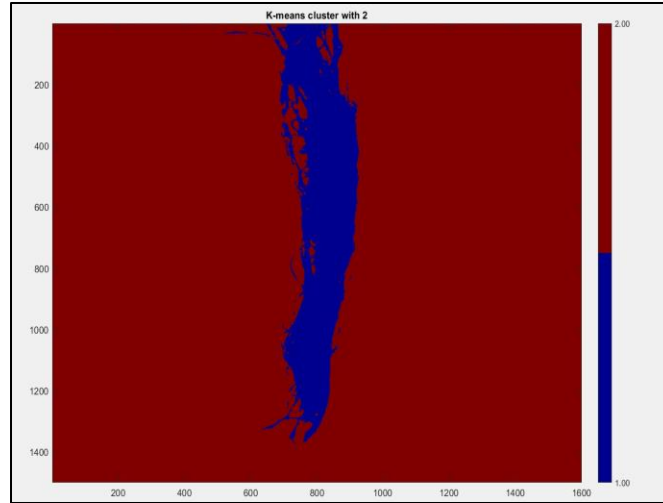


HSI Root Analysis (Results: plots and graphs)

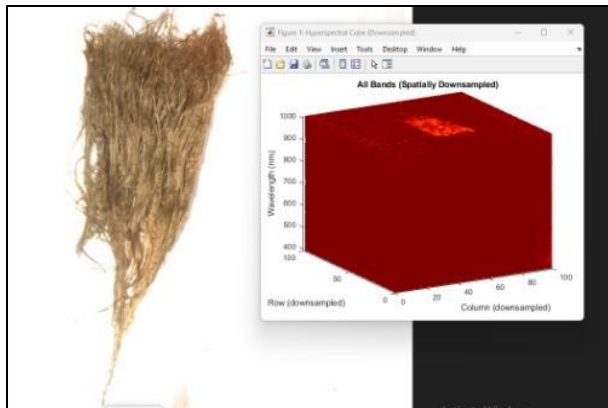
Step 1



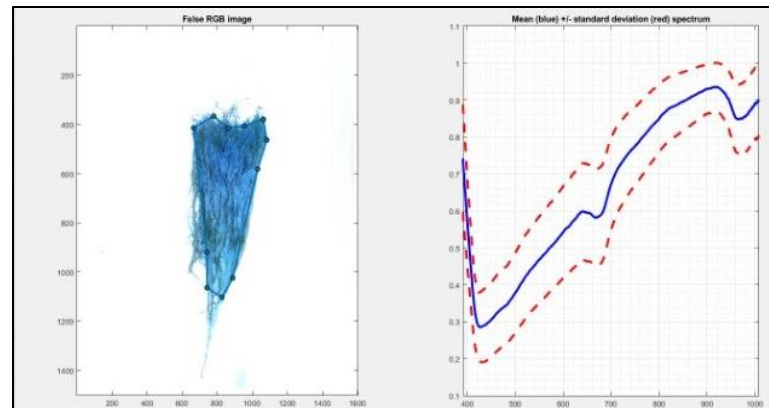
Step 3



Step 2

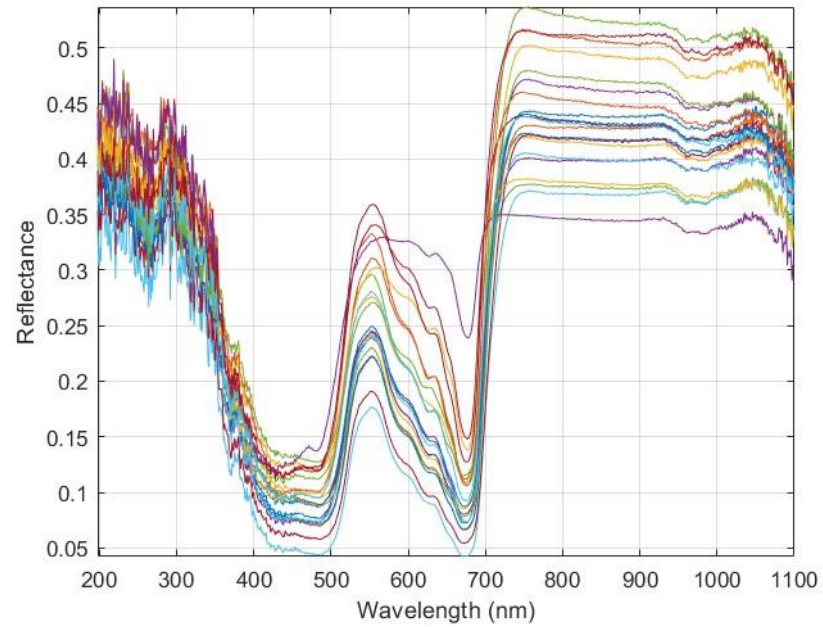


Step 4

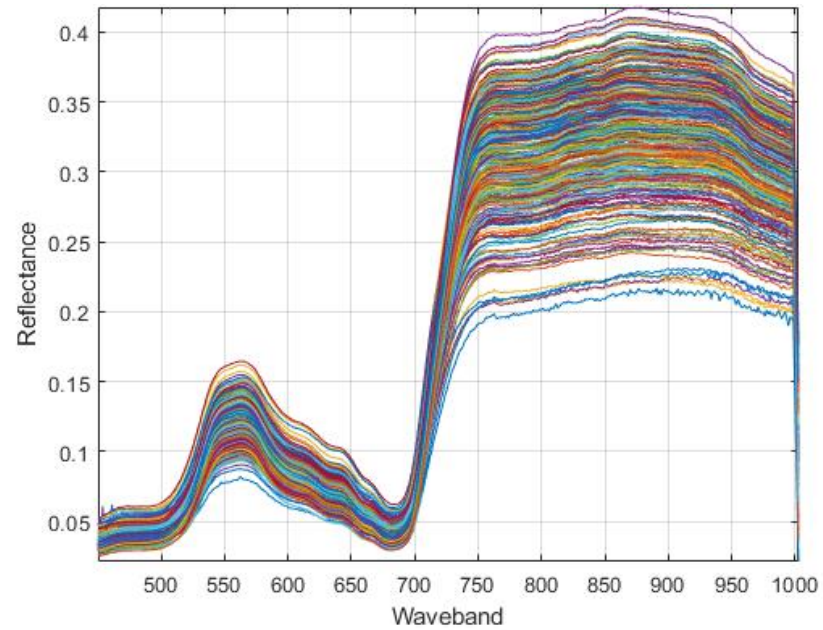


Results: plots and graphs

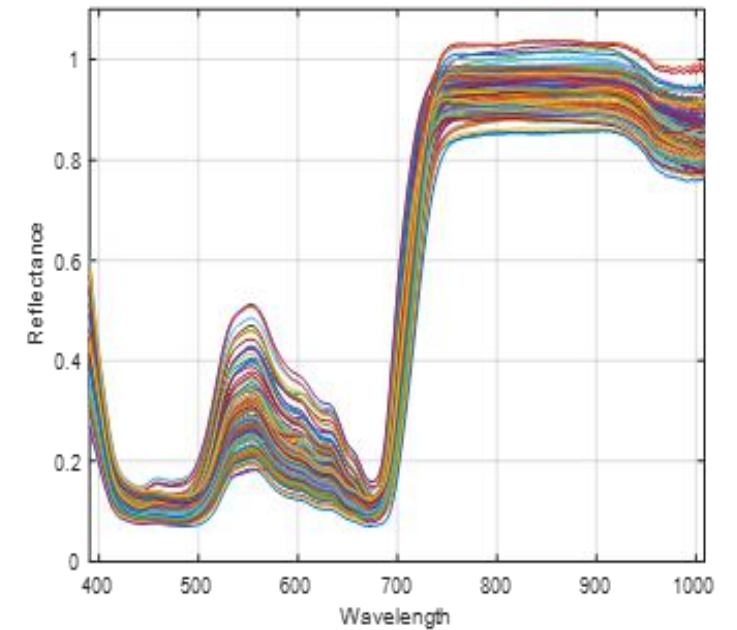
Data obtained from USDA



Data obtained from Purdue

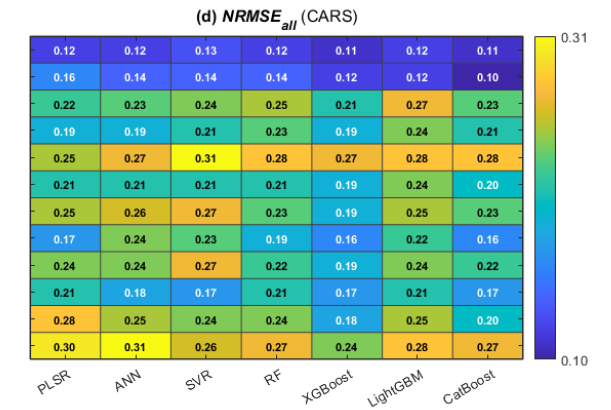
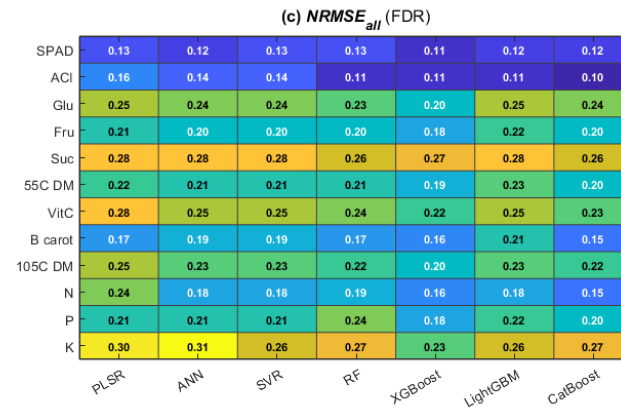
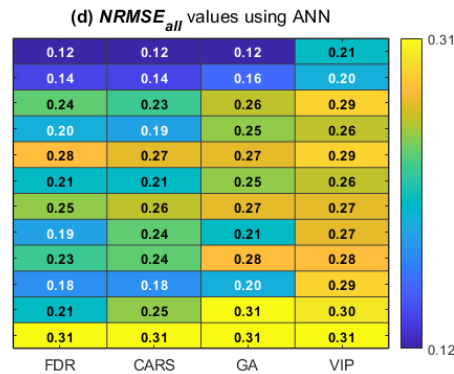
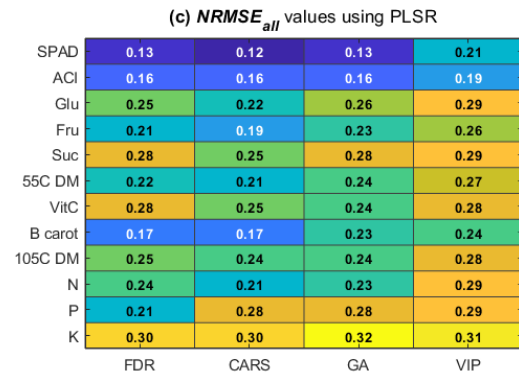
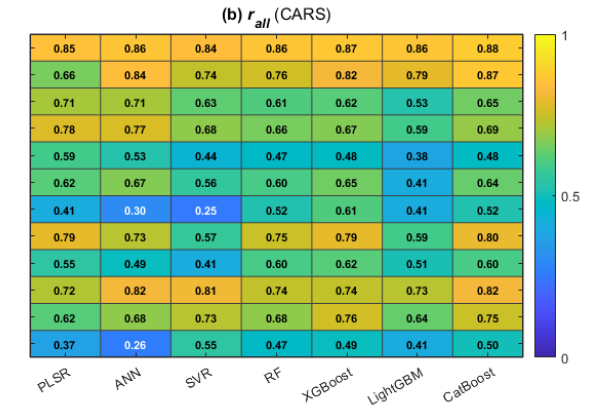
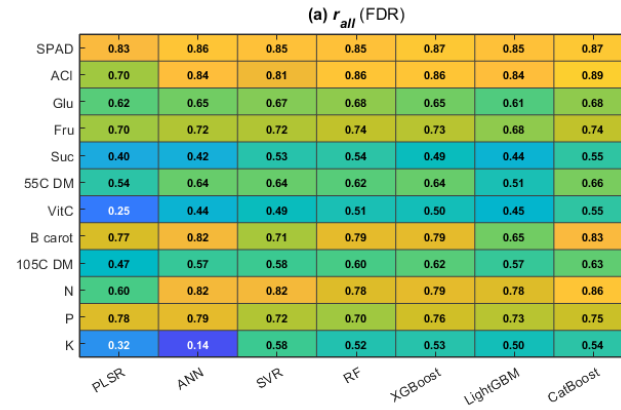
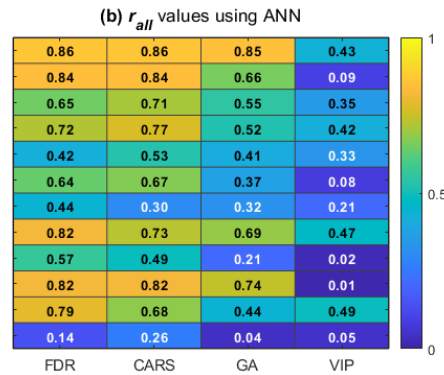
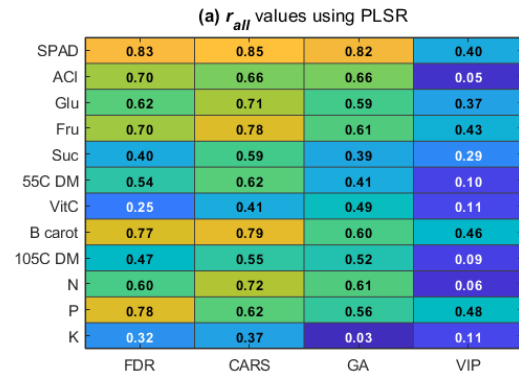


Data obtained at NDSU



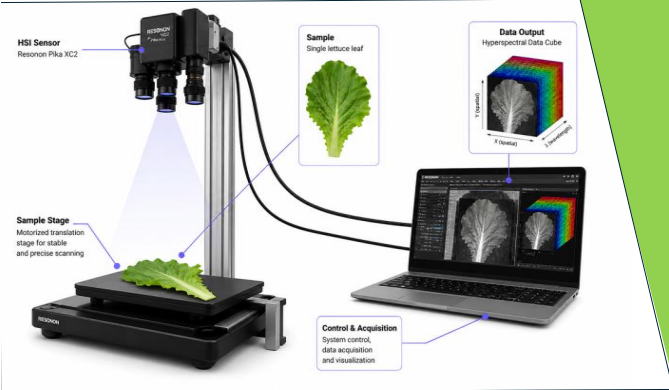
Results: plots and graphs

Results from paper: Benchmarking Feature Selection Approaches for Machine Learning Algorithms for Hyperspectral Data Modelling of Lettuce Cultivars



Conclusion

- **Proposed Framework addresses 3 core research problems related to HSI:**
 - **Leaf, Root, Whole lettuce plant** Data Integration .
 - **Feature Selection & ML Optimization** for Lettuce Growth and Nutrient Estimation
 - **Pathogenic Disease Detection** using both Hyperspectral and RGB image Data.



Prediction of Lettuce Quality Parameters Using Hyperspectral Imaging Data and Machine Learning

Mahmoud Alam Pranto

Supervisor: Dr. Sulaymon Eshkabilov

Introduction

Why Lettuce Is Important

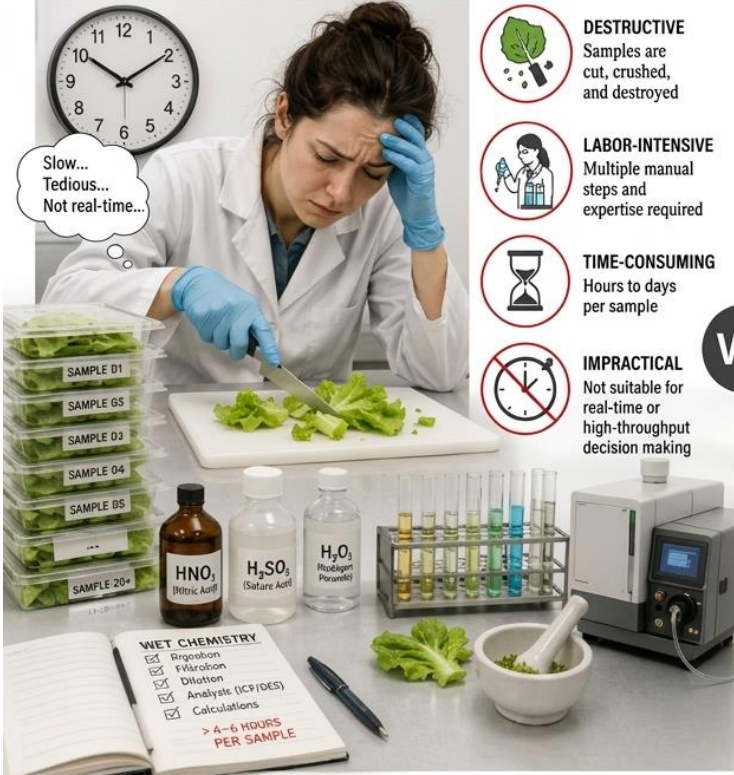
- ❑ Lettuce is the most studied and commercially prevalent crop within CEA systems [1].
- ❑ In 2022, lettuce accounted for **nearly one-fifth of the \$21.8 billion** in U.S. vegetable and melon cash receipts.
- ❑ U.S. per-capita lettuce availability was **28.6 lb in 2024** [2].

Challenges

- Disease and insect problems can develop quickly and reduce yield and quality [3].
- It is important to continue to develop and improve plant disease control tools for the reduction of potential yield loss from disease [4].

TRADITIONAL NUTRIENT ESTIMATION

DESTRUCTIVE • LABOR-INTENSIVE • TIME-CONSUMING • IMPRACTICAL



Slow...
Tedious...
Not real-time...

DESTRUCTIVE
Samples are cut, crushed, and destroyed

LABOR-INTENSIVE
Multiple manual steps and expertise required

TIME-CONSUMING
Hours to days per sample

IMPRACTICAL
Not suitable for real-time or high-throughput decision making

WET CHEMISTRY

- ☒ Weighing
- ☒ Digestion
- ☒ Dilution
- ☒ Analyze (ICP/OES)
- ☒ Calculations

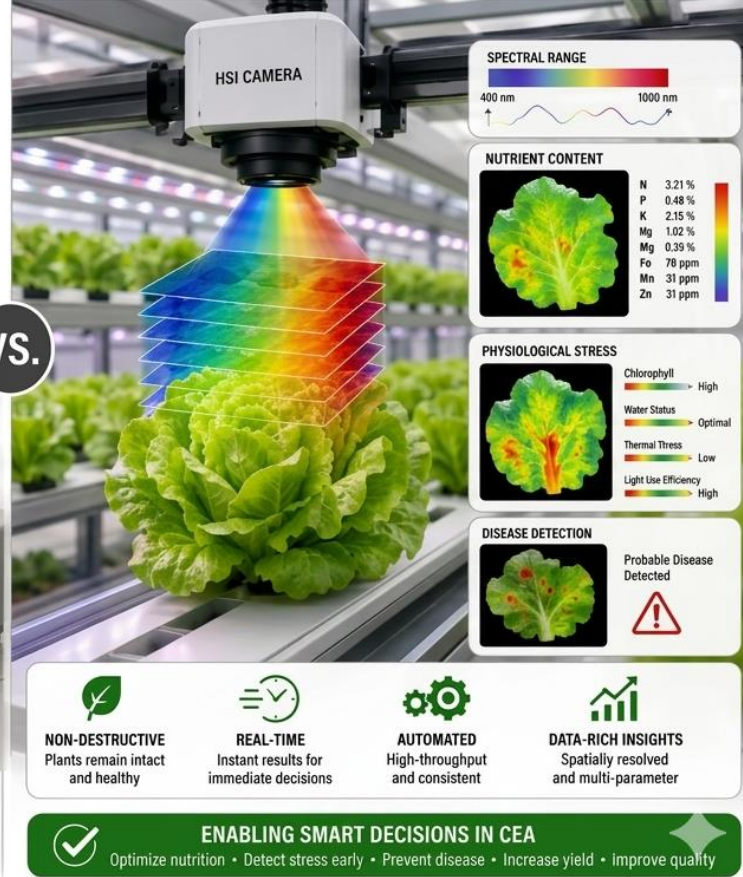
> 4-6 HOURS PER SAMPLE

LIMITATIONS

- Destructive sampling
- Laborious and error-prone
- Long turnaround time
- Not scalable for CEA operations
- Cannot capture spatial variability

HYPERSPECTRAL IMAGING (HSI) SOLUTION

NON-DESTRUCTIVE • AUTOMATED • REAL-TIME • SCALABLE



HSI CAMERA

SPECTRAL RANGE
400 nm 1000 nm

NUTRIENT CONTENT

N	3.21 %
P	0.48 %
K	2.15 %
Mg	1.02 %
Fe	0.39 %
Mn	78 ppm
Zn	31 ppm

PHYSIOLOGICAL STRESS

- Chlorophyll: High
- Water Status: Optimal
- Thermal Stress: Low
- Light Use Efficiency: High

DISEASE DETECTION
Probable Disease Detected

NON-DESTRUCTIVE
Plants remain intact and healthy

REAL-TIME
Instant results for immediate decisions

AUTOMATED
High-throughput and consistent

DATA-RICH INSIGHTS
Spatially resolved and multi-parameter

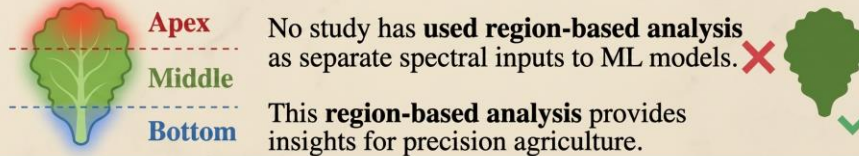
ENABLING SMART DECISIONS IN CEA
Optimize nutrition • Detect stress early • Prevent disease • Increase yield • Improve quality

Research gap & Objectives

WHAT OTHERS HAVE DONE

- ❑ Most HSI studies in crops focus on **leaf-only spectra** and commonly use **PCA, PLSR, SVM/SVR, RF, and ANN** [5][6].
- ❑ Studied effect of **pH, EC, temperature, O₂, humidity, light intensity, CO₂** [7][8][9].
- ❑ Application of **ML models such as ANN** for optimizing hydroponic lettuce [10].

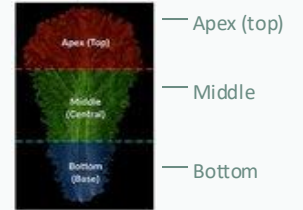
THE GAP



Objective 1

Evaluate intra-leaf spatial variability

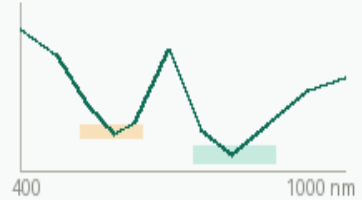
Compare ML model prediction performance across the apex, middle, bottom, and complete leaf regions using HSI data (400–1000 nm) to identify the most informative leaf zone for predicting pH, EC, NO₃⁻, Ca²⁺, and Brix.



Objective 2

Assess nutrient composition of different regions

Explore and compare spectral feature selection methods to identify the most informative wavelengths from HSI data as inputs to ML models for assessing biochemical composition across all leaf regions.



Objective 3

Develop a robust AI/ML model framework

Build, train, and validate robust machine learning regression models to achieve stable, reliable prediction of five biochemical quality parameters of lettuce grown under CEA conditions.



Methodology

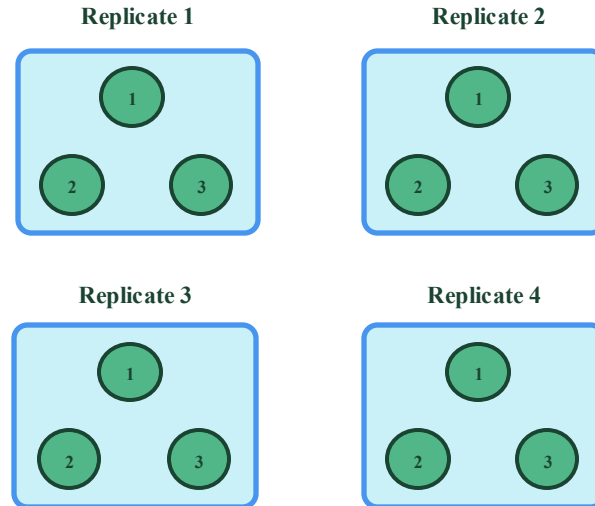
Lettuce Grown in Controlled Environment Agriculture

Lettuce type: Tacitus romaine lettuce



Physical Setup : Main & Backup Tanks

TANKS (Experimental)



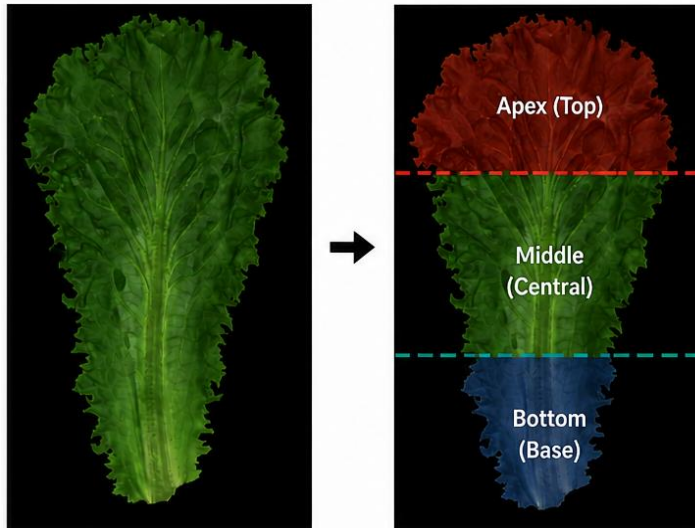
pH = 5.5-6.5

Adjust pH regularly

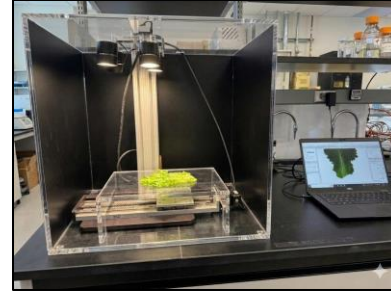
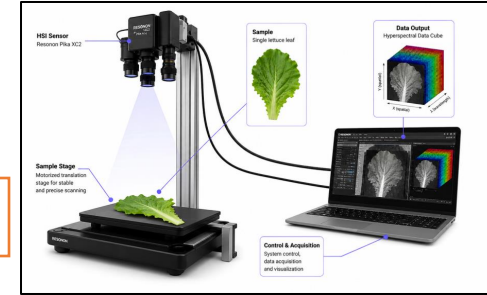
Week	Action
Week 1	Transplant 3 plants into each main tank
Week 2	Sample 1 plant from each tank (3-4 true leaves)
Week 3	Sample 2nd plant from each tank (7-8 true leaves)
Week 4	Sample 3rd (last) plant (10-15 true leaves)

Methodology Cont'd

HSI Data and Biochemical composition Collection



HSI data

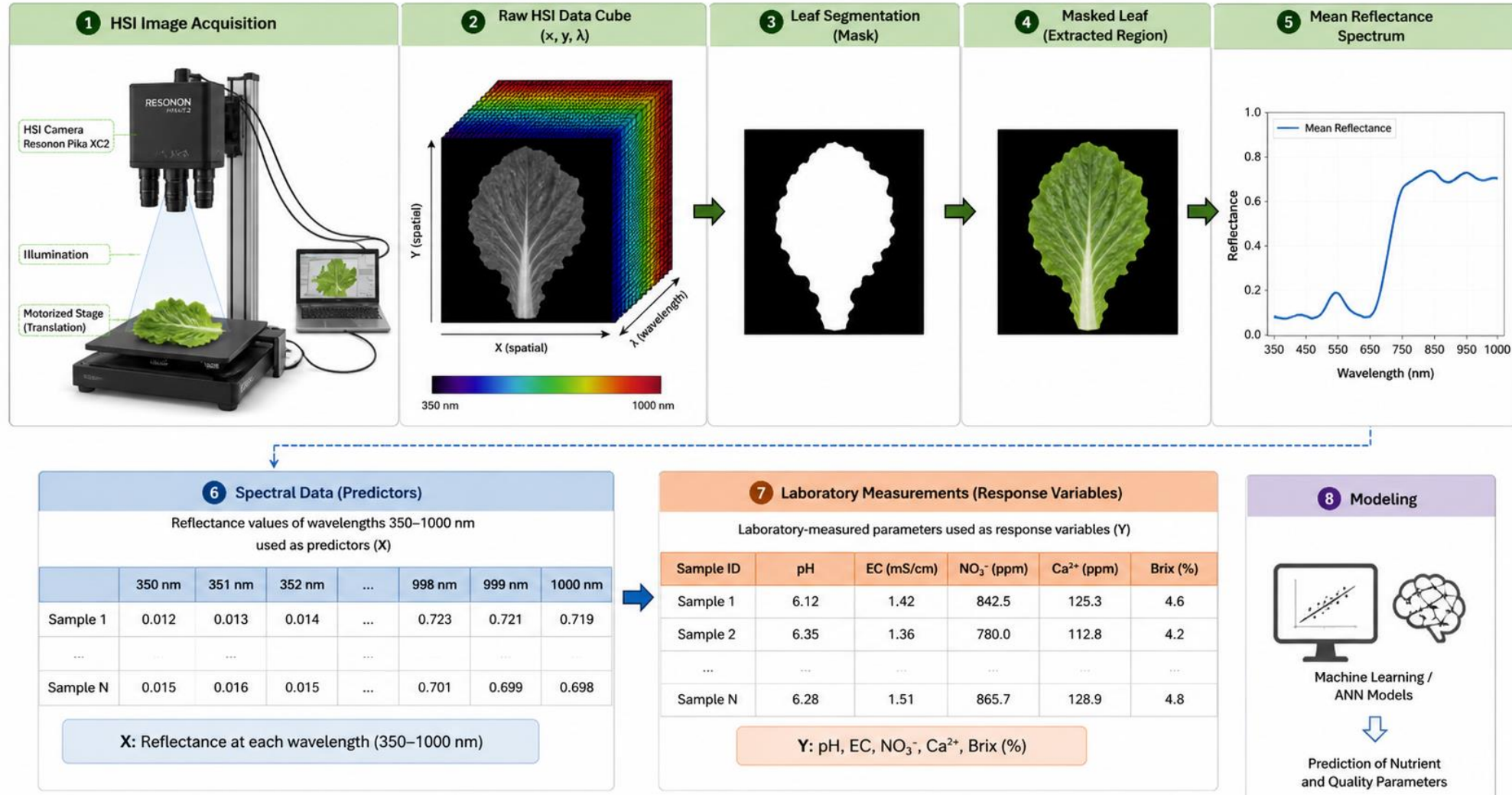


- pH
- EC
- NO_3^-
- Ca^{2+}
- Brix



Methodology Cont'd

HSI image analysis workflow

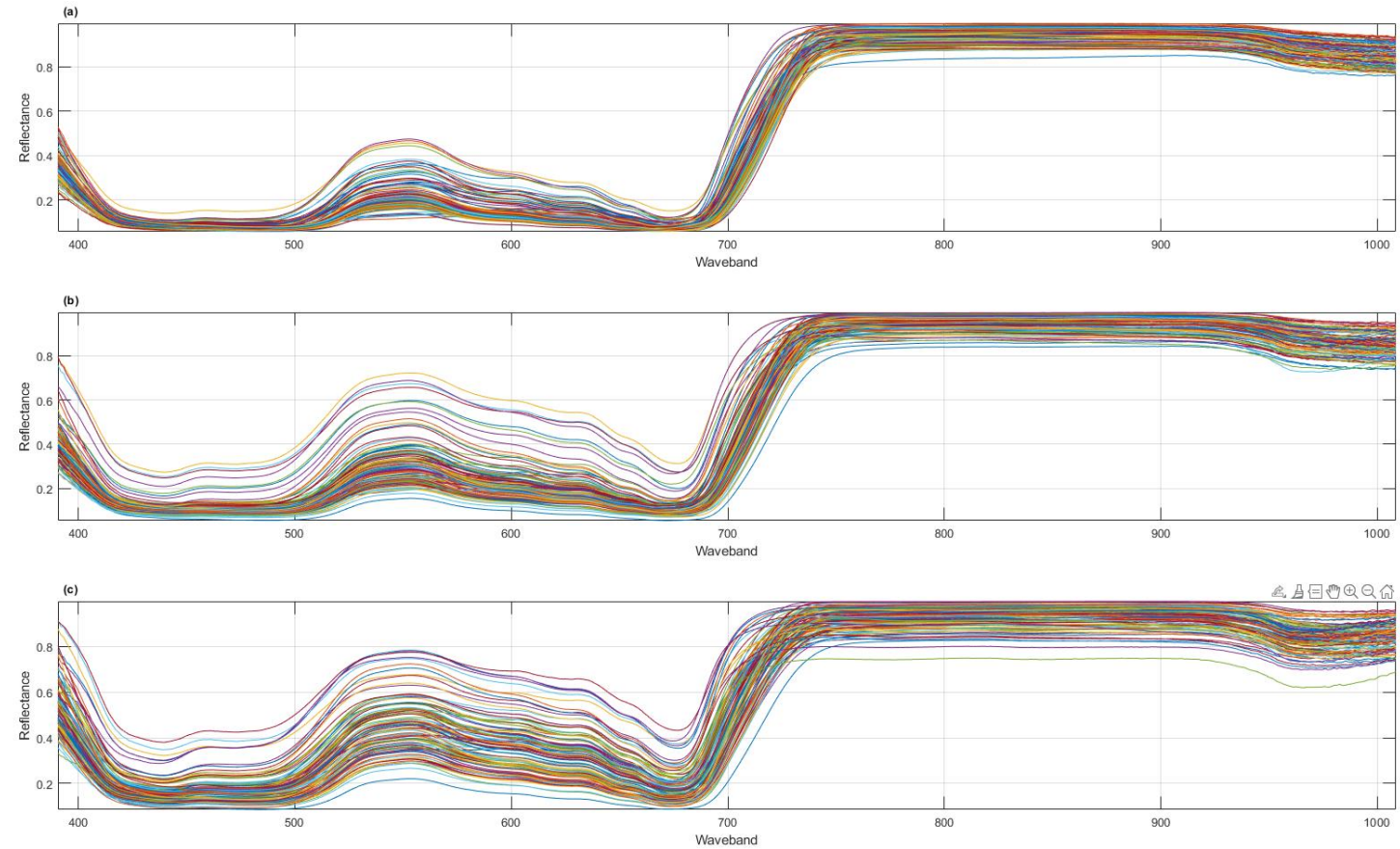
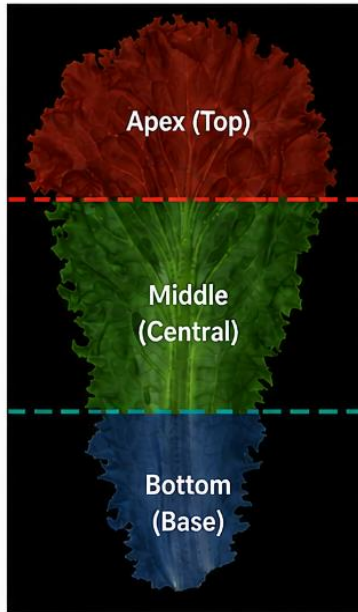


Results & Plots

Segmented leaf
(Background removed)



Leaf divided into
three regions



Conclusion & Overview

CORE OBJECTIVES

01

Evaluate intra-leaf spatial variability

Assess how Apex, Middle, Bottom, and complete leaf regions differ as spectral inputs for predicting pH, EC, NO_3^- , Ca^{2+} , and Brix.

02

Explore & compare feature selection methods

Investigate multiple spectral feature selection approaches to identify the most informative wavelengths for each leaf region — methods are not fixed and will be evaluated comparatively.

03

Explore ML models & validate nutrient assessment

Explore and compare multiple ML/AI model architectures across all leaf regions to find the most accurate, generalizable framework for non-destructive biochemical composition prediction.

WHY IT MATTERS

Replacing destructive lab tests

Methods destroy samples and take days. HSI predicts the same parameters non-destructively in real time.

Intra-leaf spatial biology is ignored

No prior HSI study separates apex, middle, and bottom zones. These regions have distinct stomatal densities and vein distributions that affect spectral signals.

Enabling real-time CEA monitoring

Controlled environment agriculture needs rapid, inline quality checks. This framework targets scalable, real-time nutrient monitoring for hydroponic production.

Drone Precision Landing Project

Md. Ibne Joha

Supervised By: Dr. Sulaymon Eshkabilov

Objectives:

1. Develop and Integrate a Vision-Assisted Precision Landing System for UAVs

To design, integrate, and validate a camera-based precision landing solution on the IF800 medium-lift UAV by establishing reliable MAVLink communication, landing-target detection, and autonomous precision landing functionality.

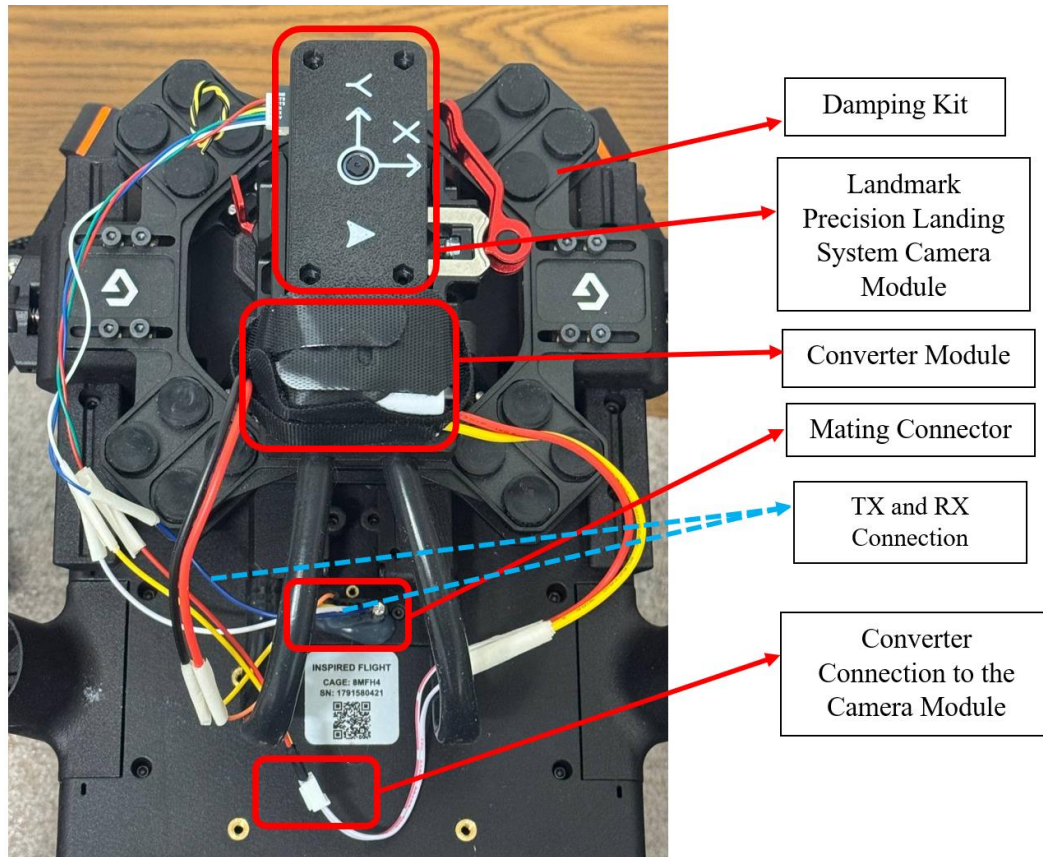
3. Conduct Data-Driven Analysis for System Validation and Optimization

To perform Python and MATLAB-based tools for flight-log extraction, 3D trajectory reconstruction and visualization, landing performance evaluation, camera behavior analysis, and failure investigation to support future system optimization and large-scale testing.

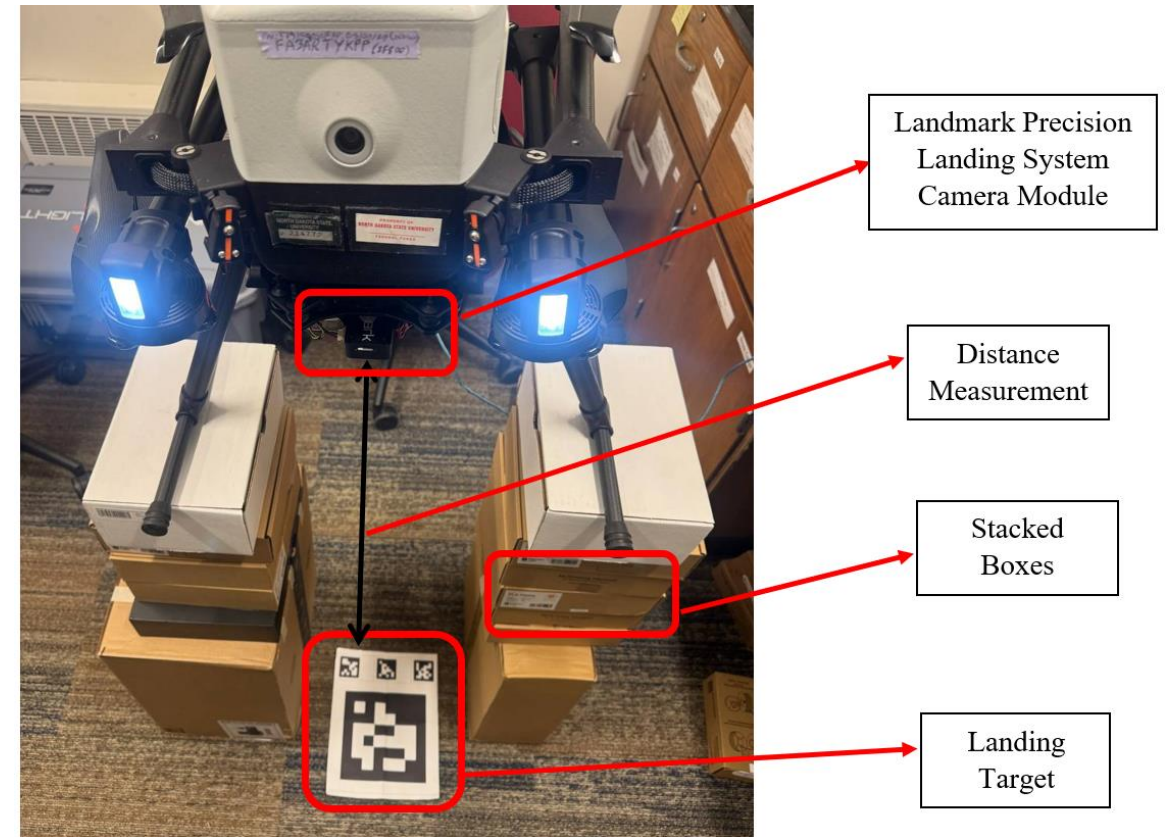
2. Evaluate Landing Accuracy and System Performance Under Real-World Conditions

To quantitatively assess the accuracy, reliability, and operational performance of the precision landing system through indoor validation, outdoor flight experiments, sensor verification, and landing error analysis using real flight log data.

Experimental Setup:



(a)



(b)

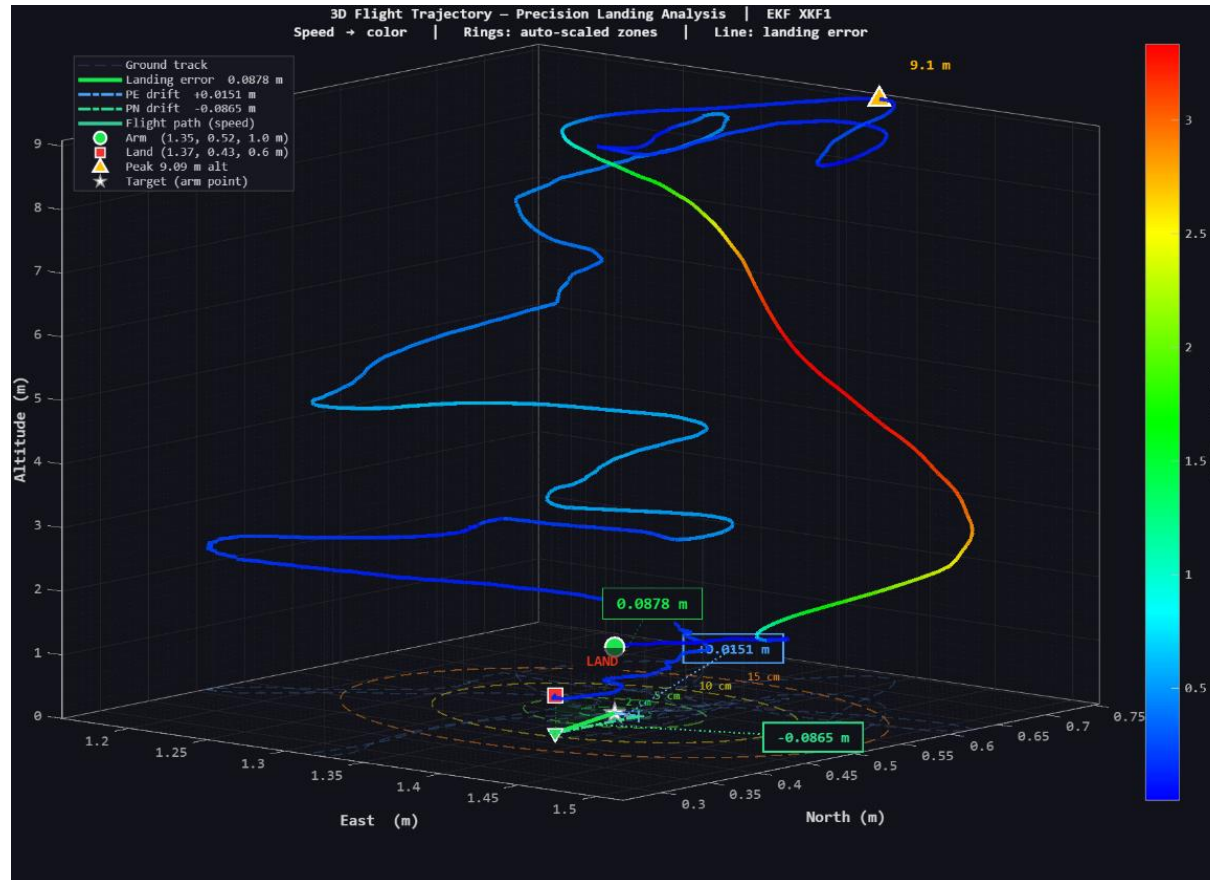
Figure 1: (a) Integrated Landmark precision landing system with the UAV (IF 800); (b) Validation of the precision landing system with the UAV positioned above a ground target at adjustable heights.

Experimental Setup :



Figure 02: (a) Verification of precision landing system functionality through telemetry monitoring in Mission Planner under real-world operational conditions; (b) Outdoor experimental assessment of the UAV precision landing capability.

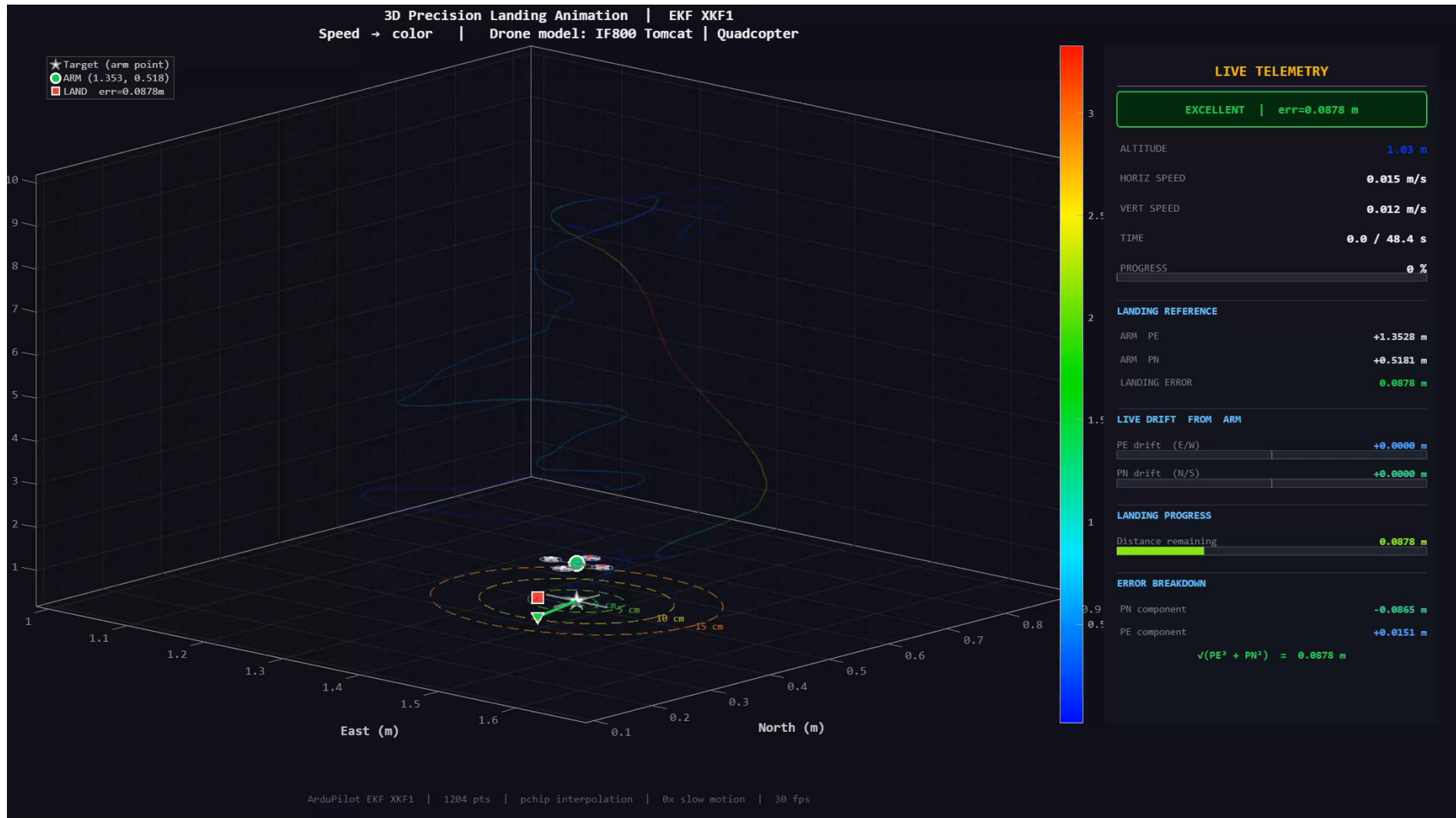
Results and Discussion:



FLIGHT SUMMARY	
EXCELLENT — < 10 cm	
LANDING ERROR	
Error (2D horiz)	0.0878 m
Error (3D total)	0.4437 m
PN drift (N/S)	-0.0865 m
PE drift (E/W)	+0.0151 m
Error bearing	170.1 deg
POSITION	
Arm point (PE,PN)	(1.353, 0.518) m
Max altitude	9.09 m
Max dist (arm)	0.26 m
PN range	0.26 to 0.76 m
PE range	1.16 to 1.52 m
VELOCITY	
Max 3D speed	3.33 m/s
Max horiz speed	0.27 m/s
Max climb rate	3.33 m/s
Max sink rate	0.61 m/s
FLIGHT	
EKF instance	0
Total time	48.4 s
Data points	1204
Sample rate	24.8 Hz
Accuracy zones (auto-scaled):	
2 cm	5 cm 10 cm 15 cm
EKF Testcase 0 1204 pts 24.8 Hz	

Figure 03: (a) MATLAB-based 3D trajectory reconstruction of a successful precision landing flight with a landing error of **8.78 cm**; (b) Key flight performance parameters, including arming position, landing position, landing error, and UAV speed.

Results and Discussion: (Landing Error = 8.78cm) (3D Animation)



Conclusion:

- ✓ The precision landing system was successfully integrated and validated, demonstrating reliable target detection and autonomous precision landing capability.
- ✓ High landing accuracy was achieved, with landing errors 8.78 cm under real world operating conditions.
- ✓ Flight-log analysis identified key factors affecting landing performance, providing valuable insights for future system improvement and large-scale testing.

Any Questions?

Thank you



Our Lab Website